

Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas/

January 25, 2019

Mr. Jonathon Travis
Ryan, LLC
2800 Post Oak Blvd, Ste 4200
Houston, TX 77056

**RE: Dahl Federal #15-11H
SWSE Sec. 11, T.153N., R.101W.
McKenzie County, North Dakota
Baker Field
Well File No. 21266
STRIPPER WELL DETERMINATION**

Dear Mr. Travis:

Oasis Petroleum North America LLC (Oasis) filed with the North Dakota Industrial Commission – Oil and Gas Division (Commission) on January 14, 2019 an application for a Stripper Well Property Determination for the above captioned well.

Information contained in the application indicates that the above mentioned well is a property pursuant to statute and rule, and Oasis has elected to designate said well as a separate property for stripper well purposes. The well produced from a well depth greater than 10000 feet. During the qualifying period, August 1, 2016 through July 31, 2017, the well produced at a maximum efficient rate or was not capable of exceeding the production threshold. The average daily production from the well was 29.9 barrels of oil per day during this period.

It is therefore determined that the above captioned well qualifies as a "Stripper Well Property" pursuant to Section 57-51.1-01 of the North Dakota Century Code. This determination is applicable only to the Bakken Pool in and under said property.

The Commission shall have continuing jurisdiction, and shall have the authority to review the matter, and to amend or rescind the determination if such action is supported by additional or newly discovered information. If you have any questions, do not hesitate to contact me.

Sincerely,

David J. McCusker
Petroleum Engineer

Cc: ND Tax Department

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

RECEIVED**OCT 25 2017**

Well File No.

21266**ND Oil & Gas Div.**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed April 16, 2015
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Well is now on rod pump

Well Name and Number Dahl 15-11H							
Footages		Qtr-Qtr		Section	Township	Range	
22 F S L	2488 F E L	SWSE	11		153 N	101 W	
Field Baker		Pool Bakken		County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective 04/16/2015, the above referenced well was equipped with a rod pump.

End of Tubing: 2-7/8" L-80 tubing @ 10170'

Pump: 2-1/2" x 2.0" x 36' insert pump @10082'

Company Oasis Petroleum North America LLC		Telephone Number 281 404-9494	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Sadie Goodrum	
Title Regulatory Specialist		Date October 23, 2017	
Email Address sgoodrum@oasispetroleum.com			

FOR STATE USE ONLY	
☒ Received	☐ Approved
Date	11-2-2017
By	
Title	JARED THUNE Engineering Technician



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

RECEIVED

OCT 27 2017

ND Oil & Gas Div.

Well File No.

21266

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date November 1, 2017
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Name Change

Well Name and Number Dahl 15-11H					
Footages 22 F S L 2488 F E L		Qtr-Qtr SWSE	Section 11	Township 153 N	Range 101 W
Field Baker		Pool Bakken		County McKenzie	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests approval to make the following change to the above referenced well:

Name change: Dahl Federal 15-11H (previously Dahl 15-11H)

The well is operated in a DSU with federal minerals. Therefore, Oasis requests to add "Federal" to the well name to clearly indicate to stakeholders the well has federal implications.

Please utilize the credit card on file for the associated fees.

cc \$25.00 11/1/17
cc \$25.00

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Jennifer Swenson	
Title Regulatory Specialist		Date October 26, 2017	
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 10-30-2017	By
Title David Burns Engineering Tech.	



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas/

September 21, 2017

OASIS PETRO NORTH AMERICA
ATTENTION: MICHAEL KUKUK
1001 FANNIN, STE 1500
HOUSTON, TX 77002

RE:

FOLEY FEDERAL 5301 43-12H
SWSE 12-153N-101W
MCKENZIE COUNTY
WELL FILE NO.: 20863

DAHL 15-11H
SWSE 11-153N-101W
MCKENZIE COUNTY
WELL FILE NO.: 21266

DAHL FEDERAL 2-15H
NWNE 15-153N-101W
MCKENZIE COUNTY
WELL FILE NO.: 21796

Dear Michael Kukuk:

A Sundry notice (Form 4) is needed for the above wells, detailing the changeover from flowing to well now on rod pump. If you have any questions, feel free to contact our office.

Sincerely,

Tom Delling
Tom Delling
Petroleum Engineer - Field Inspector

TKD/RSD/RLR



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas/

March 7, 2017

Oasis Petroleum North America, LLC
Attn: Michael Lou
1001 Fannin, Ste 1500
Houston, TX 77002

RE: CHANGE OF OPERATOR FROM SM ENERGY COMPANY TO OASIS
PETROLEUM NORTH AMERICA, LLC.
157 Wells

Dear Michael Lou:

Please find enclosed a copy of the approved Form 15, Notice of Transfer of Oil and Gas Wells, in regard to the above-referenced matter. This transfer has now been approved and subject well is now covered by Bond No. RLB0015967.

If you should have any questions, please feel free to contact this office.

Sincerely,

Rachel Morris
Administrative Assistant

Enclosure:

cc: SM Energy Company
Attn: Tim Keating
PO Box 7168
Billings, MT 59103

**NOTICE OF TRANSFER OF OIL AND GAS WELLS - FORM 15**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5782 (03-2000)

RECEIVED**JAN 17 2017****ND Oil & Gas Div.**

FOR STATE USE ONLY

NDIC Bond Number

0312

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM. PLEASE SUBMIT THE ORIGINAL AND SIX COPIES.
THIS NOTICE ALONG WITH A FEE OF \$25.00 PER WELL SHALL BE FILED AT LEAST THIRTY DAYS BEFORE THE CLOSING DATE OF TRANSFER.

TRANSFERRING OPERATOR

Name of Operator Representative Tim J. Keating			
Operator Transferring Oil and/or Gas Wells SM Energy Company			Telephone Number 406-245-6248
Address P.O. Box 7168	City Billings	State MT	Zip Code 59103
I, the above named representative, acknowledge the transfer of the oil and/or gas wells named below for the purpose of ownership and/or operation to the company named below.			
Signature 		Title (Must be an officer or power of attorney must be attached) Attorney in Fact	Date 12/1/16

Well File Number	Requested Official Well Name and Number	Location (Qtr-Qtr, S-T-R)	Assignment Date
	See Attached list of wells	Various	October 1, 2016

RECEIVING OPERATOR

Name of Operator Representative Michael H. Lou			
Operator Receiving Oil and/or Gas Wells Oasis Petroleum North America LLC			Telephone Number (281) 404-9500
Address 1001 Fannin Suite 1500	City Houston	State TX	Zip Code 77002
I, the above named representative, have read the foregoing statement and accept such transfer, also the responsibility of ownership and/or operation of said well or wells, under the said company bond, said bond being tendered to or on file with the Industrial Commission of North Dakota.			
Signature 		Title (Must be an officer or power of attorney must be attached) CEO	Date 12/1/16

SURETY COMPANY

Surety RLI Insurance Company		Telephone Number (713) 961-1300	Amount of Bond \$ 100,000.00
Address 2925 Richmond Avenue, Suite 1600	City Houston	State TX	Bond Number RLB0015967
The above named SURETY agrees that such bond shall extend to compliance with Chapter 38-08 of North Dakota Century Code and amendments and the rules and regulations of the Industrial Commission of North Dakota prescribed to govern the production of oil and gas on government and private lands within the State of North Dakota, in relation to the above stated transfer; it being further agreed and understood that the bond sum or amount is not to be considered increased because of such extension.			
Signature 		Title (Must be an officer or power of attorney must be attached)	Date 12/1/2016
Printed Name Robbie Duxbury, Attorney-in-Fact			

FOR STATE USE ONLY

Date Approved March 17, 2017
By Bruce S. Fuhs
Title Assistant Director

NDIC

Well File No.	API No.	NDIC Well Name	County	Section	Township	Range	Qtr-Qtr
7503	33105008220000	MARLEY STATE 1-36HR	WILLIAMS	36	153 N	104 W	SESE
8957	33053014010000	WILLOW 1-4	MCKENZIE	4	152 N	101 W	SWNW
9309	33053014780000	LINDVIG 1-11HR	MCKENZIE	11	153 N	101 W	SESE
9610	33053015400000	NORBY 2A-20HR	MCKENZIE	20	153 N	95 W	SESW
10088	33053016570000	ECKERT 2-6	MCKENZIE	6	152 N	101 W	SWNE
10089	33053016580000	ECKERT 2-5HR	MCKENZIE	5	152 N	101 W	NESW
10157	33053016880000	THOMPSON-FEDERAL 1-19HR	MCKENZIE	19	153 N	95 W	SENE
10320	33053017310000	FEDERAL 19-42	MCKENZIE	19	150 N	103 W	SENE
10560	33053017850000	FEDERAL 20-14	MCKENZIE	20	150 N	103 W	SWSW
10812	33053018460000	ECKERT 4-5	MCKENZIE	5	152 N	101 W	NWSE
11009	33053019070000	ECKERT 5-12B	MCKENZIE	5	152 N	101 W	LT 5
11059	33053019180000	ANDERSON 10 32	MCKENZIE	10	152 N	101 W	SWNE
11070	33053019230000	ECKERT 5-5	MCKENZIE	5	152 N	101 W	LOT 7
11151	33053019480000	LINDVIG 1-4	MCKENZIE	4	152 N	101 W	SWSW
11316	33053019980000	LINDVIG 10 1	MCKENZIE	10	152 N	101 W	NENW
11345	33053020060000	FREDRICKSON 33-33 1	MCKENZIE	33	153 N	101 W	NWSE
11492	33053020470000	STATE OF NORTH DAKOTA F 1	MCKENZIE	33	153 N	101 W	NWNE
11523	33053020540000	POWERS 1	MCKENZIE	33	153 N	101 W	NESW
11617	33053020760000	HAGEN 1-13	MCKENZIE	13	153 N	95 W	SESW
11624	33053020790000	ANDERSON 10-13	MCKENZIE	13	149 N	96 W	NWSE
11700	33053020890000	B. A. GREEN 1	MCKENZIE	5	151 N	101 W	SWSW
11913	33053021460000	BEAR DEN FEDERAL 10-25HR	MCKENZIE	25	149 N	96 W	NWSE
11990	33053021590000	INDIAN HILL STATE 1-36	MCKENZIE	36	153 N	102 W	SWSE
12025	33053021650000	STATE 2-36HR	MCKENZIE	36	153 N	102 W	NESW
12216	33053022030000	ISZLEY-USA 1	MCKENZIE	1	150 N	104 W	NWSW
12566	33053022620000	ELLETSON 33-1HR	MCKENZIE	1	150 N	104 W	NWSE
13457	33105014130000	TURMOIL 10-4	WILLIAMS	4	152 N	104 W	NWSE
15170	33053025460000	BARNES 1-2	MCKENZIE	2	152 N	102 W	NENE
15344	33053025550000	REHAB 4-33	MCKENZIE	33	153 N	101 W	NWNW
15915	33053026690000	FANCY BUTTE 14-32	MCKENZIE	32	151 N	96 W	SESW
16351	33053027540000	FROHOLM 2-24H	MCKENZIE	24	153 N	95 W	NWNE
17662	33053029440000	BEAR DEN FEDERAL 4Y-30H	MCKENZIE	30	149 N	95 W	NWNW
17681	33053029480000	ACADIA 31-25H	MCKENZIE	25	151 N	97 W	NWNE
17778	33053029670000	KIRKLAND 8-13H	MCKENZIE	13	149 N	96 W	SENE
17952	33053030010000	KLAMM 13-10H	MCKENZIE	10	150 N	97 W	SWSW
18248	33053030430000	BEAR DEN FEDERAL 4Z-30H	MCKENZIE	30	149 N	95 W	NWNW
18436	33053030760000	ANDERSON 12-14H	MCKENZIE	14	149 N	96 W	NWSW
18579	33053030940000	JORGENSEN 1-30H	MCKENZIE	30	149 N	95 W	NENE
18580	33053030950000	NELSON 4-29H	MCKENZIE	29	151 N	96 W	NWNW
18805	33053031290000	JOHNSON 16-34H	MCKENZIE	34	150 N	98 W	SESE
18814	33053031320000	WILSON 8-20H	MCKENZIE	20	153 N	95 W	SENE
18815	33053031330000	NORBY 9-20H	MCKENZIE	20	153 N	95 W	NESE
18816	33053031340000	NORBY 16-20H	MCKENZIE	20	153 N	95 W	SESE
18957	33053031540000	LEE 13-8H	MCKENZIE	8	152 N	101 W	SWSW
19049	33053031660000	JORGENSEN 1X-30H	MCKENZIE	30	149 N	95 W	NENE
19303	33053032120000	SEATTLE 1-35H	MCKENZIE	35	152 N	100 W	SWSE
19324	33053032170000	FOSSUM 15-35H	MCKENZIE	35	153 N	101 W	SWSE
19390	33053032340000	BRODERSON 13-35H	MCKENZIE	35	150 N	98 W	SWSW
19997	33053033710000	JOHNSRUD 1-1H	MCKENZIE	1	150 N	99 W	LOT 1
20022	33053033790000	JAYNES 16-12H	MCKENZIE	12	151 N	100 W	SESE
20053	33053033850000	NELSON 15-11H	MCKENZIE	11	150 N	97 W	SWSE
20266	33053034230000	WOLD 15-33H	MCKENZIE	33	153 N	97 W	SWSE
20318	33053034350000	LEISETH 16-13H	MCKENZIE	13	151 N	100 W	SESE

NDIC

Well File No.	API No.	NDIC Well Name	County	Section	Township	Range	Qtr-Qtr
20485	33053034980000	BARNES 2-2H	MCKENZIE	2	152 N	102 W	LOT2
20497	33053034990000	LEISETH 1-24H	MCKENZIE	24	151 N	100 W	NENE
20592	33053035300000	SCHMITZ 3-4H	MCKENZIE	4	152 N	102 W	LOT3
20661	33053035520000	HOEHN 2-13H	MCKENZIE	13	152 N	102 W	NWNE
20733	33053035770000	STENBERG 16-8H	MCKENZIE	8	151 N	99 W	SESE
20752	33105021920000	KAROMOJO 2-4H	WILLIAMS	4	152 N	104 W	LOT2
20867	33053036100000	STEPANEK 1-18H	MCKENZIE	18	152 N	101 W	NENE
21101	33053036620000	SYVERSON 16-34H	MCKENZIE	34	152 N	100 W	SESE
21258	33053037010000	NORA 13-9H	MCKENZIE	9	151 N	100 W	SWSW
21266	33053037030000	DAHL 15-11H	MCKENZIE	11	153 N	101 W	SWSE
21796	33053038460000	DAHL FEDERAL 2-15H	MCKENZIE	15	153 N	101 W	NWNE
21985	33053038930000	KIRKLAND 4-18H	MCKENZIE	18	149 N	95 W	NWNW
22376	33053039730000	ANDERSON 5X-14H	MCKENZIE	14	149 N	96 W	SWNW
22377	33053039740000	ANDERSON 5-14H	MCKENZIE	14	149 N	96 W	SWNW
22475	33105025470000	DISHON 1-30H	WILLIAMS	30	153 N	103 W	NENE
22520	33053040200000	ANDERSON FEDERAL 16-15H	MCKENZIE	15	149 N	96 W	SESE
22691	33105025820000	ANDERSON 14-20H	WILLIAMS	20	153 N	103 W	SESW
22700	33053040610000	NORBY 9X-20HA	MCKENZIE	20	153 N	95 W	NESE
22701	33053040620000	NORBY 9X-20HB	MCKENZIE	20	153 N	95 W	NESE
22816	33105026010000	CAPSTICK 2-4H	WILLIAMS	4	152 N	104 W	LOT2
22819	33053041000000	OAKLAND 13-31H	MCKENZIE	31	151 N	103 W	LOT4
22975	33053041440000	HOLM 14-12HB	MCKENZIE	12	150 N	99 W	SESW
22976	33053041450000	HOLM 14X-12H	MCKENZIE	12	150 N	99 W	SESW
22977	33053041460000	HOLM 14-12HA	MCKENZIE	12	150 N	99 W	SESW
23121	33053041920000	HOLM 14X-12HA	MCKENZIE	12	150 N	99 W	SESW
23122	33053041930000	HOLM 13-12H	MCKENZIE	12	150 N	99 W	SWSW
23256	33053042140000	ARNOLD 16X-12H	MCKENZIE	12	150 N	99 W	SESE
23257	33053042150000	DOROTHY 16-12H	MCKENZIE	12	150 N	99 W	SESE
23524	33053043010000	CEYNAR 4-18HA	MCKENZIE	18	151 N	99 W	LOT1
23525	33053043020000	CEYNAR 4X-18H	MCKENZIE	18	151 N	99 W	LOT1
23526	33053043030000	CEYNAR 4-18HB	MCKENZIE	18	151 N	99 W	LOT1
24039	33053044620000	BRODERSON 2X-27HB	MCKENZIE	27	150 N	98 W	NWNE
24040	33053044630000	BRODERSON 2-27H	MCKENZIE	27	150 N	98 W	NWNE
24041	33053044640000	BRODERSON 2X-27HA	MCKENZIE	27	150 N	98 W	NWNE
24042	33053044650000	DIDRICK 4-27HB	MCKENZIE	27	150 N	98 W	NENW
24043	33053044660000	DIDRICK 4X-27H	MCKENZIE	27	150 N	98 W	NWNW
24044	33053044670000	DIDRICK 4-27HA	MCKENZIE	27	150 N	98 W	NWNW
24190	33053045180000	JORGENSEN FEDERAL 14-19H	MCKENZIE	19	149 N	95 W	SWSE
24191	33053045190000	JORGENSEN FEDERAL 14X-19H	MCKENZIE	19	149 N	95 W	SWSE
24387	33053045870000	BEHAN 2-29H	MCKENZIE	29	153 N	101 W	NWNE
24402	33053045920000	KOESER 3X-26HB	MCKENZIE	26	150 N	98 W	NENW
24403	33053045930000	KOESER 3-26H	MCKENZIE	26	150 N	98 W	NENW
24404	33053045940000	KOESER 3X-26HA	MCKENZIE	26	150 N	98 W	NENW
24446	33053046060000	LUCILLE 1-27H	MCKENZIE	27	150 N	98 W	NENE
24447	33053046070000	LUCILLE 1X-27H	MCKENZIE	27	150 N	98 W	NENE
24524	33053046410000	KOESER 4X-26H	MCKENZIE	26	150 N	98 W	NWNW
24654	33053046880000	HARTEL 1-26HA	MCKENZIE	26	150 N	98 W	NENE
24655	33053046890000	HARTEL 1X-26H	MCKENZIE	26	150 N	98 W	NENE
24656	33053046900000	HARTEL 1-26HB	MCKENZIE	26	150 N	98 W	NENE
24955	33053047830000	CADE 12-19HB	MCKENZIE	19	151 N	99 W	LOT3
24956	33053047840000	CADE 12X-19H	MCKENZIE	19	151 N	99 W	LOT3
24957	33053047850000	CADE 12-19HA	MCKENZIE	19	151 N	99 W	LOT3
25000	33053047960000	ANNIE 12X-18HB	MCKENZIE	18	151 N	99 W	LOT3
25001	33053047970000	ANNIE 12-18H	MCKENZIE	18	151 N	99 W	LOT3
25002	33053047980000	ANNIE 12X-18HA	MCKENZIE	18	151 N	99 W	LOT3

NDIC

Well File No.	API No.	NDIC Well Name	County	Section	Township	Range	Qtr-Qtr
25518	33053049650000	WALLA 13X-19H	MCKENZIE	19	151 N	99 W	LOT4
25519	33053049660000	WALLA 13-19H	MCKENZIE	19	151 N	99 W	LOT4
26135	33053051770000	DORIS 1X-13H	MCKENZIE	13	151 N	100 W	NENE
26136	33053051780000	ELERY 1-13H	MCKENZIE	13	151 N	100 W	NENE
26228	33053052160000	WILSON FEDERAL 1X-20H	MCKENZIE	20	153 N	95 W	NENE
26254	33053052310000	OAKLAND FEDERAL 1-1H	MCKENZIE	1	150 N	104 W	LOT1
26374	33053052560000	LORAIN 1X-20H	MCKENZIE	20	153 N	95 W	NENE
26658	33053053560000	HANK 13X-7H	MCKENZIE	7	151 N	99 W	SWSW
26659	33053053570000	EMMA 13-7H	MCKENZIE	7	151 N	99 W	SWSW
26750	33053053870000	RICK 16X-12H	MCKENZIE	12	150 N	99 W	SESE
26851	33053054250000	BONNER 9X-12HA	MCKENZIE	12	151 N	100 W	NESE
26852	33053054260000	BONNER 9-12H	MCKENZIE	12	151 N	100 W	NESE
26853	33053054270000	BONNER 9X-12HB	MCKENZIE	12	151 N	100 W	NESE
27622	33053056720000	DONNA 12X-35H	MCKENZIE	35	152 N	100 W	NWSW
27623	33053056730000	CAROL 12-35H	MCKENZIE	35	152 N	100 W	NWSW
27624	33053056740000	DIXIE 12X-35H	MCKENZIE	35	152 N	100 W	NWSW
27767	33053057110000	TODD 13X-35H	MCKENZIE	35	152 N	100 W	SWSW
27768	33053057120000	LUELLA 13-35H	MCKENZIE	35	152 N	100 W	SWSW
27769	33053057130000	TRACY 13-35H	MCKENZIE	35	152 N	100 W	SWSW
27937	33053057550000	ELWAY 1-32H	MCKENZIE	32	152 N	100 W	NENE
27938	33053057560000	MANNING 1X-32H	MCKENZIE	32	152 N	100 W	NENE
27939	33053057570000	DAVIS 1-32H	MCKENZIE	32	152 N	100 W	NENE
28196	33053058510000	PHYLIS 14X-12H	MCKENZIE	12	151 N	100 W	SESW
28197	33053058520000	LUKE 14-12H	MCKENZIE	12	151 N	100 W	SESW
28198	33053058530000	JESSE 14X-12H	MCKENZIE	12	151 N	100 W	SESW
28200	33053058540000	CALVIN 2-13H	MCKENZIE	13	151 N	100 W	NWNE
28201	33053058550000	DALLAS 2X-13H	MCKENZIE	13	151 N	100 W	NWNE
28812	33053060930000	JAMES 16X-35HB	MCKENZIE	35	152 N	100 W	SESE
28813	33053060940000	JAMES 16-35H	MCKENZIE	35	152 N	100 W	SESE
28815	33053060960000	JAMES 16X-35HA	MCKENZIE	35	152 N	100 W	SESE
28890	33053061140000	STENEHJEM 15-9HB	MCKENZIE	9	151 N	100 W	SWSE
28891	33053061150000	STENEHJEM 15X-9H	MCKENZIE	9	151 N	100 W	SWSE
28892	33053061160000	STENEHJEM 15-9HA	MCKENZIE	9	151 N	100 W	SWSE
29402	33053062610000	SEATTLE FEDERAL 14X-35H	MCKENZIE	35	152 N	100 W	SESW
29403	33053062620000	SEATTLE FEDERAL 14-35H	MCKENZIE	35	152 N	100 W	SESW
29404	33053062630000	SEATTLE 14X-35H	MCKENZIE	35	152 N	100 W	SESW
30270	33053065940000	STENEHJEM 14X-9HA	MCKENZIE	9	151 N	100 W	SESW
30271	33053065950000	STENEHJEM 14-9H	MCKENZIE	9	151 N	100 W	SESW
30272	33053065960000	STENEHJEM 14X-9HB	MCKENZIE	9	151 N	100 W	SESW
30754	33053067430000	FOSSUM 15-35HR	MCKENZIE	35	153 N	101 W	SWSE
30790	33053067560000	STACEY 13-9H	MCKENZIE	9	151 N	100 W	SWSW
30791	33053067570000	ASHLEY 13X-9H	MCKENZIE	9	151 N	100 W	SWSW
30889	33053067920000	STENBERG 5-9H	MCKENZIE	9	151 N	99 W	SWNW
30890	33053067930000	STENBERG 4X-9H	MCKENZIE	9	151 N	99 W	NWNW
30891	33053067940000	STENBERG 4-9H	MCKENZIE	9	151 N	99 W	NWNW
31064	33053068700000	RICH 1X-16HA	MCKENZIE	16	151 N	100 W	NENE
31065	33053068710000	STEVE 1-16H	MCKENZIE	16	151 N	100 W	NENE
31066	33053068720000	RINI 1X-16HB	MCKENZIE	16	151 N	100 W	NENE
90002	33053900020000	USA 43-11	MCKENZIE	11	150 N	104 W	NESE
90261	33053902610000	CEYNAR 4-18SWD	MCKENZIE	18	151 N	99 W	LOT1

**AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE - Form 8**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)

Received**DEC 05 2016**

Well File No.

21266

NDIC CTB No.

121266

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

ND Oil & Gas Division

Well Name and Number DAHL 15-11H	Qtr-Qtr SWSE	Section 11	Township 153	Range 101	County McKenzie
--	------------------------	----------------------	------------------------	---------------------	---------------------------

Operator Oasis Petroleum North America LLC	Telephone Number (281) 404-9573	Field BAKER
--	---	-----------------------

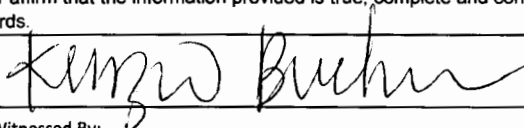
Address 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
---	------------------------	--------------------	--------------------------

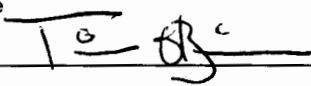
Name of First Purchaser Oasis Petroleum Marketing LLC	Telephone Number (281) 404-9627	% Purchased 100%	Date Effective December 1, 2016
Principal Place of Business 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
Field Address	City	State	Zip Code

Transporter Targa	Telephone Number 713-584-1000	% Transported 100%	Date Effective December 1, 2016
Address 1000 Louisiana, Suite 4300	City Houston	State TX	Zip Code 77002

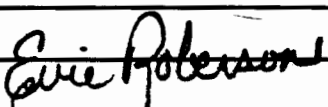
The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Comments		

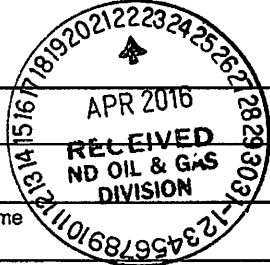
I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.		Date December 1, 2016
Signature 	Printed Name Kenzie Buchanan	Title Marketing Analyst I

Above Signature Witnessed By:		
Signature 	Printed Name Tim O'Brien	Title Marketing Analyst II

FOR STATE USE ONLY

Date Approved MAR 09 2017
By 
Title Oil & Gas Production Analyst

North Dakota Industrial Commission Follow-up Spill Report

API Number 33 - 053 - 03703								Well File or Facility No. 21266					
Operator SM Energy Company								Telephone Number 406-869-8642					
Address P O Box 7168						City Billings		State MT		Zip Code 59103			
Well Name and Number or Facility Name Dahl 15-11H						Field Baker							
Location of Well or Facility		Footages 22 F S L 2488 F E L		Qtr-Qtr SWSE		Section 11		Township 153 N		Range 101 W			
										County McKenzie			
Description of Spill Location if not on Well or Facility Site and/or Distance and Direction from Well or Facility Well Site Facility													
Directions to Site													
Release Discovered By Gary Bjerke			Date Release Discovered April 5, 2016			Time Release Discovered 12 : 00 PM			Date Release Controlled April 5, 2016			Time Release Controlled 12 : 30 PM	
Company Personnel Notified Luke Studer			How Notified Internal Reporting System						Date Notified April 5, 2016			Time Notified 12 : 30 PM	
Type of Incident Other-Described Below				Root Cause of Release Other-Described Below				Date Clean up Activities Concluded April 8, 2016					
Distance to Nearest Residence or Occupied Building 2 Miles				Distance to Nearest Fresh Water Well 2 Miles									
Piping Specifics (If Applicable)		Size (Decimal Format) "		Type				Location of Piping					
Volume of Release		Oil 0.00 Barrels		Saltwater 12.00 Barrels				Other 0.00 Barrels					
Volume of Release Recovered		Oil 0.00 Barrels		Saltwater 0.00				Other 0.00 Barrels					
Was Release Contained Within Dike			If No, Was Release Contained on Well Site Yes			If No, Was Release Contained on Facility Site or Pipeline ROW							
Areal Extent of Release if not Within Dike				Affected Medium Well/Facility Soil				General Land Use Well/Facility Site					
Describe Cause of Release or Fire and Other Type of Incidents, Root Causes of Release, Land Uses, and Released Substances 12 barrels of SW was discovered in corner of location. Was determined that the release was not from any production activity and is believed to be from an illegal dumping.													
Action Taken to Control Release and Clean Up Action Undertaken The contaminated soil was dug up and disposed of at an authorized disposal site.													
Potential Environmental Impacts													
Planned Future Action and/or Action Taken to Prevent Reoccurrence													
Where Were Recovered Liquids Disposed						Where Were Recovered Solids Disposed							
Weather Conditions		Wind Speed MPH		Wind Direction		Temperature ° F		Skies		Estimated Cleanup Cost \$		Damage Value \$	
Regulatory Agencies/Others Notified NDIC/NDDH				Person Notified on line		Date Notified April 6, 2016		Time Notified 3 : 30 PM		Notified By Luke Studer			
Fee Surface Owner													
Federal Agency				Lease Number									
BLM													
USFS													
Report Originator Lorena Griggs <i>Lorena Griggs</i>						Title Regulatory&Safety Assistant			Date April 18, 2016				
Reviewed By						Title			Date				



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (03-2004)



Well File No.

21266

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed March 25, 2015
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other Gas Analysis	

Well Name and Number Dahl 15-11H					
Footages 22 F S L 2488 F E L		Qtr-Qtr SWSE	Section 11	Township 153 N	Range 101 W
Field Baker		Pool Bakken		County McKenzie	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

SM Energy Company respectfully submits gas analysis on the DAHL 15-11H. Please see attached for further detail.

Company SM Energy Co.		Telephone Number 406-245-6248	
Address P. O. Box 7168			
City Billings		State MT	Zip Code 59103-7168
Signature 		Printed Name Cody Hoskins	
Title Engineering Tech		Date July 13, 2015	

FOR STATE USE ONLY	
☒ Received	☐ Approved
Date August 16, 2015	
By 	
Title PETROLEUM ENGINEER	

Gas Analysis Certificate Report
Hiland Partners

3/25/2015 3:59 PM

P.O.Box 5103 Enid, Ok. 73702
(O) 580.242.6040 (Fax) 580.616.2083
EnidMeasurement@kindermorgan.com

Betty Furnage

Analysis ID: 7833029		Alternate ID:		Use Contract Values: No	
Name	DAHL 15-11H	Company Name: Hiland Partners			
Effective Date:	08/01/2014 09:00	Saturated HV:	1429.9	Sample Date:	08/21/2014
Valid Thru Date:	12/31/2078 00:00	As Del. HV:		Sample ID:	
Last Update:	09/09/2014 09:33	Dry HV:	1454.5	Sample Type:	Spot
Data Acquisition:	File Transfer from Lab	Measured HV:		Sample Pressure Base:	14.730
Data Source:	Lab Analysis	WOBBE:	1553.5	Sample Temperature:	71.0
Gravity:	0.8766	Water Content:		Sample Pressure:	54.0
Status:	Active			Lab Code:	

Component	% Mol	GPM	H2S PPM
Methane	57.8075		
Ethane	22.5945	6.0728	
Propane	11.1930	3.0991	
I Butane	0.8897	0.2926	
N Butane	2.7769	0.8798	
I Pentane	0.4250	0.1562	
N Pentane	0.5247	0.1911	
Hexanes +	0.4037	0.1770	
Nitrogen	2.6152		
CO2	0.7697		
Oxygen	0.0000		
H2O	0.0000		
CO	0.0000		
H2S	0.0002		
Hydrogen	0.0000		
Helium	0.0000		
Argon	0.0000		
Total	100.0001	10.8686	

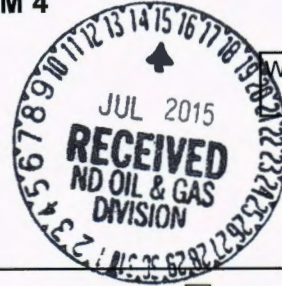
Sample Comments:

Configuration Comments:



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (03-2004)



Well File No.
21266

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed July 2, 2015
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	<u>Site Facility Diagram</u>

Well Name and Number Dahl 15-11H					
Footages 22 F S L 2499 F E L		Qtr-Qtr SWSE	Section 11	Township 153 N	Range 101 W
Field Baker		Pool Bakken		County McKenzie	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

SM Energy Company respectfully submits the site facility diagram for the above well. Please see attached for further detail.

Company SM Energy Co.		Telephone Number 406-245-6248	
Address P. O. Box 7168			
City Billings		State MT	Zip Code 59103-7168
Signature 		Printed Name Cody Hoskins	
Title Engineering Tech		Date July 13, 2015	

FOR STATE USE ONLY	
☒ Received	☐ Approved
Date 7/15/15	
By 	
Title Ruth Supina	

**SM ENERGY CO
SITE SECURITY DIAGRAM:**

WELL: DAHL 15-11H
NDIC NO: 21266
FIELD NAME: BAKER FIELD
LOCATION: SWSE SEC 11-T153N-R101W
COUNTY: DIVIDE STATE: ND
CA# NDM105420

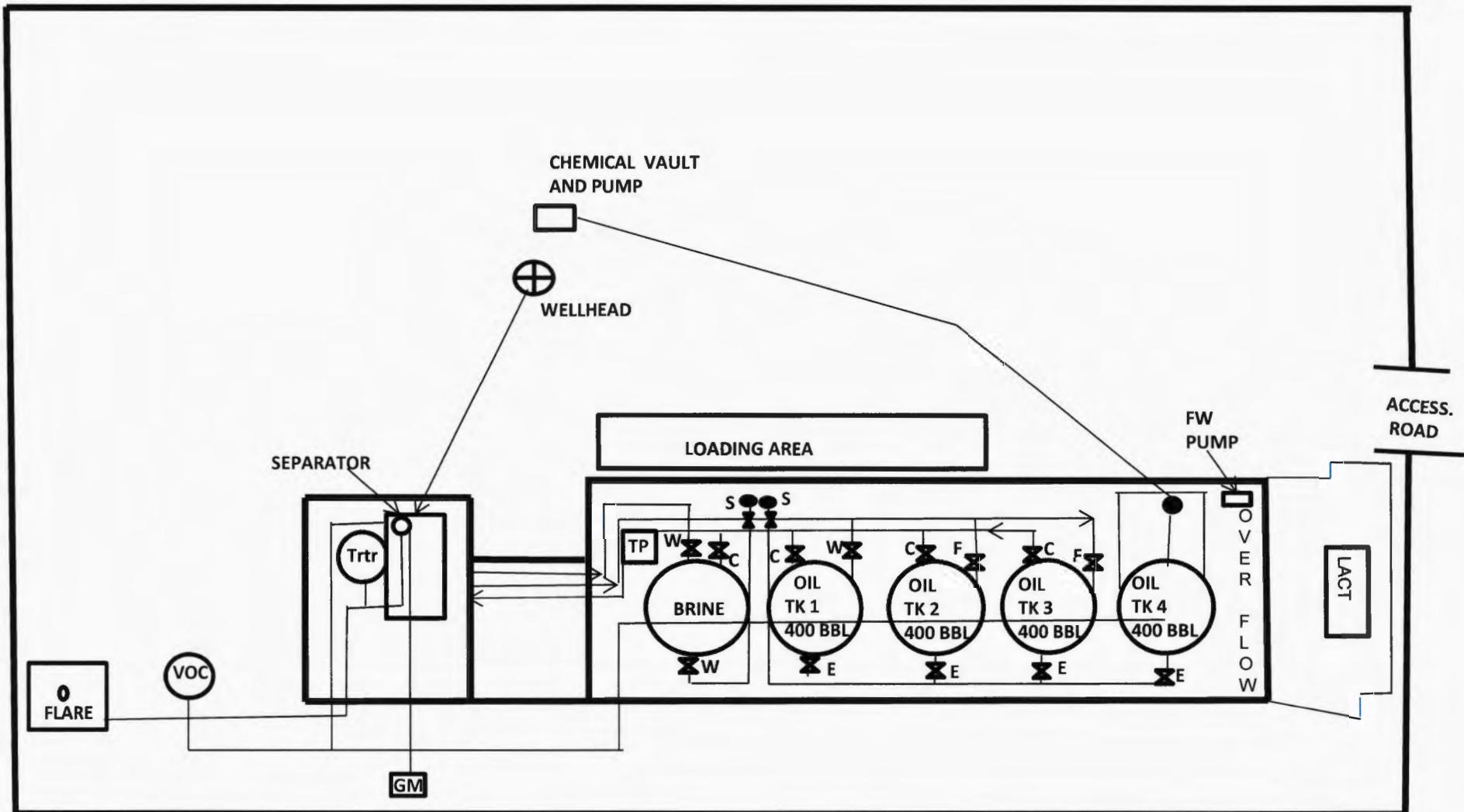
SITE FACILITY PLAN LOCATED AT:
SM ENERGY CO
550 N 31ST ST, SUITE 500
BILLINGS, MT 59103



NOT TO SCALE

VALVE SEALING DETAIL	PRODUCTION OR NORMAL OPERATIONS	RECYCLING	SALES
C=RECYCLING	O/C	O	SC
E=EQUALIZER	O/C	O/C	SC
F=PRODUCTION	O	O/C	SC
S=SALES	SC	SC	O

WHERE: O – OPEN, SO – SEALED OPEN,
C – CLOSED, SC – SEALED CLOSED,
O/C – OPEN OR SEALED



AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE FORM 8INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION

600 EAST BOULEVARD DEPT 405

BISMARCK, ND 58505-0840

SFN 5698 (03-2000)



Well File No.

21266

NDIC CTB No.

21266PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number DAHL 15-11H	Qtr-Qtr SWSE	Section 11	Township 153 N	Range 101 W	County MCKENZIE
Operator SM ENERGY COMPANY	Telephone # 406-245-6248	Field BAKER			
Address P O BOX 7168	City BILLINGS	State MT	Zip Code 59103-7168		

Name of First Purchaser Targa Badlands LLC.	Telephone # 701-842-3315	% Purchased 100	Date Effective January 31, 2014	
Principal Place of Business 1000 Louisiana St. Suite 4300	City Houston	State TX	Zip Code 77002	
Field Address 2680 109th Ave. NW	City Watford City	State ND	Zip Code 58854	
Name of Transporter Targa Badlands LLC.	Telephone Number 713-584-1061	% Transported 100	Date Effective January 31, 2014	
Address 1000 Louisiana St. Suite 4300	City Houston	State TX	Zip Code 77002	

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Comments		

I hereby swear or affirm that the information herein provided is true, complete and correct as determined from all available records.

Date

January 31, 2014

Signature 	Printed Name Devin Schmidt	Title Production Accountant
---------------	--------------------------------------	---------------------------------------

Above Signature Witnessed By

Witness Signature 	Witness Printed Name Brenda Young	Witness Title Production Accountant
-----------------------	---	---

FOR STATE USE ONLY

Date Approved FEB 11 2014
By
Title Oil & Gas Production Analyst

**WELL COMPLETION OR RECOMPLETION REPORT - FORM 6**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)



Well File No.
21266

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion					
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well	☐ Added Horizontal Leg	☐ Extended Horizontal Leg
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:		
Well Name and Number DAHL 15-11H			Spacing Unit Description Sec 2 & 11, T153N-R101W		
Operator SM ENERGY COMPANY		Telephone Number (406) 245-6248		Field BAKER	
Address P. O. BOX 7168			Pool BAKKEN		
City BILLINGS	State MT	Zip Code 59103-7168	Permit Type ☐ Wildcat ☒ Development ☐ Extension		

LOCATION OF WELL

At Surface 22 F S L	2488 F E L	Qtr-Qtr SWSE	Section 11	Township 153 N	Range 101 W	County McKenzie
Spud Date October 15, 2011	Date TD Reached November 18, 2011	Drilling Contractor and Rig Number Extreme Drilling #18		KB Elevation (Ft) 2127	Graded Elevation (Ft) 2110	
Type of Electric and Other Logs Run (See Instructions) CBL/GR/CCL						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Vertical Hole	Surface	9 5/8	0	2113	13 1/2	36			736	SURF
Vertical Hole	Intermediate	7	0	10952	8 3/4	29-32			922	2560
Lateral1	Liner	4 1/2"	10240	20835	6	11.6				

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft)		Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	20850	Open Hole	10952	20850	10286		11/18/2011			

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) 10952 - 20850						Name of Zone (If Different from Pool Name)				
Date Well Completed (SEE INSTRUCTIONS) January 11, 2012			Producing Method Flowing		Pumping-Size & Type of Pump			Well Status (Producing or Shut-In) Producing		
Date of Test 01/14/2012	Hours Tested 24	Choke Size 20 /64	Production for Test		Oil (Bbls) 494	Gas (MCF) 418	Water (Bbls) 554	Oil Gravity-API (Corr.) 41.7 °	Disposition of Gas Flared	
Flowing Tubing Pressure (PSI) 860		Flowing Casing Pressure (PSI) 0		Calculated 24-Hour Rate	Oil (Bbls) 494	Gas (MCF) 418	Water (Bbls) 554	Gas-Oil Ratio 846		

Well Specific Stimulations

Date Stimulated 01/11/2012	Stimulated Formation Bakken	Top (Ft) 10952	Bottom (Ft) 20850	Stimulation Stages 13	Volume 1822380	Volume Units Gallons
Type Treatment Sand Frac	Acid %	Lbs Proppant 1450600	Maximum Treatment Pressure (PSI) 8842		Maximum Treatment Rate (BBLS/Min) 45.5	
Details Pumped 999,800# 20/40 sand, 450,800 40/70 sand, 1,822,380 Gals total fluid in 13 stages over 4 days (1/8/11 - 1/11/12) at a maximum rate of 45.5 bpm and maximum pressure of 8842 psi.						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

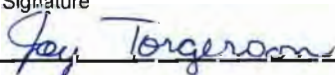
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

--	--	--

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address jtorgerson@sm-energy.com		Date 09/21/2012
	Signature 	Printed Name Joy Torgerson	Title Engineering Tech



212660
Oil and Gas Division

Lynn D. Helms - Director Bruce E. Hicks - Assistant Dir

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

September 6, 2012

RANDY CARLSON
SM ENERGY CO
PO BOX 7168
BILLINGS, MT 59103-7168

RE: See Attached List of Wells

Dear Randy:

According to our records, our office has not received a Completion Report (Form 6) for the attached list of wells.

Please note: A Completion Report is due thirty days after the completion of a well, except in the case of a wildcat well, where the Completion Report is due immediately. Please submit the Form 6's, in duplicate, at your earliest convenience. If you have any questions, feel free to contact our office.

Sincerely,

Nathaniel Erbele
Petroleum Resource Specialist

List of Completion Report Due Wells for SM. RGY CO

CARTER 9-8HW
NESE 8-163N-101W
DIVIDE COUNTY
WELL FILE NO: 22717

ADAMS 2-18H
NWNE 18-163N-100W
DIVIDE COUNTY
WELL FILE NO: 22516

GULBRANSON 2-1H
LOT2 1-163N-100W
DIVIDE COUNTY
WELL FILE NO: 22471

DAHL FEDERAL 2-15H
NWNE 15-153N-101W
MCKENZIE COUNTY
WELL FILE NO: 21796

DAHL 15-11H
SWSE 11-153N-101W
MCKENZIE COUNTY
WELL FILE NO: 21266

WOLD 15-33H
SWSE 33-153N-97W
MCKENZIE COUNTY
WELL FILE NO: 20266

NELSON 15-11H
SWSE 11-150N-97W
MCKENZIE COUNTY
WELL FILE NO: 20053

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No.

21266

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed April 14, 2012
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☒ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Clean out, Convert to rod pump

Well Name and Number Dahl 15-11H					
Footages	Qtr-Qtr	Section	Township	Range	
22 F S L 2488 F E L	SWSE	11	153 N	101 W	
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	389 Bbls
Water	Bbls	Water	301 Bbls
Gas	MCF	Gas	167 MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

On 4/14/12, SM Energy cleaned out and converted the Dahl 15-11H from flowing to rod pump. The ball seats of the frac sleeves were milled up as the well was cleaned out to TD. After cleanout, production equipment was run and the well was returned to production on rod pump.

Pump: 2-1/2" x 2-1/4" x 40' tubing barrel pump MSN set at ~10,076'

Tubing: 2-7/8" L-80 TAC set at ~10,077' EOT at ~10,165'

Pumping Unit: Rotaflex RH1100-500-306

Please submit Completion Report - Form 6

Company SM Energy Company		Telephone Number (406) 245-6248	
Address P.O. Box 7168			
City Billings	State MT	Zip Code 59103	
Signature <i>Cris Rogers</i>	Printed Name Cris Rogers		
Title Operations Engineer	Date April 19, 2012		
Email Address crogers@sm-energy.com			

FOR STATE USE ONLY	
☒ Received	☐ Approved
Date <i>May 14, 2012</i>	
By <i>[Signature]</i>	
Title PETROLEUM ENGINEER	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.

21266

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed November 30, 2011
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Reserve Pit Reclamation

Well Name and Number Dahl 15-11H					
Footages	Qtr-Qtr	Section	Township	Range	
22 F S L 2488 F E L	SWSE	11	153 N	101 W	
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Franz Construction Inc.			
Address P. O. Box 1046	City Sidney	State MT	Zip Code 59270

DETAILS OF WORK

SM Energy Company has completed the reclamation of the reserve pit as follows:

Disposed of the pit water at the following salt water disposals:

**Hjort 32-17 SWD (Permit #MTS21MD-0088)
1820' FNL, 2320' FEL (SWNE)
Sec 17-T32N-R56E
Sheridan Co., MT**

Applied 85 tons of flyash from the Rosebud Power Plant, Colstrip, MT to the drilling mud to solidify it, covered with 4' backfill and topsoil, then graded to promote surface drainage away from the reclaimed pit.

Company SM Energy Company		Telephone Number (406) 245-6248	
Address P. O. Box 7168			
City Billings		State MT	Zip Code 59103-7168
Signature 		Printed Name Lorena Griggs	
Title Regulatory & Safety Assistant		Date November 28, 2011	
Email Address rgriggs@sm-energy.com			

FOR STATE USE ONLY	
☒ Received	☐ Approved
Date 2-6-12	
By 	
Title 	

AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE - FORM 8

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)



Well File No.

21266

NDIC CTB No.

21266

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number DAHL 15-11H	Qtr-Qtr SWSE	Section 11	Township 153 N	Range 101 W	County MCKENZIE
Operator SM ENERGY COMPANY	Telephone # 406-245-6248	Field BAKER			
Address P O BOX 7168	City BILLINGS	State MT	Zip Code 59103-7168		

Name of First Purchaser Plains Marketing, LP	Telephone # 713-646-4100	% Purchased 100	Date Effective January 13, 2012
Principal Place of Business P O Box 4648	City Houston	State TX	Zip Code 77002
Field Address 621 - 17th St., Ste 825	City Denver	State CO	Zip Code 80293
Name of Transporter Plains Marketing, L.P.	Telephone Number 713-993-5900	% Transported 100	Date Effective January 13, 2012
Address P O Box 4648	City Houston	State TX	Zip Code 77210-4648

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Comments		

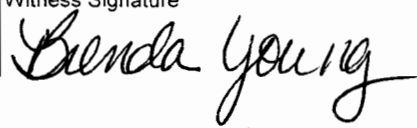
I hereby swear or affirm that the information herein provided is true, complete and correct as determined from all available records.

Date

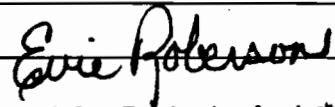
January 13, 2012

Signature 	Printed Name Will Burruss	Title Production Accountant
--	-------------------------------------	---------------------------------------

Above Signature Witnessed By

Witness Signature 	Witness Printed Name Brenda Young	Witness Title Production Accountant
---	---	---

FOR STATE USE ONLY

Date Approved JAN 19 2012
By 
Title Oil & Gas Production Analyst

Industrial Commission of North Dakota
Oil and Gas Division
Verbal Approval To Purchase and Transport Oil

Well or Facility No

21266

Tight Hole **Yes**



OPERATOR

Operator SM ENERGY COMPANY	Representative Lynnette Watts	Rep Phone
--------------------------------------	---	-----------

WELL INFORMATION

Well Name DAHL 15-11H					Inspector Marc Binns
Well Location	QQ	Sec	Twp	Rng	County
	SWSE	11	153 N	101 W	MCKENZIE
Footages	22	Feet From the S Line			Field BAKER
	2488	Feet From the E Line			Pool
Date of First Production Through Permanent Wellhead					This Is Not The First Sales

PURCHASER / TRANSPORTER

Purchaser PLAINS MARKETING, L.P.	Transporter PLAINS MARKETING, L.P.
--	--

TANK BATTERY

Single Well Tank Battery Number :

SALES INFORMATION This Is Not The First Sales

ESTIMATED BARRELS TO BE SOLD		ACTUAL BARRELS SOLD	DATE
10000	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	

DETAILS

Frac set to start 1/7/12 Verbal to Purchase and Transport Oil issued on condition that date of first production, date of first sales, and amount of first sales are reported to commission, also that sales do not exceed the barrels approved prior to submission of the Form 8. 43-02-03-81. AUTHORIZATION TO TRANSPORT OIL FROM A WELL OR CENTRAL PRODUCTION FACILITY. Before any crude oil is transported from a well or central production facility, the operator of the well or central production facility shall file with the director, and obtain the director's approval, an authorization to purchase
--

Start Date	1/10/2012
Date Approved	1/10/2012
Approved By	Jessica Gilkey

21266
JLW**Llewellyn, Linda B.**

From: oilandgasinfo@nd.gov
Sent: Tuesday, January 10, 2012 4:59 PM
To: Wollan, Glenn L.; Fetzer, Annette M.; Holweger, Todd L.; Suggs, Richard A.; Llewellyn, Linda B.; Joersz, Kyle W.; Gilkey, Jessica L.; Binns, Marc L.; lwatts@sm-energy.com
Subject: "CONFIDENTIAL"-Verbal Approval Procedure PTO - 21266

Type of Verbal Procedure : **PTO**
 NDIC file no : **21266**
 Well Confidential : **Yes**
 Well name : **DAHL 15-11H**
 Well operator : **SM ENERGY COMPANY**
 Company Representative : **Lynnette Watts**
 Representative Phone Number : **()-**
 Field name : **BAKER**
 Pool name :
 County : **MCKENZIE**
 Location STR : **11-153N-101W**
 Feet From N/S : **22 S**
 Feet From E/W : **2488 E**
 Purchaser : **PLAINS MARKETING, L.P.**
 Transporter : **PLAINS MARKETING, L.P.**
 First Sales Y/N : **No**
 First Prod Dt :
 CTB No :
 CTB Type : **SingleWellTB**

Est ! Barrels	Act Barrels	Date
10000		

Details :Frac set to start 1/7/12 Verbal to Purchase and Transport Oil issued on condition that date of first production, date of first sales, and amount of first sales are reported to commision, also that sales do not exceed the barrels approved prior to submission of the Form 8. 43-02-03-81. AUTHORIZATION TO TRANSPORT OIL FROM A WELL OR CENTRAL PRODUCTION FACILITY. Before any crude oil is transported from a well or central production facility, the operator of the well or central production facility shall file with the director, and obtain the director's approval, an authorization to purchase and transport oil from a well or central production facility (form 8). All forms, reports, logs, and other information required by this section shall be submitted within thirty days after the completion of such work, although a completion report shall be filed immediately after the completion or recompletion of a well in a pool or reservoir not then covered by an order of the commission.

Date Approved ! :Jan 10 2012
Approved By :JLG
Date Start :Jan 10 2012
Inspector :MLB

1/18/2012



19510 Oil Center Blvd
Houston, TX 77073
Bus 281.443.1414
Fax 281.443.1676

Tuesday, November 22, 2011

State of North Dakota

Subject: **Surveys**

Re: **SM Energy**
Dahl 15-11 H
McKenzie, ND

Enclosed, please find the original and one copy of the survey performed on the above-referenced well by Ryan Energy Technologies U.S.A., Inc. (Operator No. 740022). Other information required by your office is as follows:

<i>Surveyor Name</i>	<i>Surveyor Title</i>	<i>Borehole Number</i>	<i>Start Depth</i>	<i>End Depth</i>	<i>Start Date</i>	<i>End Date</i>	<i>Type of Survey</i>	<i>TD Straight Line Projection</i>
Sanford, Rory	Directional Driller	O.H.	2113'	20798'	10/17/11	11/18/11	MWD	20850'

A certified plat on which the bottom hole location is oriented both to the surface location and to the lease lines (or unit lines in case of pooling) is attached to the survey report. If any other information is required please contact the undersigned at the letterhead address or phone number.

A handwritten signature in black ink that reads "Douglas Hudson". The signature is fluid and cursive, with the first name "Douglas" and last name "Hudson" clearly distinguishable.

Douglas Hudson
Well Planner



19510 Oil Center Blvd
Houston, TX 77073
Bus 281.443.1414
Fax 281.443.1676

Tuesday, November 22, 2011

State of North Dakota

Subject: **Survey Certification Letter**

Re: **SM Energy**
Dahl 15-11 H
McKenzie, ND

I, Rory Sanford, certify that; I am employed by Ryan Energy Technologies U.S.A., Inc.; that I did on the conduct or supervise the taking of the following MWD surveys:

on the day(s) of 10/17/2011 thru 11/18/2011 from a depth of 2113' MD to a depth of 20798' MD and
Straight line projection to TD 20850' MD;

that the data is true, correct, complete, and within the limitations of the tool as set forth by Ryan Energy Technologies U.S.A., Inc.; that I am authorized and qualified to make this report; that this survey was conducted at the request of SM Energy for the Dahl 15-11 H; in McKenzie, ND.

Rory Sanford

Rory Sanford

Directional Driller

Ryan Energy Technologies U.S.A., Inc.



Ryan Energy Technologies

COMPLETION REPORT



Company: SM Energy Company	Date: 11/22/2011	Time: 11:01:21	Page: 1
Field: McKenzie Co., ND	Co-ordinate(NE) Reference:	Site: Dahl 15-11H, True North	
Site: Dahl 15-11H	Vertical (TVD) Reference:	SITE 2127.0	
Well: 15-11H	Section (VS) Reference:	Well (0.00N,0.00E,359.66Azi)	
Wellpath: OH	Survey Calculation Method:	Minimum Curvature	Db: Sybase

Field: McKenzie Co., ND

Map System: US State Plane Coordinate System 1983
Geo Datum: GRS 1980
Sys Datum: Mean Sea Level

Map Zone: North Dakota, Northern Zone
Coordinate System: Site Centre
Geomagnetic Model: IGRF2010

Site: Dahl 15-11H
 Sec. 11-T153N-R101W
 22' FSL & 2488' FEL

Site Position:	Northing: 410403.41 ft	Latitude: 48 4 56.376 N
From: Geographic	Easting: 1202212.98 ft	Longitude: 103 38 9.239 W
Position Uncertainty: 0.00 ft		North Reference: True
Ground Level: 2110.00 ft		Grid Convergence: -2.33 deg

Well: 15-11H

Slot Name:

Well Position: +N/-S 0.00 ft	Northing: 410403.41 ft	Latitude: 48 4 56.376 N
+E/-W 0.00 ft	Easting: 1202212.98 ft	Longitude: 103 38 9.239 W
Position Uncertainty: 0.00 ft		

Wellpath: OH

Current Datum: SITE	Height 2127.00 ft	Drilled From: Surface
Magnetic Data: 10/7/2011		Tie-on Depth: 0.00 ft
Field Strength: 56737 nT		Above System Datum: Mean Sea Level
Vertical Section: Depth From (TVD)	+N/-S	Declination: 8.72 deg
ft	ft	Mag Dip Angle: 73.12 deg
		+E/-W ft
		Direction deg
0.00	0.00	359.66

Survey Program for Definitive Wellpath

Date: 11/22/2011	Validated: No	Version: 0
Actual From ft	To ft	Toolcode
2113.00	20798.00	Survey #1 (2113.00-20798.00)
20850.00	20850.00	Survey #2 (20850.00-20850.00)
		MWD Projection
		MWD Projected to TD

Survey

MD ft	Incl deg	TVD ft	VS ft	Azim deg	N/S ft	E/W ft	Clsr Dist. ft	Clsr Azi. deg	Dog Leg deg/100ft	Tool
2113.00	0.00	2113.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	TIE LINE
2167.00	1.20	2167.00	-0.52	203.90	-0.52	-0.23	0.57	203.90	2.22	MWD
2259.00	1.40	2258.97	-2.40	206.60	-2.40	-1.12	2.65	205.04	0.23	MWD
2351.00	1.60	2350.94	-4.45	215.00	-4.46	-2.36	5.05	207.91	0.32	MWD
2443.00	2.20	2442.89	-7.20	195.10	-7.22	-3.56	8.05	206.25	0.96	MWD
2534.00	2.60	2533.81	-10.90	190.70	-10.93	-4.40	11.78	201.91	0.48	MWD
2625.00	2.80	2624.71	-15.15	184.10	-15.18	-4.94	15.96	198.03	0.41	MWD
2716.00	3.00	2715.59	-19.71	189.70	-19.74	-5.50	20.49	195.57	0.38	MWD
2807.00	3.00	2806.47	-24.36	194.10	-24.40	-6.48	25.24	194.88	0.25	MWD
2897.00	2.50	2896.37	-28.55	192.80	-28.60	-7.49	29.56	194.68	0.56	MWD
2988.00	3.10	2987.26	-32.93	185.50	-32.98	-8.17	33.98	193.91	0.77	MWD
3079.00	3.00	3078.13	-37.75	185.40	-37.80	-8.63	38.77	192.85	0.11	MWD
3162.00	3.00	3161.01	-42.07	184.20	-42.13	-8.99	43.08	192.04	0.08	MWD
3245.00	3.00	3243.90	-46.41	183.90	-46.46	-9.30	47.38	191.31	0.02	MWD
3328.00	2.70	3326.80	-50.52	185.00	-50.58	-9.61	51.48	190.76	0.37	MWD
3411.00	2.40	3409.71	-54.19	185.30	-54.25	-9.94	55.16	190.39	0.36	MWD
3495.00	3.20	3493.61	-58.28	184.80	-58.34	-10.30	59.24	190.02	0.95	MWD
3578.00	2.70	3576.50	-62.53	184.90	-62.60	-10.66	63.50	189.67	0.60	MWD
3661.00	3.10	3659.40	-66.72	181.90	-66.79	-10.91	67.67	189.27	0.52	MWD
3744.00	3.00	3742.28	-71.13	183.90	-71.20	-11.13	72.06	188.88	0.18	MWD
3828.00	2.80	3826.17	-75.37	183.90	-75.44	-11.42	76.30	188.61	0.24	MWD
3910.00	2.50	3908.08	-79.15	183.70	-79.22	-11.67	80.08	188.38	0.37	MWD



Ryan Energy Technologies COMPLETION REPORT



Company: SM Energy Company
Field: McKenzie Co., ND
Site: Dahl 15-11H
Well: 15-11H
Wellpath: OH

Date: 11/22/2011
Co-ordinate(NE) Reference: Site: Dahl 15-11H, True North
Vertical (TVD) Reference: SITE 2127.0
Section (VS) Reference: Well (0.00N,0.00E,359.66Azi)
Survey Calculation Method: Minimum Curvature
Db: Sybase

Page: 2

Survey

MD ft	Incl deg	TVD ft	VS ft	Azim deg	N/S ft	E/W ft	Clsr Dist. ft	Clsr Azi. deg	Dog Leg deg/100ft	Tool
3994.00	3.20	3991.98	-83.32	181.70	-83.39	-11.86	84.23	188.09	0.84	MWD
4077.00	3.30	4074.85	-88.02	182.80	-88.09	-12.04	88.91	187.78	0.14	MWD
4160.00	3.00	4157.72	-92.57	185.30	-92.64	-12.36	93.46	187.60	0.40	MWD
4244.00	2.30	4241.63	-96.41	191.50	-96.48	-12.90	97.34	187.61	0.90	MWD
4327.00	2.40	4324.56	-99.74	191.20	-99.82	-13.57	100.74	187.74	0.12	MWD
4401.00	2.00	4398.51	-102.49	197.50	-102.57	-14.26	103.56	187.91	0.63	MWD
4492.00	2.50	4489.44	-105.89	197.60	-105.98	-15.33	107.08	188.23	0.55	MWD
4576.00	2.50	4573.36	-109.32	202.30	-109.42	-16.58	110.67	188.62	0.24	MWD
4659.00	2.30	4656.28	-112.57	198.40	-112.67	-17.80	114.07	188.98	0.31	MWD
4742.00	2.40	4739.21	-115.80	197.20	-115.92	-18.84	117.44	189.23	0.13	MWD
4825.00	2.40	4822.14	-119.10	199.20	-119.22	-19.92	120.87	189.49	0.10	MWD
4908.00	2.50	4905.06	-122.47	196.30	-122.60	-21.00	124.38	189.72	0.19	MWD
4991.00	1.60	4988.01	-125.33	192.00	-125.47	-21.75	127.34	189.83	1.10	MWD
5075.00	2.20	5071.96	-128.04	193.70	-128.18	-22.38	130.12	189.90	0.72	MWD
5158.00	2.70	5154.89	-131.51	190.40	-131.65	-23.11	133.66	189.95	0.63	MWD
5241.00	2.10	5237.82	-134.95	184.10	-135.09	-23.57	137.13	189.90	0.79	MWD
5324.00	3.00	5320.73	-138.63	177.10	-138.78	-23.57	140.76	189.64	1.15	MWD
5408.00	3.00	5404.62	-143.03	182.10	-143.17	-23.54	145.09	189.34	0.31	MWD
5490.00	2.10	5486.54	-146.67	184.80	-146.81	-23.74	148.72	189.19	1.11	MWD
5574.00	1.90	5570.48	-149.53	197.40	-149.67	-24.28	151.63	189.22	0.57	MWD
5657.00	1.60	5653.45	-151.90	203.20	-152.05	-25.15	154.12	189.39	0.42	MWD
5740.00	2.50	5736.39	-154.72	193.70	-154.87	-26.04	157.05	189.54	1.16	MWD
5823.00	2.50	5819.31	-158.26	188.50	-158.42	-26.73	160.66	189.58	0.27	MWD
5906.00	2.10	5902.25	-161.57	183.50	-161.73	-27.09	163.98	189.51	0.54	MWD
5989.00	1.60	5985.20	-164.14	204.20	-164.31	-27.66	166.62	189.56	1.00	MWD
6073.00	1.10	6069.18	-165.92	207.30	-166.09	-28.51	168.52	189.74	0.60	MWD
6156.00	0.50	6152.17	-166.97	198.20	-167.14	-28.99	169.64	189.84	0.74	MWD
6239.00	0.40	6235.17	-167.56	210.70	-167.74	-29.25	170.27	189.89	0.17	MWD
6322.00	0.30	6318.17	-167.96	225.80	-168.14	-29.56	170.72	189.97	0.16	MWD
6405.00	0.10	6401.17	-168.15	240.10	-168.33	-29.77	170.94	190.03	0.25	MWD
6488.00	0.20	6484.17	-168.04	8.20	-168.22	-29.82	170.84	190.05	0.33	MWD
6571.00	0.30	6567.17	-167.68	355.80	-167.86	-29.81	170.48	190.07	0.14	MWD
6655.00	0.20	6651.17	-167.31	2.70	-167.49	-29.82	170.13	190.10	0.12	MWD
6738.00	0.30	6734.16	-166.96	345.20	-167.14	-29.87	169.79	190.13	0.15	MWD
6821.00	0.30	6817.16	-166.56	328.20	-166.74	-30.04	169.43	190.21	0.11	MWD
6904.00	0.40	6900.16	-166.10	345.10	-166.28	-30.23	169.00	190.30	0.17	MWD
6987.00	0.60	6983.16	-165.41	337.80	-165.60	-30.47	168.38	190.43	0.25	MWD
7071.00	0.70	7067.15	-164.56	329.30	-164.75	-30.90	167.62	190.62	0.17	MWD
7154.00	0.60	7150.15	-163.69	2.40	-163.88	-31.14	166.81	190.76	0.46	MWD
7237.00	0.60	7233.14	-162.83	8.60	-163.01	-31.05	165.94	190.79	0.08	MWD
7320.00	0.70	7316.14	-161.92	19.50	-162.11	-30.82	165.01	190.76	0.19	MWD
7403.00	0.80	7399.13	-160.91	23.30	-161.10	-30.42	163.94	190.69	0.13	MWD
7486.00	0.80	7482.12	-159.84	19.30	-160.02	-30.00	162.80	190.62	0.07	MWD
7569.00	0.70	7565.12	-158.81	16.70	-158.98	-29.66	161.73	190.57	0.13	MWD
7653.00	0.70	7649.11	-157.83	17.30	-158.00	-29.36	160.71	190.53	0.01	MWD
7736.00	0.60	7732.10	-157.01	40.50	-157.19	-28.93	159.83	190.43	0.34	MWD
7819.00	1.10	7815.10	-156.11	43.10	-156.28	-28.10	158.78	190.19	0.60	MWD
7902.00	1.30	7898.08	-154.77	36.30	-154.94	-27.00	157.27	189.89	0.30	MWD
7985.00	1.00	7981.06	-153.41	32.30	-153.56	-26.06	155.76	189.63	0.37	MWD
8068.00	0.80	8064.05	-152.26	22.60	-152.42	-25.45	154.53	189.48	0.30	MWD
8152.00	0.90	8148.04	-151.14	28.10	-151.29	-24.91	153.33	189.35	0.15	MWD
8235.00	1.00	8231.03	-150.06	44.40	-150.20	-24.10	152.12	189.11	0.35	MWD
8318.00	0.90	8314.02	-148.98	30.30	-149.12	-23.26	150.92	188.87	0.31	MWD



Ryan Energy Technologies COMPLETION REPORT



Company: SM Energy Company
Field: McKenzie Co., ND
Site: Dahl 15-11H
Well: 15-11H
Wellpath: OH

Date: 11/22/2011
Co-ordinate(NE) Reference: Site: Dahl 15-11H, True North
Vertical (TVD) Reference: SITE 2127.0
Section (VS) Reference: Well (0.00N,0.00E,359.66Azi)
Survey Calculation Method: Minimum Curvature
Db: Sybase

Page: 3

Survey

MD ft	Incl deg	TVD ft	VS ft	Azim deg	N/S ft	E/W ft	Clsr Dist. ft	Clsr Azi. deg	Dog Leg deg/100ft	Tool
8401.00	0.80	8397.01	-147.97	39.00	-148.11	-22.57	149.82	188.66	0.20	MWD
8484.00	0.80	8480.00	-147.03	31.10	-147.16	-21.90	148.78	188.47	0.13	MWD
8567.00	0.80	8562.99	-145.96	7.00	-146.09	-21.53	147.67	188.39	0.40	MWD
8650.00	0.80	8645.99	-144.81	351.50	-144.94	-21.55	146.54	188.46	0.26	MWD
8734.00	0.80	8729.98	-143.65	349.30	-143.79	-21.75	145.42	188.60	0.04	MWD
8817.00	1.00	8812.97	-142.36	355.30	-142.49	-21.91	144.17	188.74	0.27	MWD
8900.00	1.00	8895.95	-140.92	355.80	-141.05	-22.02	142.76	188.87	0.01	MWD
8984.00	1.00	8979.94	-139.45	1.70	-139.59	-22.06	141.32	188.98	0.12	MWD
9066.00	1.10	9061.93	-137.96	352.10	-138.09	-22.14	139.86	189.11	0.25	MWD
9149.00	1.00	9144.91	-136.45	5.60	-136.58	-22.18	138.37	189.22	0.32	MWD
9233.00	1.30	9228.90	-134.77	2.60	-134.90	-22.07	136.69	189.29	0.36	MWD
9313.00	1.00	9308.88	-133.16	0.40	-133.30	-22.02	135.10	189.38	0.38	MWD
9396.00	1.10	9391.87	-131.65	9.10	-131.79	-21.89	133.59	189.43	0.23	MWD
9479.00	0.70	9474.86	-130.42	27.50	-130.55	-21.53	132.31	189.36	0.59	MWD
9562.00	0.80	9557.85	-130.24	117.30	-130.36	-20.78	132.01	189.06	1.28	MWD
9646.00	1.70	9641.83	-131.64	154.80	-131.76	-19.73	133.23	188.52	1.39	MWD
9729.00	3.00	9724.76	-134.93	182.40	-135.05	-19.30	136.42	188.13	2.03	MWD
9812.00	3.60	9807.62	-139.69	185.80	-139.81	-19.65	141.18	188.00	0.76	MWD
9895.00	3.40	9890.47	-144.72	187.10	-144.84	-20.22	146.25	187.95	0.26	MWD
9978.00	3.20	9973.33	-149.46	187.30	-149.58	-20.82	151.02	187.92	0.24	MWD
10061.00	3.10	10056.21	-153.90	196.70	-154.03	-21.76	155.56	188.04	0.63	MWD
10145.00	2.20	10140.12	-157.58	200.30	-157.72	-22.97	159.38	188.29	1.09	MWD
10228.00	1.40	10223.08	-159.98	205.90	-160.12	-23.96	161.91	188.51	0.99	MWD
10274.00	0.90	10269.07	-160.57	255.40	-160.72	-24.56	162.59	188.69	2.31	MWD
10316.00	4.90	10311.02	-158.86	358.20	-159.01	-24.93	160.95	188.91	12.32	MWD
10356.00	12.60	10350.52	-152.78	359.70	-152.93	-25.01	154.96	189.29	19.26	MWD
10398.00	18.50	10390.97	-141.53	359.40	-141.68	-25.10	143.88	190.05	14.05	MWD
10440.00	24.80	10429.98	-126.04	359.80	-126.19	-25.21	128.68	191.30	15.00	MWD
10481.00	28.50	10466.62	-107.65	359.40	-107.80	-25.34	110.74	193.23	9.03	MWD
10523.00	32.40	10502.82	-86.37	358.90	-86.52	-25.66	90.25	196.52	9.31	MWD
10564.00	35.30	10536.87	-63.54	359.90	-63.69	-25.89	68.75	202.12	7.20	MWD
10606.00	39.30	10570.27	-38.09	0.60	-38.25	-25.77	46.12	213.97	9.58	MWD
10648.00	43.60	10601.74	-10.31	1.70	-10.46	-25.20	27.29	247.47	10.38	MWD
10689.00	47.60	10630.43	18.95	2.80	18.81	-24.04	30.53	308.03	9.94	MWD
10731.00	52.40	10657.42	51.02	4.90	50.89	-21.86	55.39	336.75	12.05	MWD
10772.00	59.10	10680.48	84.77	4.20	84.66	-19.19	86.81	347.23	16.40	MWD
10814.00	65.80	10699.89	121.87	4.40	121.77	-16.39	122.87	352.33	15.96	MWD
10855.00	71.80	10714.71	159.95	4.20	159.87	-13.53	160.44	355.16	14.64	MWD
10897.00	77.50	10725.83	200.37	1.30	200.30	-11.60	200.64	356.69	15.12	MWD
10936.00	83.50	10732.26	238.80	1.30	238.74	-10.73	238.98	357.43	15.38	MWD
10988.00	84.80	10737.56	290.52	0.20	290.47	-10.05	290.64	358.02	3.27	MWD
11080.00	86.80	10744.30	382.27	359.70	382.21	-10.13	382.35	358.48	2.24	MWD
11172.00	89.60	10747.19	474.21	0.80	474.16	-9.73	474.26	358.82	3.27	MWD
11263.00	90.20	10747.35	565.18	1.00	565.14	-8.30	565.21	359.16	0.70	MWD
11355.00	90.70	10746.63	657.17	359.80	657.14	-7.66	657.18	359.33	1.41	MWD
11447.00	91.20	10745.10	749.16	0.10	749.12	-7.74	749.16	359.41	0.63	MWD
11539.00	90.70	10743.58	841.14	0.60	841.11	-7.18	841.14	359.51	0.77	MWD
11630.00	91.60	10741.75	932.10	0.90	932.08	-5.99	932.10	359.63	1.04	MWD
11722.00	92.20	10738.70	1024.02	1.30	1024.01	-4.22	1024.02	359.76	0.78	MWD
11814.00	89.60	10737.25	1115.96	1.40	1115.97	-2.05	1115.97	359.89	2.83	MWD
11905.00	89.20	10738.21	1206.93	0.90	1206.95	-0.23	1206.95	359.99	0.70	MWD
11996.00	89.80	10739.00	1297.91	0.30	1297.94	0.72	1297.94	0.03	0.93	MWD



Ryan Energy Technologies COMPLETION REPORT



Company: SM Energy Company
Field: McKenzie Co., ND
Site: Dahl 15-11H
Well: 15-11H
Wellpath: OH

Date: 11/22/2011
Co-ordinate(NE) Reference: Site: Dahl 15-11H, True North
Vertical (TVD) Reference: SITE 2127.0
Section (VS) Reference: Well (0.00N,0.00E,359.66Azi)
Survey Calculation Method: Minimum Curvature
Page: 4
Db: Sybase

Survey

MD ft	Incl deg	TVD ft	VS ft	Azim deg	N/S ft	E/W ft	Clsr Dist. ft	Clsr Azi. deg	Dog Leg deg/100ft	Tool
12087.00	90.50	10738.76	1388.90	0.00	1388.93	0.96	1388.94	0.04	0.84	MWD
12177.00	90.00	10738.37	1478.90	0.60	1478.93	1.43	1478.93	0.06	0.87	MWD
12268.00	90.10	10738.29	1569.89	0.60	1569.93	2.39	1569.93	0.09	0.11	MWD
12358.00	89.60	10738.53	1659.88	359.90	1659.93	2.78	1659.93	0.10	0.96	MWD
12449.00	90.50	10738.45	1750.88	0.10	1750.92	2.78	1750.93	0.09	1.01	MWD
12533.00	90.20	10737.93	1834.86	1.40	1834.91	3.88	1834.92	0.12	1.59	MWD
12616.00	89.40	10738.22	1917.84	0.40	1917.90	5.18	1917.91	0.15	1.54	MWD
12699.00	89.80	10738.80	2000.82	1.10	2000.89	6.27	2000.90	0.18	0.97	MWD
12782.00	90.90	10738.30	2083.80	0.60	2083.88	7.50	2083.89	0.21	1.46	MWD
12865.00	90.90	10736.99	2166.77	0.90	2166.86	8.59	2166.88	0.23	0.36	MWD
12948.00	91.00	10735.62	2249.74	0.70	2249.84	9.75	2249.86	0.25	0.27	MWD
13032.00	91.40	10733.86	2333.71	0.80	2333.82	10.84	2333.84	0.27	0.49	MWD
13115.00	90.40	10732.55	2416.69	0.60	2416.80	11.86	2416.83	0.28	1.23	MWD
13198.00	90.50	10731.90	2499.68	0.30	2499.79	12.51	2499.82	0.29	0.38	MWD
13281.00	90.60	10731.11	2582.66	1.00	2582.78	13.45	2582.82	0.30	0.85	MWD
13364.00	91.40	10729.66	2665.62	1.30	2665.75	15.12	2665.80	0.32	1.03	MWD
13447.00	91.30	10727.70	2748.57	1.00	2748.71	16.78	2748.76	0.35	0.38	MWD
13530.00	89.90	10726.83	2831.55	359.00	2831.70	16.78	2831.75	0.34	2.94	MWD
13614.00	89.50	10727.27	2915.55	359.20	2915.69	15.46	2915.73	0.30	0.53	MWD
13697.00	89.60	10727.92	2998.54	359.10	2998.68	14.23	2998.71	0.27	0.17	MWD
13780.00	90.30	10728.00	3081.54	0.00	3081.68	13.58	3081.71	0.25	1.37	MWD
13863.00	90.90	10727.13	3164.53	359.80	3164.67	13.44	3164.70	0.24	0.76	MWD
13946.00	91.30	10725.53	3247.52	0.40	3247.65	13.58	3247.68	0.24	0.87	MWD
14029.00	90.90	10723.94	3330.50	359.90	3330.64	13.80	3330.67	0.24	0.77	MWD
14113.00	90.40	10722.99	3414.49	358.40	3414.62	12.55	3414.64	0.21	1.88	MWD
14196.00	90.60	10722.26	3497.46	358.40	3497.59	10.23	3497.60	0.17	0.24	MWD
14279.00	90.50	10721.47	3580.41	357.20	3580.52	7.05	3580.53	0.11	1.45	MWD
14362.00	90.90	10720.45	3663.36	358.30	3663.45	3.79	3663.45	0.06	1.41	MWD
14445.00	90.10	10719.73	3746.31	357.40	3746.38	0.68	3746.38	0.01	1.45	MWD
14528.00	90.40	10719.37	3829.26	357.80	3829.31	-2.80	3829.31	359.96	0.60	MWD
14612.00	89.70	10719.29	3913.18	356.60	3913.21	-6.90	3913.21	359.90	1.65	MWD
14695.00	88.70	10720.45	3996.06	356.70	3996.06	-11.75	3996.07	359.83	1.21	MWD
14778.00	89.30	10721.90	4078.97	357.90	4078.95	-15.66	4078.98	359.78	1.62	MWD
14861.00	91.00	10721.68	4161.95	359.60	4161.92	-17.47	4161.96	359.76	2.90	MWD
14944.00	90.90	10720.31	4244.94	359.50	4244.91	-18.12	4244.95	359.76	0.17	MWD
15027.00	91.50	10718.57	4327.92	0.00	4327.89	-18.49	4327.93	359.76	0.94	MWD
15111.00	91.60	10716.30	4411.89	359.40	4411.86	-18.93	4411.90	359.75	0.72	MWD
15194.00	89.50	10715.50	4494.88	358.50	4494.84	-20.45	4494.88	359.74	2.75	MWD
15277.00	89.40	10716.30	4577.85	358.40	4577.80	-22.69	4577.86	359.72	0.17	MWD
15360.00	89.30	10717.24	4660.83	358.30	4660.76	-25.08	4660.83	359.69	0.17	MWD
15443.00	90.00	10717.74	4743.82	359.30	4743.74	-26.82	4743.82	359.68	1.47	MWD
15526.00	90.80	10717.16	4826.81	359.30	4826.73	-27.83	4826.81	359.67	0.96	MWD
15609.00	89.20	10717.16	4909.80	358.60	4909.71	-29.35	4909.80	359.66	2.10	MWD
15693.00	89.90	10717.82	4993.78	358.40	4993.68	-31.55	4993.78	359.64	0.87	MWD
15776.00	88.40	10719.06	5076.73	357.40	5076.61	-34.59	5076.73	359.61	2.17	MWD
15859.00	87.50	10722.02	5159.62	357.70	5159.48	-38.14	5159.62	359.58	1.14	MWD
15942.00	88.40	10724.99	5242.52	358.00	5242.37	-41.25	5242.53	359.55	1.14	MWD
16025.00	88.50	10727.24	5325.47	358.60	5325.30	-43.71	5325.48	359.53	0.73	MWD
16108.00	89.40	10728.76	5408.45	359.00	5408.27	-45.45	5408.46	359.52	1.19	MWD
16192.00	89.90	10729.27	5492.44	359.00	5492.26	-46.92	5492.46	359.51	0.60	MWD
16275.00	89.20	10729.93	5575.43	358.60	5575.23	-48.66	5575.45	359.50	0.97	MWD
16358.00	88.80	10731.37	5658.40	358.70	5658.20	-50.61	5658.43	359.49	0.50	MWD
16441.00	88.90	10733.04	5741.37	358.40	5741.16	-52.71	5741.40	359.47	0.38	MWD



Ryan Energy Technologies COMPLETION REPORT



Company: SM Energy Company
Field: McKenzie Co., ND
Site: Dahl 15-11H
Well: 15-11H
Wellpath: OH

Date: 11/22/2011
Co-ordinate(NE) Reference: Site: Dahl 15-11H, True North
Vertical (TVD) Reference: SITE 2127.0
Section (VS) Reference: Well (0.00N,0.00E,359.66Azi)
Survey Calculation Method: Minimum Curvature
Db: Sybase

Page: 5

Survey

MD ft	Incl deg	TVD ft	VS ft	Azim deg	N/S ft	E/W ft	Clsr Dist. ft	Clsr Azi. deg	Dog Leg deg/100ft	Tool
16524.00	89.30	10734.34	5824.34	358.50	5824.11	-54.96	5824.37	359.46	0.50	MWD
16607.00	90.00	10734.85	5907.32	358.60	5907.09	-57.06	5907.36	359.45	0.85	MWD
16691.00	91.10	10734.04	5991.30	358.60	5991.06	-59.11	5991.35	359.43	1.31	MWD
16774.00	89.40	10733.68	6074.29	359.10	6074.03	-60.77	6074.34	359.43	2.13	MWD
16857.00	89.50	10734.48	6157.28	359.00	6157.02	-62.15	6157.33	359.42	0.17	MWD
16940.00	89.90	10734.91	6240.27	358.80	6240.00	-63.74	6240.33	359.41	0.54	MWD
17023.00	89.80	10735.13	6323.27	359.50	6322.99	-64.97	6323.33	359.41	0.85	MWD
17106.00	89.80	10735.42	6406.26	359.00	6405.99	-66.06	6406.33	359.41	0.60	MWD
17190.00	90.40	10735.27	6490.26	359.60	6489.98	-67.09	6490.32	359.41	1.01	MWD
17273.00	90.90	10734.33	6573.26	0.00	6572.97	-67.38	6573.32	359.41	0.77	MWD
17356.00	90.90	10733.03	6656.25	359.50	6655.96	-67.74	6656.31	359.42	0.60	MWD
17439.00	92.20	10730.78	6739.21	0.00	6738.93	-68.10	6739.27	359.42	1.68	MWD
17522.00	90.30	10728.97	6822.18	1.20	6821.90	-67.23	6822.23	359.44	2.71	MWD
17605.00	91.60	10727.60	6905.12	1.60	6904.86	-65.20	6905.17	359.46	1.64	MWD
17688.00	88.50	10727.52	6988.09	0.50	6987.83	-63.68	6988.12	359.48	3.96	MWD
17772.00	88.30	10729.87	7072.05	0.60	7071.80	-62.88	7072.08	359.49	0.27	MWD
17855.00	89.20	10731.68	7155.00	1.40	7154.76	-61.43	7155.03	359.51	1.45	MWD
17938.00	89.10	10732.91	7237.94	1.90	7237.72	-59.04	7237.96	359.53	0.61	MWD
18021.00	88.90	10734.36	7320.89	1.00	7320.68	-56.94	7320.90	359.55	1.11	MWD
18104.00	89.40	10735.59	7403.87	359.60	7403.67	-56.50	7403.88	359.56	1.79	MWD
18187.00	89.60	10736.32	7486.87	0.10	7486.66	-56.72	7486.88	359.57	0.65	MWD
18271.00	90.00	10736.61	7570.87	359.30	7570.66	-57.16	7570.88	359.57	1.06	MWD
18354.00	90.20	10736.46	7653.86	359.00	7653.65	-58.39	7653.87	359.56	0.43	MWD
18437.00	90.70	10735.81	7736.86	359.10	7736.64	-59.77	7736.87	359.56	0.61	MWD
18520.00	89.80	10735.45	7819.85	358.60	7819.62	-61.43	7819.86	359.55	1.24	MWD
18603.00	89.20	10736.17	7902.83	359.00	7902.60	-63.17	7902.85	359.54	0.87	MWD
18686.00	89.00	10737.48	7985.81	358.10	7985.56	-65.27	7985.83	359.53	1.11	MWD
18770.00	89.30	10738.72	8069.77	358.10	8069.50	-68.06	8069.79	359.52	0.36	MWD
18853.00	89.60	10739.52	8152.73	357.90	8152.45	-70.95	8152.76	359.50	0.43	MWD
18936.00	90.00	10739.81	8235.67	357.10	8235.37	-74.57	8235.71	359.48	1.08	MWD
19019.00	91.40	10738.80	8318.61	358.60	8318.30	-77.69	8318.66	359.46	2.47	MWD
19102.00	91.20	10736.91	8401.59	359.30	8401.26	-79.21	8401.64	359.46	0.88	MWD
19185.00	90.60	10735.61	8484.57	359.80	8484.25	-79.86	8484.63	359.46	0.94	MWD
19269.00	90.10	10735.10	8568.57	0.60	8568.25	-79.57	8568.62	359.47	1.12	MWD
19352.00	88.80	10735.89	8651.55	0.60	8651.24	-78.70	8651.59	359.48	1.57	MWD
19435.00	89.20	10737.34	8734.53	0.60	8734.22	-77.83	8734.57	359.49	0.48	MWD
19518.00	89.60	10738.21	8817.51	0.50	8817.21	-77.03	8817.55	359.50	0.50	MWD
19601.00	89.00	10739.23	8900.50	358.70	8900.20	-77.61	8900.54	359.50	2.29	MWD
19684.00	89.20	10740.53	8983.48	358.50	8983.16	-79.64	8983.52	359.49	0.34	MWD
19767.00	89.60	10741.40	9066.46	359.10	9066.14	-81.38	9066.51	359.49	0.87	MWD
19851.00	90.40	10741.40	9150.46	359.50	9150.13	-82.40	9150.50	359.48	1.06	MWD
19934.00	89.40	10741.54	9233.46	359.80	9233.13	-82.91	9233.50	359.49	1.26	MWD
20017.00	88.60	10742.99	9316.45	359.40	9316.12	-83.49	9316.49	359.49	1.08	MWD
20100.00	88.70	10744.95	9399.42	359.40	9399.09	-84.36	9399.47	359.49	0.12	MWD
20183.00	88.80	10746.76	9482.40	359.50	9482.06	-85.16	9482.45	359.49	0.17	MWD
20266.00	89.40	10748.06	9565.39	359.40	9565.05	-85.95	9565.44	359.49	0.73	MWD
20350.00	89.40	10748.94	9649.39	0.10	9649.04	-86.32	9649.43	359.49	0.83	MWD
20433.00	89.40	10749.81	9732.38	359.90	9732.04	-86.32	9732.42	359.49	0.24	MWD
20516.00	88.60	10751.26	9815.37	359.40	9815.02	-86.83	9815.41	359.49	1.14	MWD
20599.00	88.80	10753.14	9898.34	0.10	9898.00	-87.19	9898.39	359.50	0.88	MWD
20682.00	88.90	10754.81	9981.33	359.80	9980.98	-87.26	9981.37	359.50	0.38	MWD
20765.00	88.80	10756.47	10064.31	0.20	10063.97	-87.26	10064.35	359.50	0.50	MWD



Ryan Energy Technologies COMPLETION REPORT



Company: SM Energy Company	Date: 11/22/2011	Time: 11:01:21	Page: 6
Field: McKenzie Co., ND	Co-ordinate(NE) Reference:	Site: Dahl 15-11H, True North	
Site: Dahl 15-11H	Vertical (TVD) Reference:	SITE 2127.0	
Well: 15-11H	Section (VS) Reference:	Well (0.00N,0.00E,359.66Azi)	
Wellpath: OH	Survey Calculation Method:	Minimum Curvature	Db: Sybase

Survey

MD ft	Incl deg	TVD ft	VS ft	Azim deg	N/S ft	E/W ft	Clsr Dist. ft	Clsr Azi. deg	Dog Leg deg/100ft	Tool
20798.00	88.80	10757.16	10097.30	0.40	10096.96	-87.09	10097.34	359.51	0.61	MWD
20850.00	88.80	10758.25	10149.28	0.40	10148.95	-86.72	10149.32	359.51	0.00	PROJECTED

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
21266

TH

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date December 10, 2011	☐ Drilling Prognosis	☐ Spill Report
☐ Report of Work Done	Date Work Completed	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☒ Other	Reserve Pit Reclamation

Well Name and Number Dahl 15-11H							
Footages		Qtr-Qtr	Section	Township	Range		
22 F S L 2488 F E L		SWSE	11	153 N	101 W		
Field Baker		Pool Bakken		County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Franz Construction Inc.			
Address P. O. Box 1046		City Sidney	State MT
		Zip Code 59270	

DETAILS OF WORK

SM Energy Company requests approval to reclaim the reserve pit at the Bagley 4-30H as follows:

Dispose of pit water at one of the following salt water disposals:

Hjort 32-17 SWD (Permit #MTS21MD-0088)	Wolter 16-23 SWD (UIC #W0207S06590)
1820' FNL, 2320' FEL (SWNE)	250' FSL, 240' FEL (SESE)
Sec 17-T32N-R56E	Sec 23-T163N-R100W
Sheridan Co., MT	Divide Co., ND

Apply flyash to the drilling mud to solidify it, cover with 4' of backfill and topsoil, then grade to promote surface drainage away from the reclaimed pit.

The surface owner is: Kirby A. Dahl Linda Prochazka Dahl Rev Tst 1-26-07 Bruce E. Dahl
817 S 3rd St. 2005 Wynstone Circle NE 4715 Cliff Dr
Cold Springs, MN 56320 North Canto, OH 44720 Rapid City, SD 57702

Sharon Eilers	Carmelina Lee	James R Lee	Larry D Lee	Doris Lindvig Life Estate
2200 SW Cole Ct	3840 Rimrock Rd#3206	3957 Palisades Pk Dr	1559 E Omaha Dr	515 12th St W
Tualatin, OR 87062	Billings, MT 59102	Billings, MT 59102	Bismarck, ND 58504	Williston, ND 58801

Company SM Energy Company		Telephone Number (406) 245-6248	
Address P. O. Box 7168			
City Billings		State MT	Zip Code 59103-7168
Signature 		Printed Name Lorena Griggs	
Title Regulatory & Safety Assistant		Date November 28, 2011	
Email Address rgriggs@sm-energy.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 12-8-11	
By 	
Title Recl	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 6**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
21266

TH

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed November 21, 2011
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☒ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Drilling Progress

Well Name and Number Dahl 15-11H							
Footages		Qtr-Qtr		Section	Township	Range	
22 F S	L	2488 F	E L	SWSE	11	153 N	101 W
Field Baker		Pool Bakken		County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Nabors Drilling USA LP			
Address P. O. Box 2396		City Williston	State ND
		Zip Code 58802	

DETAILS OF WORK

Completed rig repairs, then continued drilling the 6" lateral from 11,044', reaching 20,850' TD on 11/18/11. Ran 4 1/2" 11.6# HCP-110 casing liner from 10,240' - 20,835', installed the tubing head and released the rig on 11/21/11.

Company SM Energy Company		Telephone Number (406) 245-6248	
Address P. O. Box 7168			
City Billings		State MT	Zip Code 59103-7168
Signature <i>Joy Torgerson</i>		Printed Name Joy Torgerson	
Title Engineering Tech		Date November 22, 2011	
Email Address jtorgerson@sm-energy.com			

FOR STATE USE ONLY	
☒ Received	☐ Approved
Date <i>December 2, 2011</i>	
By <i>Daniel J. M. Miller</i>	
Title PETROLEUM ENGINEER	

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4
SFN 5749

1. Approval shall be obtained prior to perforating or recompleting a well in a reservoir other than the reservoir in which the well is currently completed, prior to plug back of a well, prior to temporary abandonment of a well, prior to abandonment of a well, prior to reclamation of a well site, prior to reclamation of a reserve pit, and prior to beginning a workover project, which may qualify for a tax exemption pursuant to NDCC Section 57-51.1-03. Please refer to Section 43-02-03-16 of the North Dakota Administrative Code (NDAC) regarding recompleting a well in a reservoir other than the reservoir in which the well is currently completed or plugging back of a well, to Section 43-02-03-55 NDAC regarding temporary abandonment of a well, to Section 43-02-03-33 or Section 43-02-05-08 NDAC regarding abandonment of wells, to Section 43-02-03-19 NDAC regarding reclamation, and to Section 43-02-09-03 NDAC regarding workover projects.

2. Upon the completion of any remedial work, or attempted remedial work such as plugging back, drilling deeper, acidizing, shooting, formation fracturing, squeezing operations, setting liner, fishing operations, repair work, perforating, reperforating, or other similar operations not specifically covered herein, a report on the operation shall be filed on a Sundry Notice - Form 4 (SFN 5749) with the Director. The report shall present a detailed account of all work done and the date of such work; the daily production of oil, gas, and water both prior to and after the operation; the shots per foot, size, and depth of perforations; the quantity of sand, crude, chemical, or other materials employed in the operation; and any other pertinent information or operations which affect the original status of the well and are not specifically covered herein. Please refer to Section 43-02-03-31 NDAC.

3. Upon the completion of a workover project, which may qualify for a tax exemption pursuant to NDCC Section 57-51.1-03, a report on the operation shall be filed on a Sundry Notice - Form 4 (SFN 5749) detailing the work done. Include the dates during which the workover rig was in service actually performing work on the workover project and the date the workover was completed, a detailed list identifying all labor, services, and materials used and equipment replaced during the workover project, the cost of each item, and whether the replacement equipment was new or used. The value of all equipment removed from service must be listed. The average daily oil production from the well during the first two months after completion of the project must be included if the costs of the project did not exceed sixty-five thousand dollars. All gauge tickets of oil produced in incomplete months during the first two months after completion of the workover and the volume of oil stored on the well premises immediately prior to commencement of the workover project must also be included. Please refer to Section 43-02-09-04 NDAC.

4. Upon the initial installation of pumping equipment, or change in type or depth of pumping equipment designed to increase productivity in a well, the operator shall file a Sundry Notice - Form 4 (SFN 5749) of such installation. The notice shall include all pertinent information on the pump and the operation thereof including the date of such installation, and the daily production of the well prior to and after the pump has been installed. Please refer to Section 43-02-03-31 NDAC.

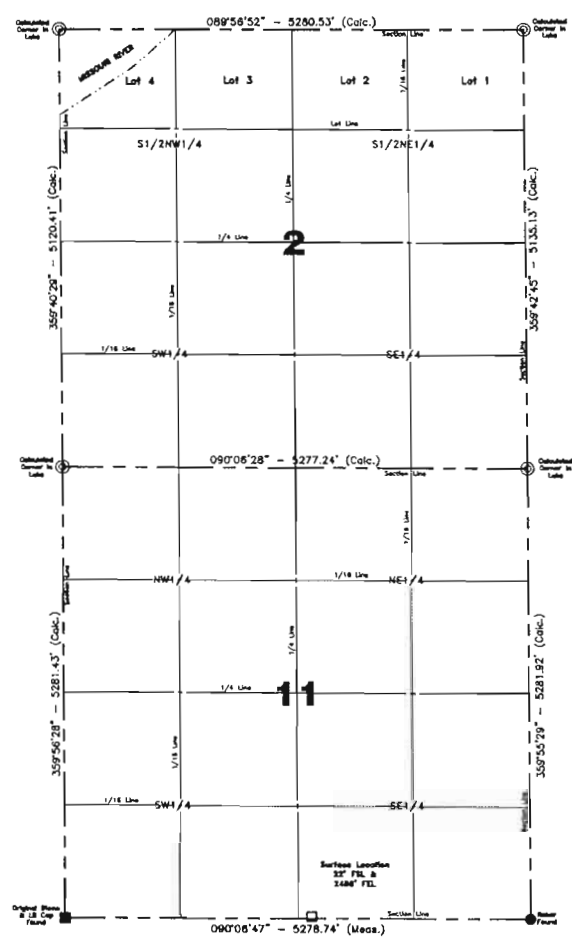
5. The well file number, well name and number, well location, field, pool, and county shall coincide with the official records on file with the Commission.

6. The original and one copy of this report shall be filed with the Industrial Commission of North Dakota, Oil and Gas Division, 600 East Boulevard, Dept. 405, Bismarck, ND 58505-0840.

21266
TH



SM Energy
Dahl 15-11H
 22' FSL & 2,488' FEL
 SW SE Section 11, T153N-R101W
 Baker Field / Bakken
 McKenzie County, North Dakota



BOTTOM HOLE LOCATION:
 10,148.96' North & 86.49' West of surface location or approx.
 238.49' FNL & 2574.49' FEL, Lot 2 Section 2, T153N-R101W

Prepared for:
 Dean Richmond
 SM Energy
 P.O.Box 7168
 Billings, MT 59103-7168

Prepared by:
 Murray Dighans Jr., Vimal Pradhan, Andy Oksendahl
 P.O.Box 51297; Billings, MT 59105
 (406) 259-4124
 geology@sunburstconsulting.com
 www.sunburstconsulting.com



04/15/2012

WELL EVALUATION



Figure 1. Extreme 18 drilling SM Energy Company Dahl 15-11H during November 2011 in McKenzie County, North Dakota (Murray Dighans, Sunburst Consulting Geologist).

INTRODUCTION

SM Energy Dahl 15-11H [SW SE Sec. 11 T153N R101W] is located 6 miles south of Williston in McKenzie County, North Dakota. Dahl 15-11H is a two section horizontal Middle Bakken well planned to be drilled north across sections 11 and 2 (Figure 2). Directional drilling and geo-steering was used to land and maintain exposure to the Middle Bakken reservoir.

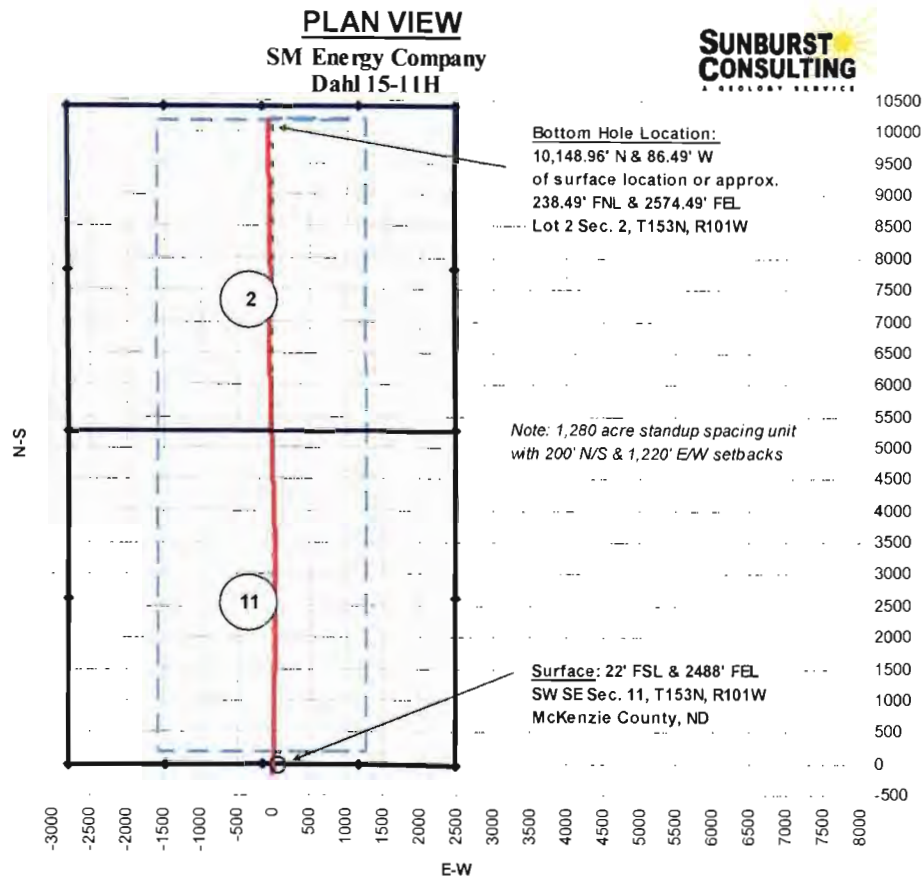


Figure 2. Plan view of Dahl 15-11H spacing unit and well path.

OFFSET WELLS

Three offset wells were used to estimate formation tops and landing target on Dahl 15-11H. These wells were the SM Energy Co., *Lindvig 1-11HR*, located 0.38 miles northeast, the SM Energy Co., *M&G 14-2HR* located 1.18 miles northwest, and the SM Energy Co., *Rosebud 22-11* located 0.7 miles northwest. The *Lindvig 1-11HR* was closest and most helpful in estimating Middle Bakken entry during curve operations.

ENGINEERING

The **Dahl 15-11H** was spud on October 14, 2011 by Extreme 18. A 13 ½" hole was drilled with fresh water to 2,133' and cased with 9 ⅝" casing cemented to surface. A Hughes 8 ¾" PDC bit was used to drill the first part of the vertical. This assembly was tripped out for a new bit and motor at 9,366'. Gamma was also picked up at this time. The second assembly was also a Hughes PDC bit. The vertical was drilled in 119.5 hrs. Diesel invert drilling fluid with a target weight of 10.0 ppg was used to drill the vertical hole.

Ryan Drilling Technologies and Solutions provided the personnel and equipment for the measurement while drilling (MWD) services as well as the night directional driller. Directional drilling was supervised by the company men during the day. Throughout this project, steering decisions were made through a combined effort of the directional drillers, MWD, and Sunburst

Consulting contractors, assisted by continual communication. A Security FXD55M PDC was used at a kick-off point (KOP) of 10,286'. After drilling the curve in 19.5 hours, intermediate casing was set at 10,952 measured depth (MD).

Once casing was set lateral operations began. Two assemblies were used to drill the lateral both utilizing Hughes DP505FX PDC bits in front of Ryan motors. The first assembly was replaced at 11,044' after an MWD tool failure. This assembly drilled 62' in 1 hour. Following the trip in for the MWD failure the rig experienced difficulties with the draw works. Parts were shipped for repair and Sunburst Geologists were placed on stand by until the parts returned. This cost ten days of rig downtime and an extra day on top of that to trip for a new BHA. The second assembly successfully reached TD drilling 9,806' in 168 hrs. Bottom Hole Location: 10,148.96' N & 86.49' W of surface location or approximately 238.49' FNL & 2,574.49' FEL; Lot 2 Sec. 2, T153N, R101W

GEOLOGY

Methods

Sunburst Consulting, Inc provided on site geologic representation for **Dahl 15-11H**. Drilling parameters such as depth, drilling rate, pump strokes and total gas were broadcast on location through Pason's electronic data recorder system. The gas was linked from Sunburst's digital gas chromatograph. Hydrocarbon constituents (C₁-C₄) are expressed in part-per-million concentrations. The chromatograph is fed by 1/4" polyflow from an agitation system set up at the shakers. Under the instruction of Sunburst geologists, rig crews collected lagged samples in 30' intervals. Intermediate samples were also collected during intervals requiring extra scrutiny. Rock cuttings were analyzed in wet and dry conditions under a 10x45 power binocular microscope (for detailed lithologic descriptions see appendix). Cuttings were sent to SM Energy and the North Dakota Geological Survey. Along with sample lithology, MWD gamma ray data and rate of penetration (ROP) were used to assist in landing the curve as well as steer through the lateral.

Lithology and Hydrocarbon Show

Sample analysis began at 8,340' MD in the Charles Formation [Mississippian Madison Group]. Samples were interbedded salt, anhydrite, and limestone. *No gas or oil shows were observed.*

The Base of the Last Salt was logged at 9,226' TVD -7,099' SS. The Ratcliffe interval was logged at 9,286' TVD -7,159' SS based on returns of limestone with trace algal material and traces of anhydrite. *The Ratcliffe interval had one gas show of 24.8 units.*

The Mission Canyon Formation [Mississippian Madison Group] was logged at 9,446' TVD -7,319' SS. Typical lithologies were tan to light brown and grayish brown, occasional dark brown, microcrystalline lime mudstone and wackestone with minor interbedded anhydrite. *Several gas shows were recorded in the Mission Canyon from 30 to 70 units increasing with depth. Intermittent traces of spotty brown oil staining were observed in samples. Background gas held from 20-40 units.*

The Lodgepole Formation [Mississippian Madison Group] was encountered at 10,018' TVD -7,891' SS. This interval contained brown to tan, white to off white argillaceous lime mudstone and wackestone. This facies was microcrystalline and earthy in texture, with varying amounts of pyrite. *Background gas increased throughout the vertical from 25 to 40+ units depleting to 20+ units after the trip for curve assembly. At a measured depth of 10,598' a possible fracture greatly increased background gas to 100+ units. Another possible fracture at 10,836' near the top of Scallion interval raised background gas 800+ units with shows as high as 2,271 units, trace amounts of spotty brown oil stain were observed.*

The top of the Bakken Formation [Mississippian-Devonian] was logged at 10,712' TVD -8,585' SS with sample returns of black carbonaceous and petroliferous shale typical of the Upper Shale Member and characterized by very high gamma (255+ api) and elevated background gas of 800-1,500 units.

The Middle Bakken was logged at 10,728' TVD, -8,601' SS. Near the completion of curve operations, below the Upper Bakken Shale and above the target zone, a medium to light gray brown siltstone was observed with interbedded limestone. It was described as, blocky, earthy, with common amounts of pyrite. Beneath the siltstone, the samples were medium to light gray with common white to off white. They were fine grained, friable, sub angular, and calcite cemented. Samples also contained varying amounts of pyrite and *oil staining*. *Oil staining was most prevalent in this zone often increasing from spotty to even oil staining. Gas values often increased in the upper portion of the preferred porosity.* These samples correlated with a ratty gamma signature ranging from 65-80 api counts.

The center of the preferred porosity zone was marked by lower gamma values as low as 40 api that typically correlated with cleaner and more calcareous samples. This zone ranged from 3-5' thick. Often these samples contained lesser amounts of pyrite. In the lower half of target samples were light to medium brown and gray samples typically containing *less oil staining*. At the base of target was a "ratty" cool signature which ranged from 45-60 api that was attributed to stringers of cleaner, heavily calcite-cemented sandstone. Below the target interval drill cuttings observed showed diminished oil staining and were darker in color. Gamma counts increased as the well bore dropped in formation correlating with the darker coloring of silty sandstone.

Perhaps the most perplexing portion of the well was at 15,000'. The wellbore traveled from gamma and samples that correlated with being below the preferred porosity to gamma values and samples that were indicative of being in the upper half of target. In this transition the cool "ratty" gamma just above the base of target was likely faulted out.

Gas values increased from below 300 units until 12,582' where a possible fracture increased gas values to 300+ units. Around 15,000' gas values began to greatly increase to 500-800 units with shows as high as 1,200+ units. This is likely through a combined effort of being higher in the preferred porosity and the possible and subsequent fracture seen at 15,000'. After 16,200' the wellbore began to drop in formation and as a result gas values fell back to 500-800 units with few gas shows reaching 1,000+ units.

From 17,220' to TD samples were compromised as a result of large amounts of lubricant used.

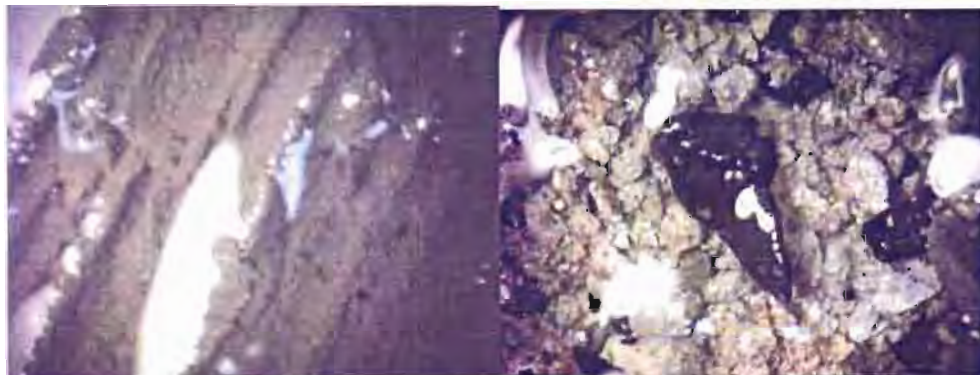


Figure 3 and 4. Lubricant contaminated sample (left); Middle Bakken gray brown siltstone (trace upper shale) (right).

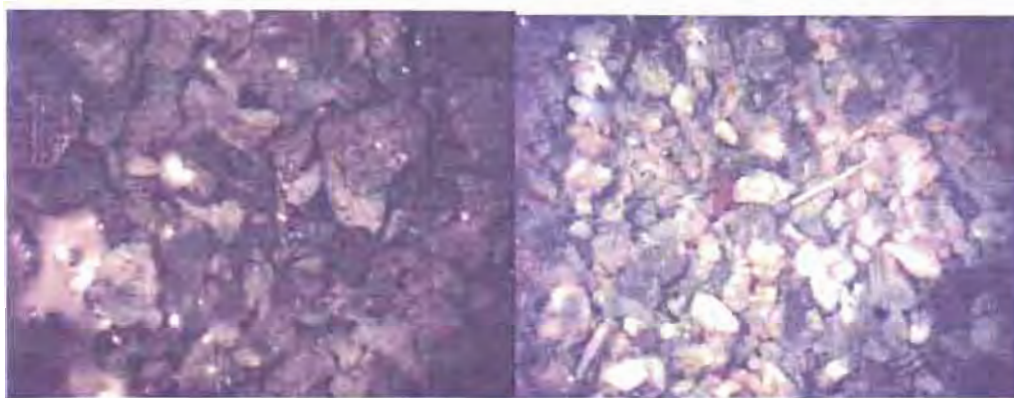


Figure 5 and 6. Gray-brown and tan silty sandstone (left) and off white silty sandstone (right) of the Middle Bakken.

Gas tracking and visible oil show in both cuttings and over the shakers were monitored to assist in evaluating the viability of the reservoir. *No oil was observed at the shakers throughout the lateral. Greater amounts of oil staining was observed in the upper half of the preferred porosity zone which varied from trace to common spotty to even brown oil stain.*

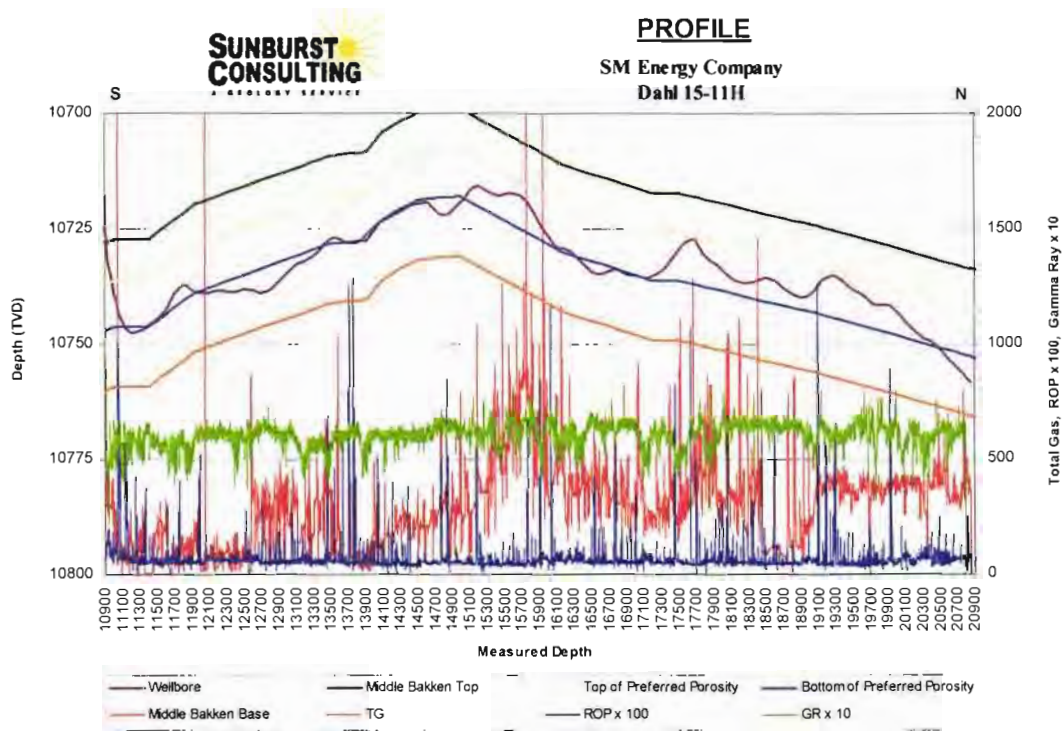


Figure 7. Cross-sectional interpretation of the Dahl 15-11H.

Geo-Steering

While drilling the lateral it was found that just below the preferred porosity zone was the ideal place to drill. Favorable drilling speeds allowed the wellbore to rotate at 120+ ft/hr and a gentle upward rotation minimized the need to slide. The cool gamma at the center of target and the ratty

gamma at the base of target were difficult to drill and often the wellbore deflected when coming into these areas. This made slides more frequent within the preferred porosity. Underneath the preferred porosity this deflection often was anticipated and allowed for smaller slides to keep the wellbore tracking with dip. Geo-steering was successful in maintaining well bore exposure to the Middle Bakken 100% of the lateral. The overall apparent dip was at 0.03 down.

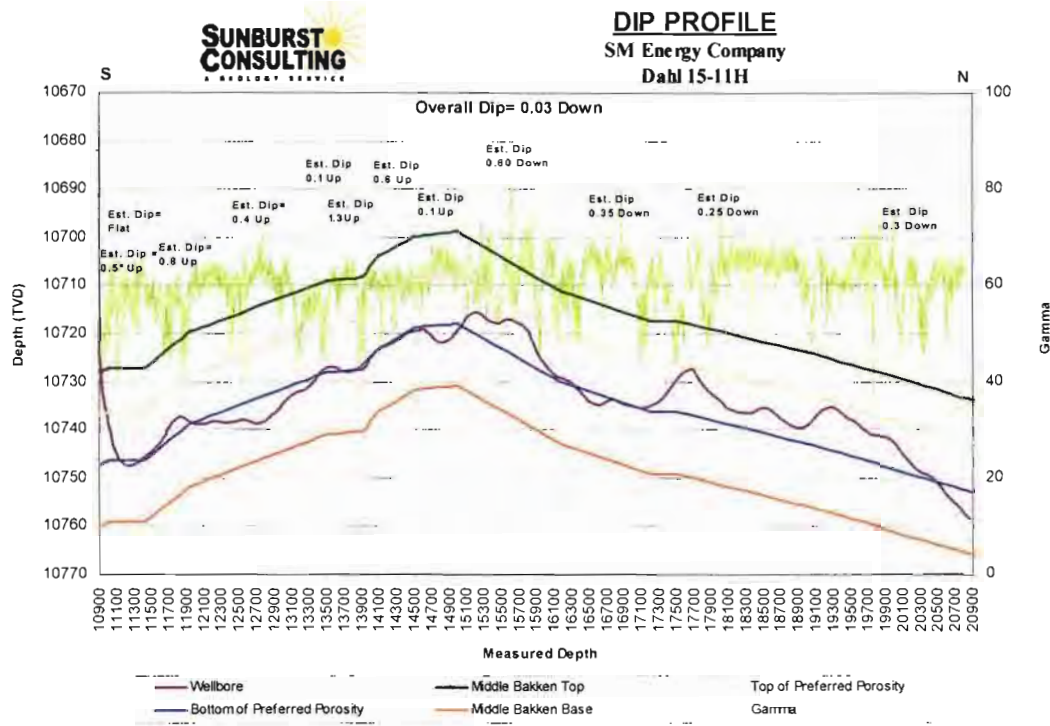


Figure 8. Cross-sectional interpretation with dip estimates of the Dahl 15-11H borehole based upon lithology, MWD gamma ray, and drill rates.

SUMMARY

Dahl 15-11H was spud on October 14, 2011. The vertical was drilled using two PDC bits in 119.5 hrs and the curve was drilled in 19.5 hrs. The most favorable place to be in the Middle Bakken due to *higher oil staining, gas shows*, ability to alleviate slides, and favorable drilling speeds was just below the preferred porosity zone. Samples contaminated with lubricant made correlations based on samples difficult later in the lateral. Oil shows consisted of *common to rare, spotty-even brown oil stain* observed in sample. The lateral was drilled in 169 hrs. One trip at 11,044' was the result of a failed MWD tool. Third party personnel were put on stand-by after issues with the rig's draw works caused a down time issue in excess of a week. The total depth was at 20,850' MD on November 18, 2011 at 1:20 a.m. CST. The SM Energy Dahl 15-11H awaits completion operations to determine its ultimate production potential.

Respectfully submitted,
Murray Dighans Jr.
 c/o Sunburst Consulting, Inc.
 19 November 2011

WELL DATA SUMMARY

<u>OPERATOR:</u>	SM Energy
<u>ADDRESS:</u>	P.O.Box 7168 Billings, MT 59103-7168
<u>WELL NAME:</u>	Dahl 15-11H
<u>API #:</u>	33-053-03703-00-00
<u>WELL FILE #:</u>	21266
<u>SURFACE LOCATION:</u>	22' FSL & 2,488' FEL SW SE Section 11, T153N-R101W
<u>FIELD/ PROSPECT:</u>	Baker Field/ Bakken
<u>COUNTY, STATE</u>	McKenzie County, North Dakota
<u>BASIN:</u>	Williston
<u>WELL TYPE:</u>	Middle Bakken Horizontal
<u>ELEVATION:</u>	GL: 2,110' KB: 2,127'
<u>SPUD/ RE-ENTRY DATE:</u>	October 14, 2011
<u>BOTTOM HOLE LOCATION:</u>	10,148.96' North & 86.49' West of surface location or approx. 238.49' FNL & 2574.49' FEL, Lot 2 Section 2, T153N-R101W
<u>CLOSURE COORDINATES:</u>	Closure Direction Azimuth: 359.512° Closure Distance: 10,459.32'
<u>TOTAL DEPTH / DATE:</u>	20,850' at 1:20 Hrs CST on 3 October 2011 100% Exposure to Middle Bakken
<u>TOTAL DRILLING DAYS:</u>	35 days
<u>STATUS OF WELL:</u>	Open-hole completion/ preparing to open-hole frac
<u>CONTRACTOR:</u>	Extreme 18

<u>PUMPS:</u>	Emsco FB1600 (stroke length - 12")
<u>TOOLPUSHERS:</u>	Larry Siegel, James Hancock
<u>FIELD SUPERVISORS:</u>	Craig Selvig, Randy Selvig
<u>CHEMICAL COMPANY:</u>	NOV Fluid Control
<u>MUD ENGINEER:</u>	Regan Taylor
<u>MUD TYPE:</u>	Oil based Mud (Vertical & Curve)/ Salt Water (Lateral)
<u>MUD LOSSES:</u>	Invert Mud: 595 bbls, Salt Water: 0 bbls
<u>PROSPECT GEOLOGIST:</u>	Dean Richmond
<u>WELLSITE GEOLOGISTS:</u>	Murray Dighans Jr., Vimai Pradhan, Andy Oksendahl
<u>GEOSTEERING SYSTEM:</u>	Sunburst Digital Wellsite Geological System
<u>ROCK SAMPLING:</u>	30' from 9,120' - 20,850'
<u>SAMPLE EXAMINATION:</u>	Binocular microscope & fluoroscope
<u>SAMPLE CUTS:</u>	Entron
<u>GAS DETECTION:</u>	MSI (Mudlogging Systems, Inc.) TGC - total gas with chromatograph ML-335 from 9,120' - 20,850' Extractor Motor/ Serial #: GE Motor/ BW212264
<u>DIRECTIONAL DRILLERS:</u>	Ryan Drilling Technologies and Solutions/ Field Supervisors Rory Sanford (Night)/ Randy Selvig, Craig Selvig (Day)
<u>MWD:</u>	Ryan Drilling Technologies and Solutions Tate Chatfield, Brad Hughes
<u>CASING:</u>	Surface: 9 5/8" Set to 2,113' Intermediate: 7" Set to 10,952'

KEY OFFSET WELLS:

SM Energy Company

Lindvig #1-11-3C

SE SE 612 FSL 793 FEL

McKenzie, ND

SM Energy Company

M&G 14-2HR

SW SW 566 FSL 575 FWL

McKenzie, ND

SM Energy Company

Rosebud 22-11

SE NW 1745 FNL 1755 FWL

McKenzie County, ND

DISTRIBUTION:

North Dakota Industrial Commission

600 East Boulevard

Bismarck, ND 58505

Oasis Petroleum North America

1001 Fannin, Suite 202

Houston, TX 77002

Lone Rider Trading Company

PO Box 1414

Dickinson, ND 58602-1414

Stuber Minerals Resources, LLC

PO Box 56

Bowman, ND 58623

Deep Creek Exploration, LLC

PO Box 1404

Dickinson, ND 58602

Missouri River Royalty Corporation

Kirkwood Office Tower

919 South 7th Street, Suite 405

Bismarck, ND 58504

WELL LOCATION PLAT

SM Energy Company
P.O. Box 7168 Billings, MT 59103

Dahl 15-11H

22 feet from the south line and 2488 feet from the east line (surface location)

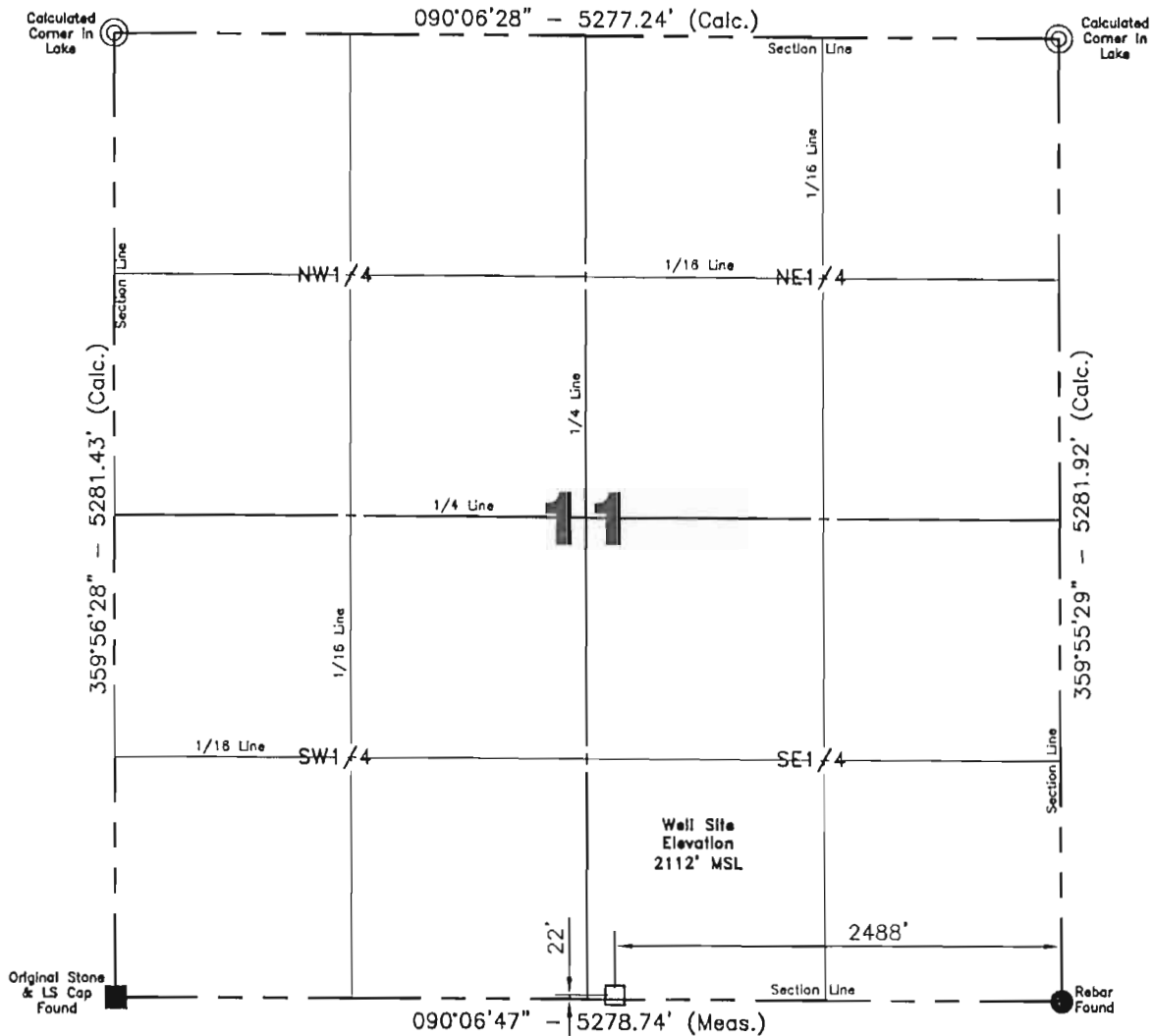
Section 11, T. 153 N., R. 101 W., 5th P.M.

McKenzie County, North Dakota

Surface owner @ well site - Sharon D. Eilers et al

Latitude 48°04'56.376" North; Longitude 103°38'09.239" West (surface location)

[Derived from OPUS Solution NAD-83(CORS96)]



Confidentiality Notice: The information contained on this plat is legally privileged and confidential information intended only for the use of recipients. If you are not the intended recipients, you are hereby notified that any use, dissemination, distribution or copying of this information is strictly prohibited.

NOTE: All land corners are assumed unless otherwise noted. The well location shown hereon is not an as-built location.

Scale 1"=1000'

I, Donald Leischner, Professional Land Surveyor, N.D. No. 7103, do hereby certify that the survey plat shown hereon was made by me, or under my direction, from notes made in the field, and the same is true and correct to the best of my knowledge and belief.

Jeff McElhiney

06/14/2011

Surveyed By

Date

Vertical Control Datum Used
Sea-Level Datum of NAVD 88
Based on elevation derived from OPUS Solution on
GPS* Q462 Located a distance of 9501.84' on an
azimuth of 203°09'37" from the SW corner of
Section 11 T.153N., R.101W., 5th P.M. being at
2099.99' Elevation MSL

Project No. 8711180

Book 0-89 Pg. 74-79 Staking

Professional Consulting Engineers
and Surveyors
Registered in
North Dakota, South Dakota
Montana, Wyoming & Minnesota
Tele-Fax No. 701-572-2019
Bus. Phone No. 701-572-6352
222 Airport Road
Williston, North Dakota 58801-2976
Certificate of Authorization #C-051



Kadmas
Lee &
Jackson
Engineers Surveyors
Planners

HORIZONTAL SECTION PLAT

SM Energy Company
P.O. Box 7168 Billings, MT 59103

Dahl 15-11H

22 feet from the south line and 2488 feet from the east line (surface location)

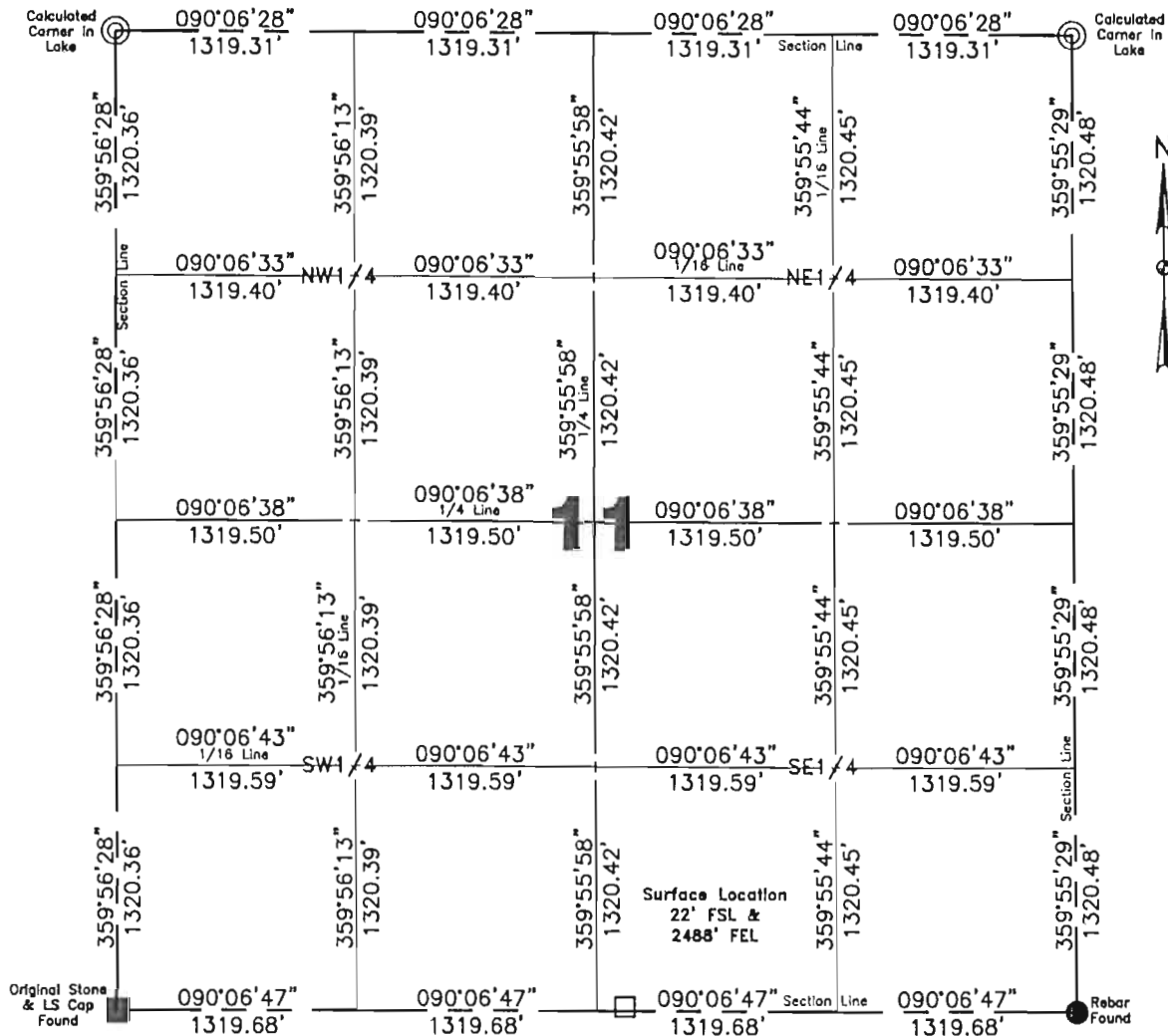
Section 11, T. 153 N., R. 101 W., 5th P.M.

McKenzie County, North Dakota

Surface owner @ well site - Sharon D. Eilers et al

Latitude 48°04'56.376" North; Longitude 103°38'09.239" West (surface location)

[Derived from OPUS Solution NAD-83(CORS96)]



Confidentiality Notice: The information contained on this plat is legally privileged and confidential information intended only for the use of recipients. If you are not the intended recipients, you are hereby notified that any use, dissemination, distribution or copying of this information is strictly prohibited.

Scale 1"=1000'

I, Donald Leischner, Professional Land Surveyor, N.D. No. 7103, do hereby certify that the survey plat shown hereon was made by me, or under my direction, from notes made in the field, and the same is true and correct to the best of my knowledge and belief.

All corners shown on this plat were found in the field during SM Energy, Dahl 15-11H oil well survey on June 14, 2010. Distances to all others are calculated. All azimuths are based on the south line of Section 11, being on an azimuth of 090°06'47".

Surveyed By	Field Book
J. McElhiney	0-89
Computed & Drawn By	Project No.
M. Meisner	8711180



Kadmas
Lee &
Jackson
Engineers Surveyors
Planners

HORIZONTAL SECTION PLAT

SM Energy Company
P.O. Box 7168 Billings, MT 59103

Dahl 15-11H

22 feet from the south line and 2488 feet from the east line (surface location)

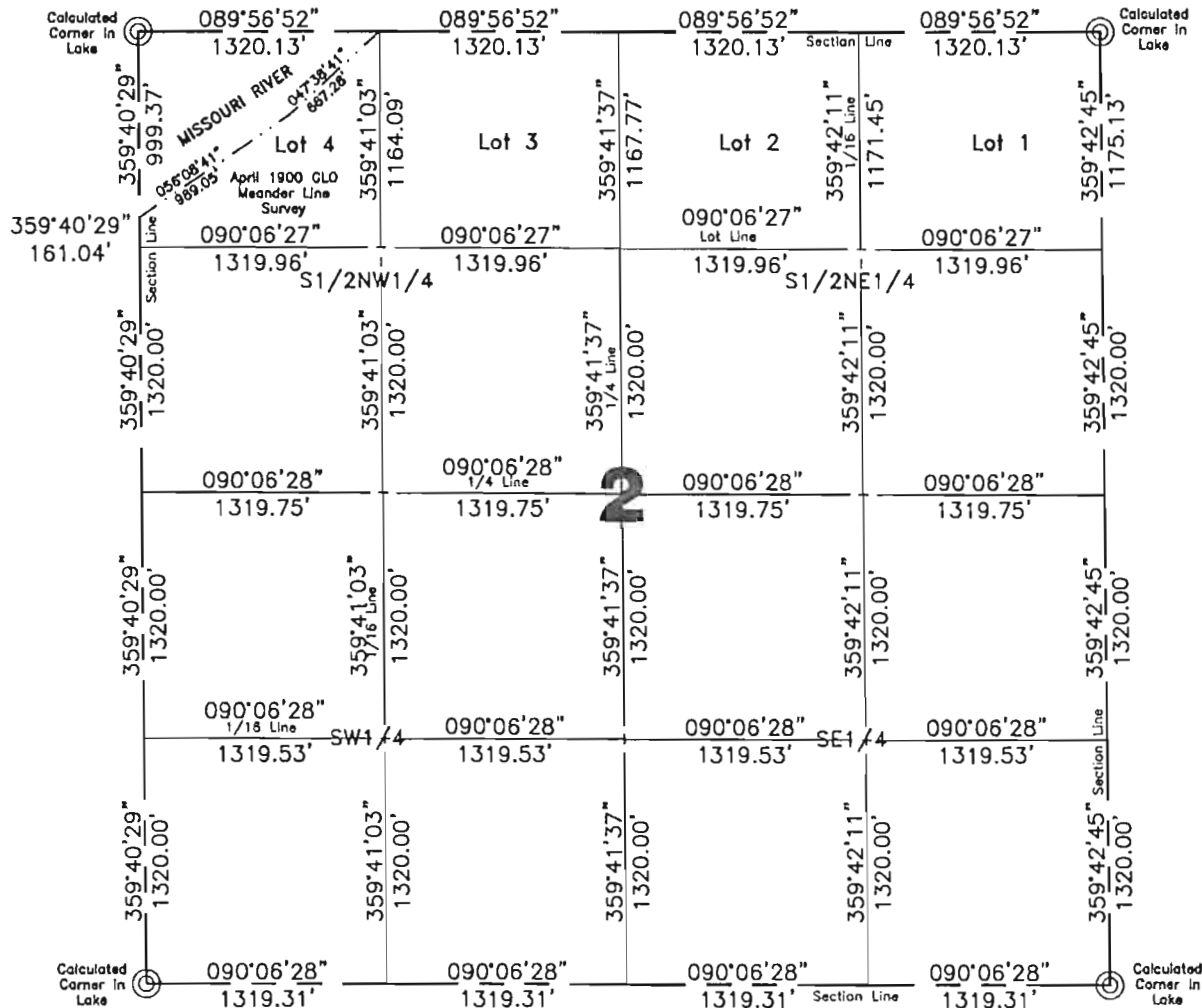
Section 11, T. 153 N., R. 101 W., 5th P.M.

McKenzie County, North Dakota

Surface owner @ well site - Sharon D. Eilers et al

Latitude 48°04'56.376" North; Longitude 103°38'09.239" West (surface location)

[Derived from OPUS Solution NAD-83(CORS96)]



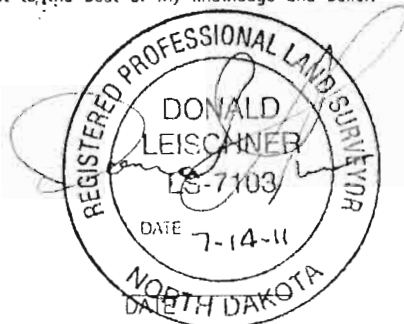
Confidentiality Notice: The information contained on this plat is legally privileged and confidential information intended only for the use of recipients. If you are not the intended recipients, you are hereby notified that any use, dissemination, distribution or copying of this information is strictly prohibited.

Scale 1"=1000'

I, Donald Leischner, Professional Land Surveyor, N.D. No. 7103, do hereby certify that the survey plat shown hereon was made by me, or under my direction, from notes made in the field, and the same is true and correct to the best of my knowledge and belief.

All corners shown on this plat were found in the field during SM Energy, Dahl 15-11H oil well survey on June 14, 2010. Distances to all others are calculated. All azimuths are based on the south line of Section 11, being on an azimuth of 090°06'47".

Surveyed By	Field Book
J. McElhiney	0-89
Computed & Drawn By	Project No.
M. Meisner	8711180



Kadmas
Lee &
Jackson
Engineers Surveyors
Planners

BOTTOM HOLE LOCATION PLAT

SM Energy Company
P.O. Box 7168 Billings, MT 59103

Dahl 15-11H

22 feet from the south line and 2488 feet from the east line (surface location)

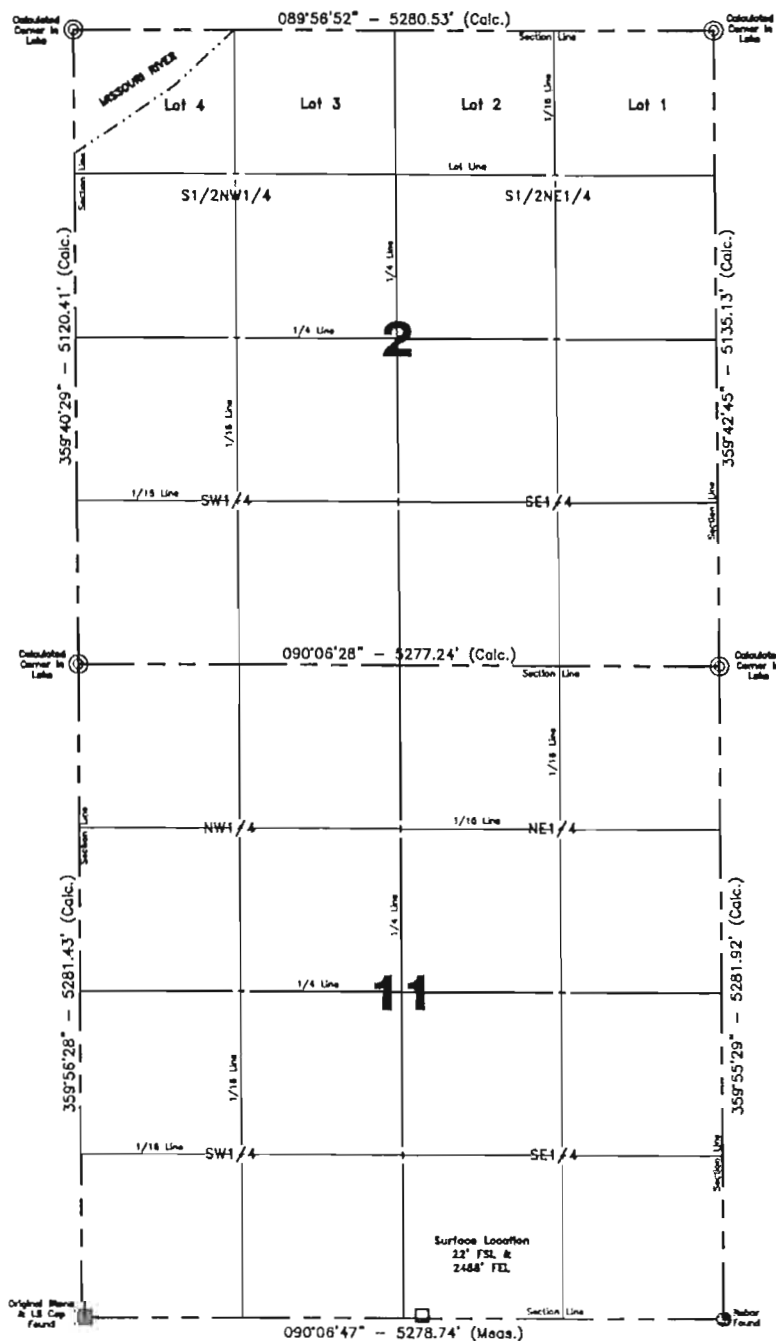
Section 11, T. 153 N., R. 101 W., 5th P.M.

McKenzie County, North Dakota

Surface owner @ well site - Sharon D. Eilers et al

Latitude 48°04'56.376" North; Longitude 103°38'09.239" West (surface location)

[Derived from OPUS Solution NAD-83(CORS96)]



Scale 1"=1500'

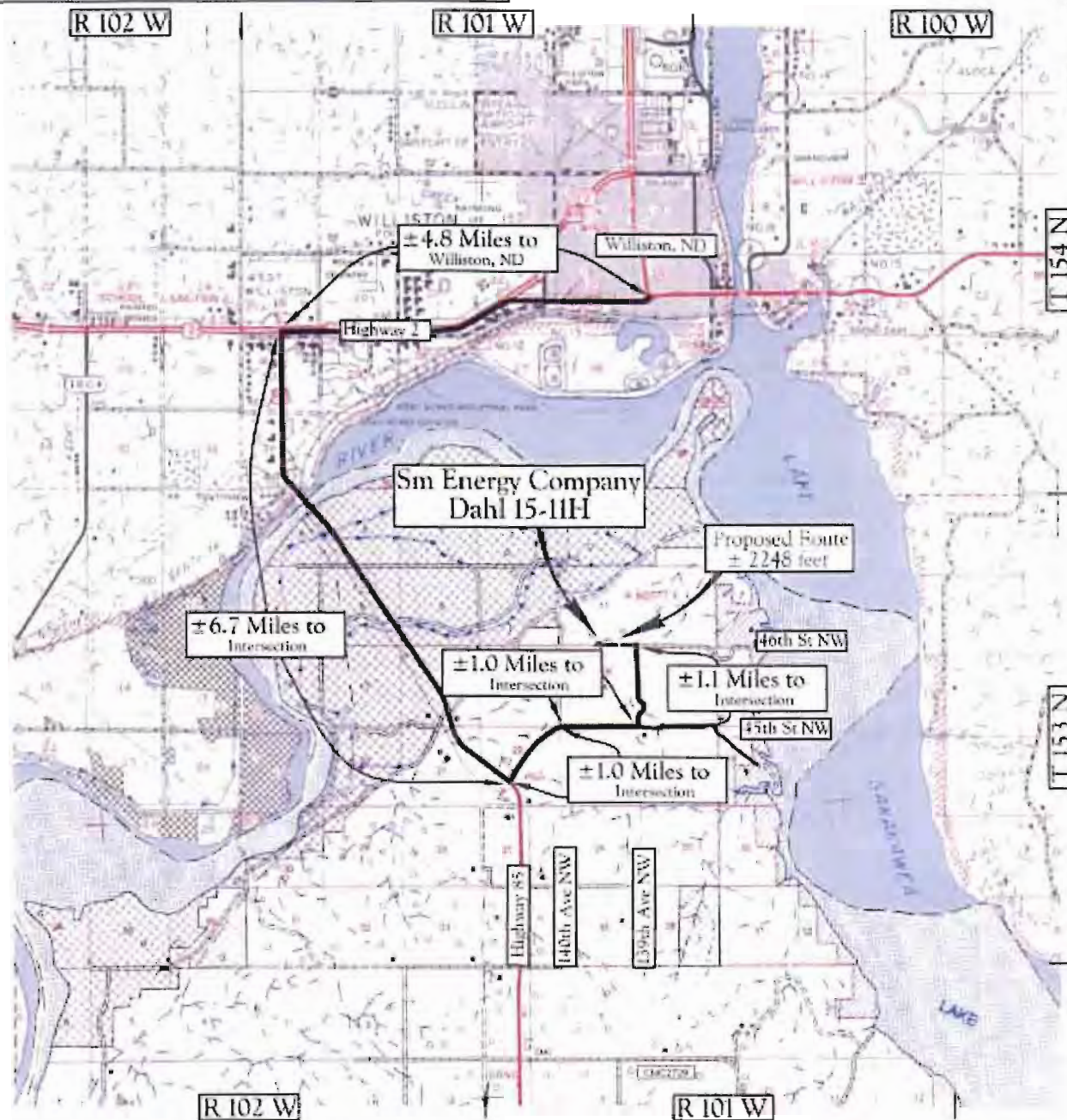
Confidentiality Notice:
The information contained on this plat is legally privileged and confidential information intended only for the use of recipients. If you are not the intended recipient, you are hereby notified that any use, dissemination, distribution or copying of this information is strictly prohibited.

Computed & Drawn By M. Meisner	Surveyed By J. McElhiney	Approved By D. Leischner	Scale 1"=1500'	Date 07/01/2011
Field Book O-89	Material B.H. Layout	Revised —	Project No. 8711180	Drawing No. 4

Kadmas
Lee &
Jackson
Engineers Surveyors
Planners

SM Energy Company
 Dahl 15-11H
 22' FSL & 2488' FEL
 SW1/4SE1/4 Section 11
 T. 153 N., R.101W., 5th P.M.
 McKenzie County, ND

Confidentiality Notice: The information contained on this plot is legally privileged and confidential information intended only for the use of recipients. If you are not the intended recipient, you are hereby notified that any use, dissemination, distribution or copying of this information is strictly prohibited.



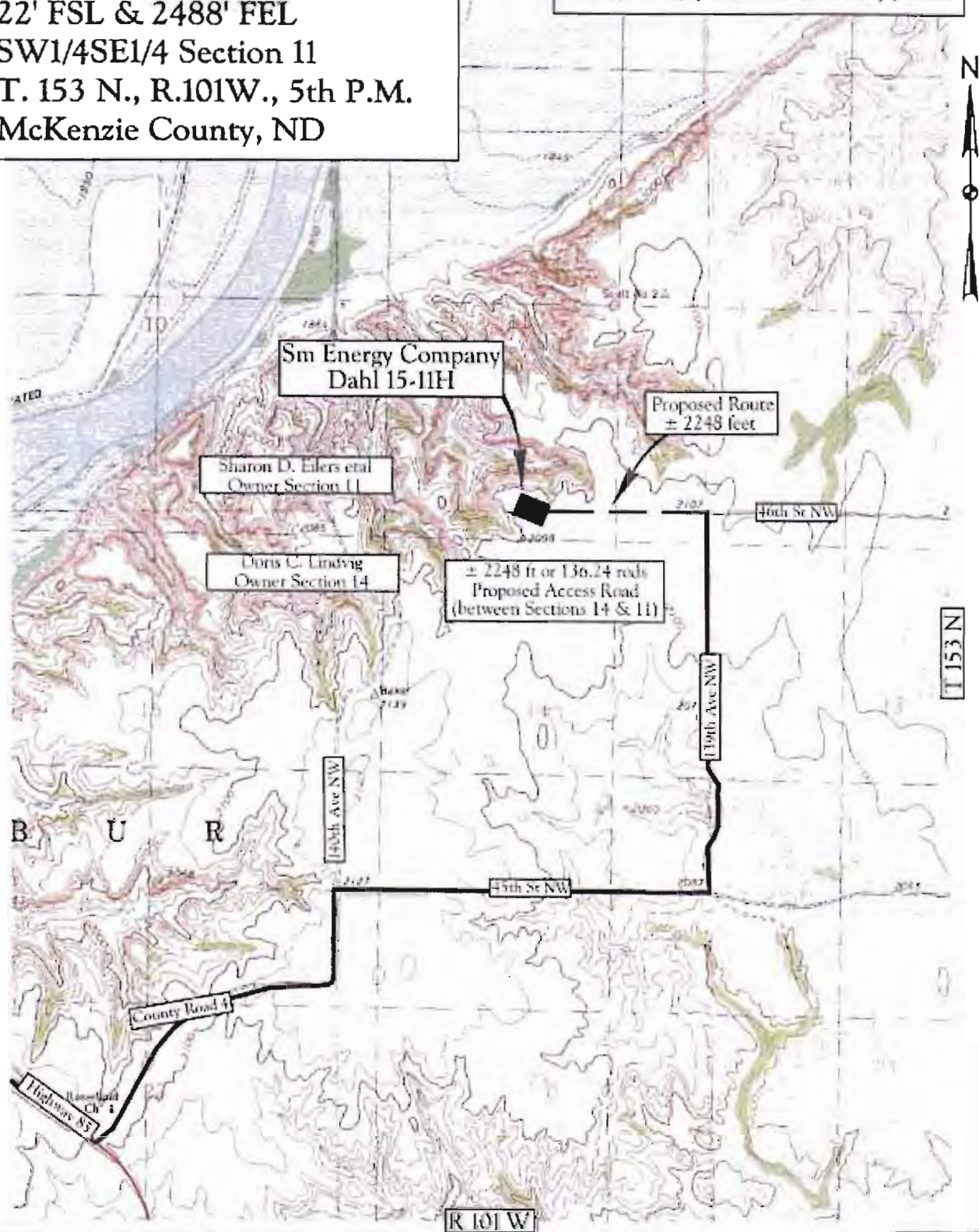
Map "A"
 County Access Route

Legend
 Existing Roads ———
 Proposed Roads - - - - -

Kadmas
 Lee &
 Jackson
 Engineers Surveyors
 Planners

SM Energy Company
 Dahl 15-11H
 22' FSL & 2488' FEL
 SW1/4SE1/4 Section 11
 T. 153 N., R.101W., 5th P.M.
 McKenzie County, ND

Confidentiality Notice: The information contained on this plot is legally privileged and confidential information intended only for the use of recipients. If you are not the intended recipient, you are hereby notified that any use, dissemination, distribution or copying of this information is strictly prohibited.



Map "B"
 Quad Access Route

Legend
 Existing Roads —————
 Proposed Roads - - - - -

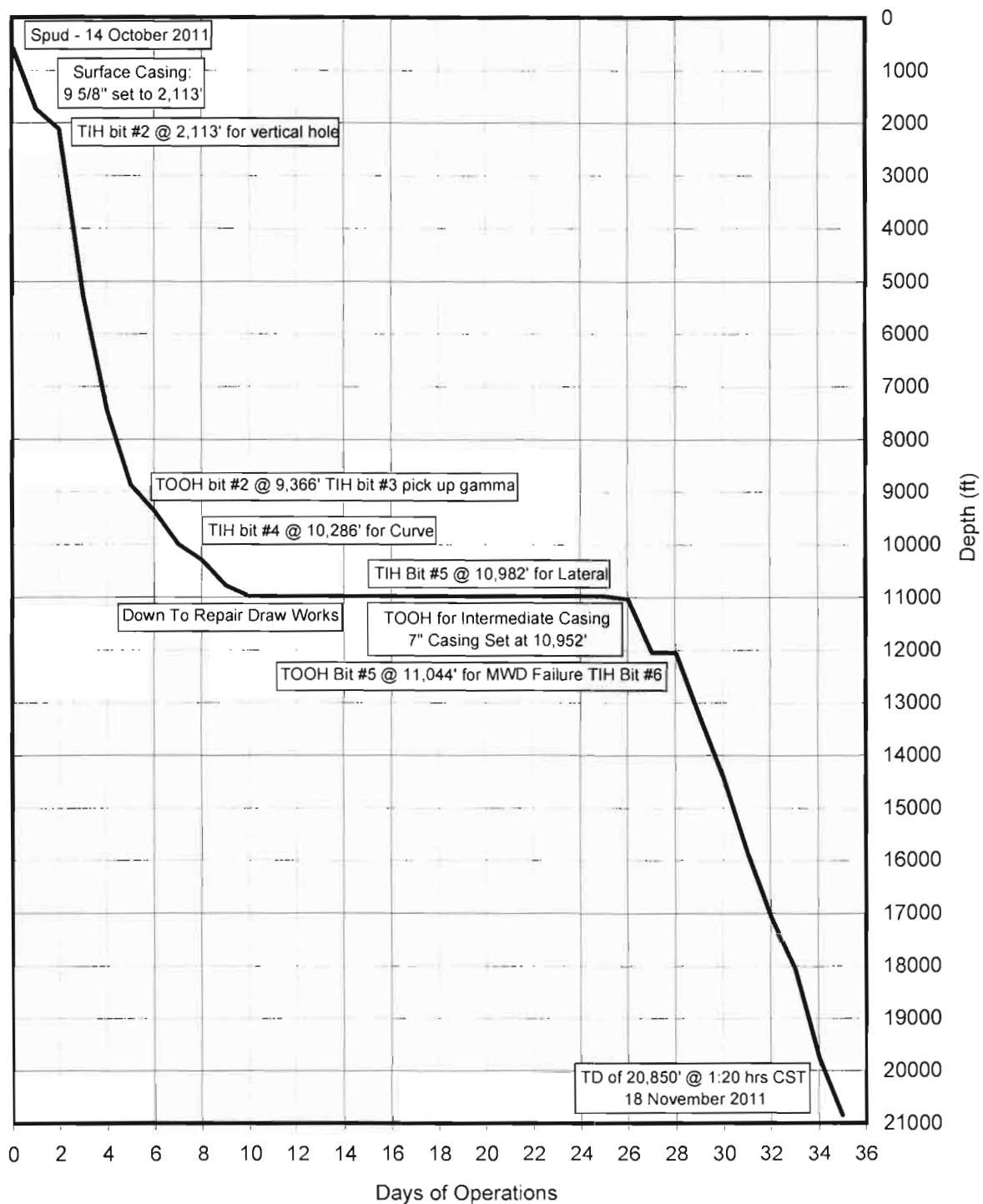
Scale 1" = 2000'

Kadmas
 Lee &
 Jackson
 ENGINEERS, SURVEYORS
 PLANNERS

TIME VS DEPTH

SM Energy

Dahl 15-11H



DAILY DRILLING SUMMARY

Day	Date	Depth (0600 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	WOB (Klbs) MM	RPM (RT)	RPM (MM)	PP	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
0	10/14	535'	-	-	-	-	-	-	-	-	-	-	Condition mud & circulate. Cement 30 min safety meeting. Cement - Rig down cementers. Cut off conductor pipe. Weld off well, head, Nipple up B.O.P., Test B.O.P., Test pipe rams & floor valve 2500 PSI, Annular 1500 PSI, Choke & blinds 5000 PSI, Set wear bushing, Dir. works, P/U BHA, Trip in hole, Tag float collar at 2092, Drill cement	-
1	10/15	1,729'	1194	1	30	-	-	-	1700	110	110	-	Condition mud & circulate. Cement 30 min safety meeting. Cement - Rig down cementers. Cut off conductor pipe. Weld off well, head, Nipple up B.O.P., Test B.O.P., Test pipe rams & floor valve 2500 PSI, Annular 1500 PSI, Choke & blinds 5000 PSI, Set wear bushing, Dir. works, P/U BHA, Trip in hole, Tag float collar at 2092, Drill cement	-
2	10/16	2,113'	384	1	-	-	-	-	-	-	-	-	Condition mud & circulate. Cement 30 min safety meeting. Cement - Rig down cementers. Cut off conductor pipe. Weld off well, head, Nipple up B.O.P., Test B.O.P., Test pipe rams & floor valve 2500 PSI, Annular 1500 PSI, Choke & blinds 5000 PSI, Set wear bushing, Dir. works, P/U BHA, Trip in hole, Tag float collar at 2092, Drill cement	-
3	10/17	5,291'	3178	2	18	-	-	-	3900	70	70	-	Drill actual from 2113 to 4296. Lubricate rig, Grease down works, crown, top drive, Check C.O.M., Drill actual from 4292 to 4792, Lubricate rig, Check C.O.M., Repair top drive, diagnose, replace top drive, Encoder, Drill actual from 4792 to 5291	-
4	10/18	7,453'	2162	2	20	-	-	-	3800	62	62	-	Drill actual from 5291 to 5873. Lubricate rig, Grease top drive, down works, crown, Check C.O.M., Drill actual from 5873 to 6538, Drill actual from 6538 to 7287, Lubricate rig, Grease crown, Top drive and Down works, Check C.O.M., Drill actual from 7287 to 7453.	-
5	10/19	8,867'	1414	2	26	-	-	-	3800	62	62	-	Drill actual from 7453 to 8119, Lubricate rig, Drill actual from 8119 to 8202, Drill actual from 8202 to 8782, Lubricate rig, Check C.O.M., Grease crown, Top drive and Down works, Drill actual from 8782 to 8867.	-
6	10/20	9,366'	499	2	28	-	40	-	3600	60	60	-	Drill actual from 8867 to 9116, Lubricate rig, Grease Top drive, Down works, Crown, Check C.O.M., Drill actual from 9116 to 9283, Drill actual from 9283 to 9366, Trip out of hole, Direction work, Change bit & Service MWD, tool for logging, Trip in hole.	-
7	10/21	9,985'	619	3	31	-	54	237	3800	60	60	-	Drill actual from 8867-9170, Lubricate rig, Grease top drive, down works and crown, Check C.O.M., Drill actual from 9116-9283, Drill actual from 9283-9366, Trip out of hole, Directional work, Change bit and service MWD tool for logging, Trip in hole	Ratcliffe

DAILY DRILLING SUMMARY

Day	Date	Depth (0600 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	WOB (Klbs) MM	RPM (RT)	RPM (MM)	PP	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
8	10/22	10,286'	301	4	28	18	55	233	3800	59	59	415	Drill actual from 9985 to 10194, Lubricate rig, Grease Down drive, Top drive, Crown, Check C.O.M., Drill actual from 10194 to 10276, Lubricate rig, Check C.O.M., Drill actual from 10276 to 10286, Trip out of hole through salts, Pump dry pill, Trip out of hole, Directional work, Change BHA for curve.	Lodgepole
9	10/23	10,777'	491	4	41	32	50	221	3900	56	56	394	Trip in hole, Curve assembly, Drill actual - Sliding, Drill actual from 10362 to 10777, Lubricate rig, Grease Crown and Down works, Check, C.O.M.	Lodgepole
10	10/24	10,982'	205	4	23	42	38	221	3980	56	56	394	Drill actual from 10777 to total depth 10982 reached @10:30, Short trip, Condition mud and circulate rig service, Trip out of hole, Dir. work lay down tools, Change bails ram clamps, Lubricate rig, Casing S/M, Rig up casers, Run intermediate casing.	Middle Bakken
11	10/25	10,982'	0	4	-	-	-	-	-	-	-	-	Run intermediate casing, Condition mud and circulate bottoms up, Rig service, Run intermediate casing, Wash to bottom, Work stuck casing, Run intermediate casing, Circulate and wash to bottom.	Middle Bakken
12	10/26	10,982'	0	4	-	-	-	-	-	-	-	-	Work intermediate casing to bottom, Condition mud and circulate, Rig up cementers, Cement first stage, Circulate for second stage cement job, S/M, Rig up cementers, Cement, Rig down cementers, Clean tanks, Change out bails and elevators Lubricate rig, Set pack off tool, Dir. work pick up tools, Trips in hole.	Middle Bakken
13	10/27	10,982'	0	4	-	-	-	-	-	-	-	-	Trips in hole, Cut off drilling line, Lubricate rig, Grease Skate/T.D./D.W./ Crown, Test casing, Trips in hole, Drill out DV tool, Test casing, Trips in hole, Drill actual, Condition mud and circulate, Pump pill, Repair top drive, Fix grabber, Trips out of hole.	Middle Bakken
14	10/28	10,982'	0	5	-	-	-	-	-	-	-	-	Trouble shoot drawworks, Trip out of hole for MWD failure, Dir. Work, Lay down MWD, Pick up new one, Trip in hole, Repair rig, Down works down.	Middle Bakken
15	10/29	10,982'	0	-	-	-	-	-	-	-	-	-	Repair Draw works	Middle Bakken
16	10/30	10,982'	0	5	-	-	-	-	-	-	-	-	Repair Draw works, Repair rig, Down works down	Middle Bakken
17	10/31	10,982'	0	5	-	-	-	-	-	-	-	-	Repair Draw works	Middle Bakken
18	11/1	10,982'	0	5	-	-	-	-	-	-	-	-	Repair Draw works	Middle Bakken
19	11/2	10,982'	0	5	-	-	-	-	-	-	-	-	Repair Draw works	Middle Bakken

DAILY DRILLING SUMMARY

Day	Date	Depth (6000 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	WOB (Klbs) MM	RPM (RT)	RPM (MM)	PP	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
20	11/3	10,982'	0	5	-	-	-	-	-	-	-	-	Repair Draw works	Middle Bakken
21	11/4	10,982'	0	5	-	-	-	-	-	-	-	-	Repair Draw works, BOP drill	Middle Bakken
22	11/5	10,982'	0	5	-	-	-	-	-	-	-	-	Repair Draw works	Middle Bakken
23	11/6	10,982'	0	5	-	-	-	-	-	-	-	-	Repair Draw works	Middle Bakken
24	11/7	10,982'	0	5	-	-	-	-	-	-	-	-	Repair Draw works, Reinstall Draw works package, Repair draw works, Circulate out gas.	Middle Bakken
25	11/8	10,982'	0	6	-	-	-	-	-	-	-	-	Repair rig, Pump kill mud, Repair Draw works, Repair rig, Trip out of hole, Repair rig, L/D tools, Repair rig, Set bridge plug/Test BOP, Testing BOP, Repair rig/Pull bridge plug, Rig down, C/K C.O.M., Repair rig/Pick up BHA, Repair rig, Trip in hole.	Middle Bakken
26	11/9	11,044'	62	6	12	28	58	158	2475	80	0	282	Repair rig, Trip in hole, Condition mud and circulate, Drill actual from 11044 to 11164, Drill actual from 11164 to 11715, Lubricate rig, Grease Down works, Top drive, Check C.O.M.	Middle Bakken
27	11/10	12,055'	1011	6	13	18	59	158	2500	80	0	282	Drill actual from 11715 to 12055, Trouble shoot high pressure, Condition mud and circulate kill mud, Trip out of hole, Dir. Work, Change BHA, Trips in hole	Middle Bakken
28	11/11	12,055'	0	6	-	-	-	-	-	-	-	-	Drill actual from 11715 to 12055, Trouble shoot high pressure, Condition mud and circulate kill mud, Trip out of hole, Dir. Work, Change BHA, Trip in hole	Middle Bakken
29	11/12	13,250'	1195	6	15	18	63	158	2900	0	80	282	Repair drawworks, drill 12560'-13000', lubricate rig-grease TD, DW, check C.O.M., drill line, brakes, drill f/1300'-13250', drill f/13250'-	Middle Bakken
30	11/13	14,430'	1180	6	15	27	65	158	3030	80	0	282	Lubricate rig, C/K C.O.M., inspect drill line, repair drawworks, rewire RTD, Drill f/13250'-13764', drill 13764'-13998', change out rotating head, drill f/1399-14430	Middle Bakken
31	11/14	15,869'	1439	6	15	45	65	158	2605	80	0	282	Drill f/14430'-14912', lubricate rig, chg C.O.M. grease crown, T.D. D.W. drill f/ 14912'-15163', drill 15163'-15869	Middle Bakken
32	11/15	17,078'	1209	6	15	52	50	158	3140	0	80	282	Drill f/15869'-16242', lubricate rig grease top drive, crown, check C.O.M., drill 16242'-16411', Drill f/ 16411'-16991', Lubricate rig grease drawworks, top drive, ck C.O.M. inspect drill line. Drill f/16991'-17078'	Middle Bakken
33	11/16	18,047'	969	6	15	42	50	158	3240	0	80	282	Drill f/17078'-17658', Install new rotating rubber. Lubricate rig, ck C.O.M. grease TD, crown, drawworks. Drill f/17658'-17700', Drill f/17700'-18323', Lubricate rig, grease td, dw, inspect drill line, check C.O.M. drill f/ 18323'-18407'.	Middle Bakken

DAILY DRILLING SUMMARY

Day	Date	Depth (0600 Hrs)	24 Hr Footage	WOB (Klbs) RT	WOB (Klbs) MM	RPM (RT)	RPM (MM)	PP	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
34	11/17	19,755'	1708	15	72	50	158	3400	80	0	282	Drill f/18407'-18988', safty stand down-do JSA'S for each job task no cutting corners, lubricate rig grease TD, DW, crown, drill f/18988'-19133'. Drill f/ 19133'-19279'. Safty stand down-discussed the JSA program and how to properly utilize it. Drill f/19279'-19755'	Middle Bakken
35	11/18	20,850'	1095	15	72	50	158	3520	0	80	282	Drill f/19755'-20316', lubricate rig CK C.O.M, grease crown, TD DW. Drill f/20316'-20402'. Drill f/20402'-20850' TD	Middle Bakken

DAILY MUD SUMMARY

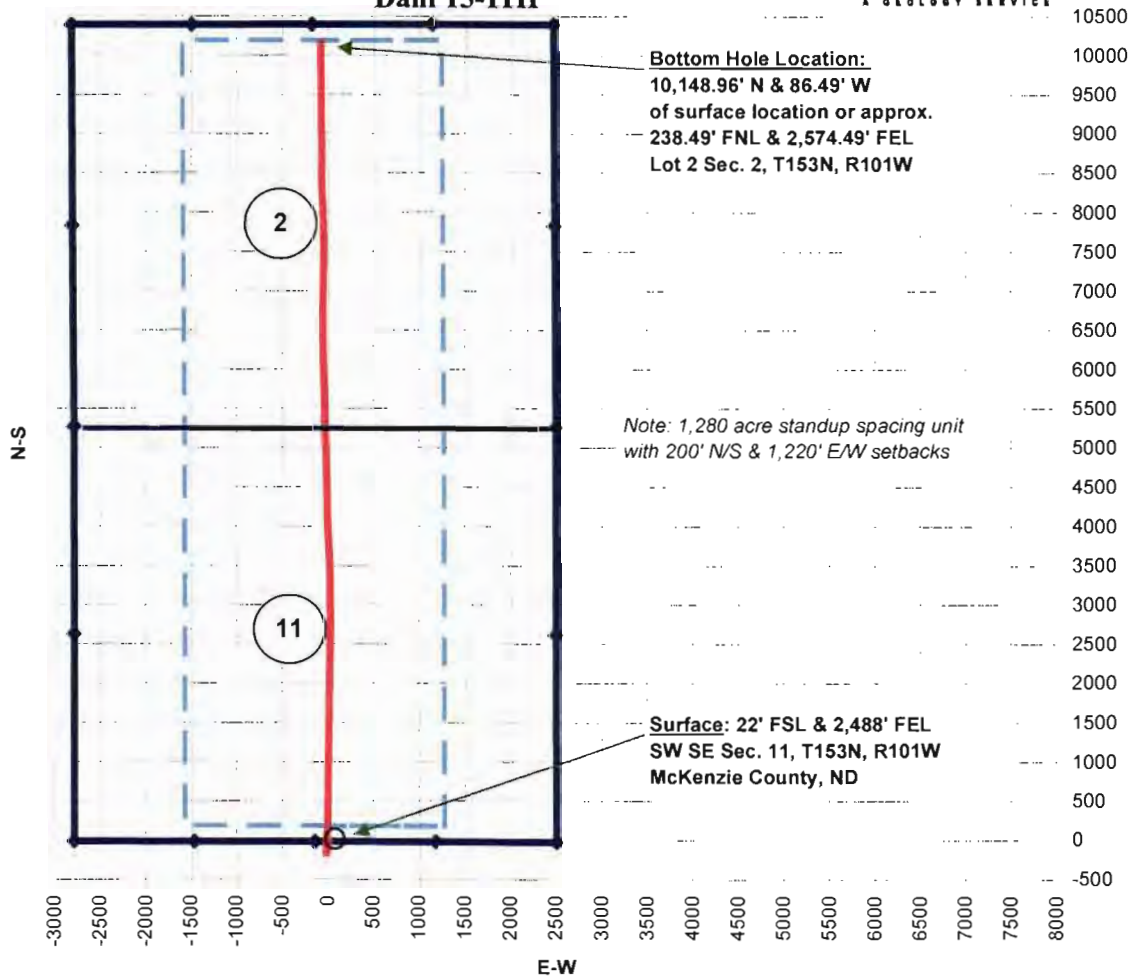
Day	Date 2011	Mud Depth	Mud WT (ppg)	VIS (sec)	PV (cP)	YP (lbs/100 ft ²)	Gels (lbs/100 ft ²)	600/300	NAP/ H ₂ O (ratio)	NAP/ H ₂ O (% by vol)	Cake (API/HTHP)	Cor. Solids (%)	Alk	Excess Lime (lb/bbl)	Cl ⁻ (mg/l)	LGS/ HGS (%)	Electrical Stability	Gain/ Loss (bbls)
7	10/20	9,266'	9.90	46	20	6	10/16	46/26	80.9/19.1	70/16.5	2	13.5	2.6	3.4	48k	4.2/7.2	564	-107
8	10/21	9,609'	10.00	50	20	10	10/16	50/30	77.3/22.7	66.5/19.5	2	14	2.7	3.5	50k	4.5/6.9	511	-20
9	10/22	10,085'	10.00	48	18	9	9/16	45/27	79.1/20.9	68/18	2	14	2.7	3.5	51k	4.8/6.8	520	-37
10	10/23	10,286'	10.20	58	18	8	9/15	44/26	80.2/19.8	69/17	2	14	2.6	3.4	50k	4.2/7.6	572	-70
11	10/24	10,880'	10.10	45	-	-	-	-	-	-	-	-	-	-	-	-	-	-
12	10/25	10,982'	10.10	57	-	-	-	-	-	-	-	-	-	-	-	-	-	-
13	10/26	10,982'	10.10	57	-	-	-	-	-	-	-	-	-	-	-	-	-	-
14	10/27	10,982'	10.10	57	-	-	-	-	-	-	-	-	-	-	-	-	-	-
15	10/28	10,982'	10.10	57	-	-	-	-	-	-	-	-	-	-	-	-	-	-
16	10/29	10,982'	10.10	57	-	-	-	-	-	-	-	-	-	-	-	-	-	-
17	10/30	10,982'	10.10	57	-	-	-	-	-	-	-	-	-	-	-	-	-	-
18	10/31	10,982'	10.10	57	-	-	-	-	-	-	-	-	-	-	-	-	-	-
19	11/01	10,982'	10.10	57	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20	11/02	10,982'	10.10	57	-	-	-	-	-	-	-	-	-	-	-	-	-	-
21	11/03	10,982'	10.10	57	-	-	-	-	-	-	-	-	-	-	-	-	-	-
22	11/04	10,982'	10.10	57	-	-	-	-	-	-	-	-	-	-	-	-	-	-
23	11/05	10,982'	10.10	57	-	-	-	-	-	-	-	-	-	-	-	-	-	-
24	11/06	10,982'	10.10	57	-	-	-	-	-	-	-	-	-	-	-	-	-	-
30	11/07	10,982'	10.10	57	-	-	-	-	-	-	-	-	-	-	-	-	-	-
26	11/08	10,982'	10.10	57	-	-	-	-	-	-	-	-	-	-	-	-	-	-
27	11/09	11,066'	9.90	30	-	-	-	-	-	-	-	-	-	-	-	-	-	-
28	11/10	12,040'	9.90	28	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	11/11	12,040'	9.90	28	-	-	-	-	-	-	-	-	-	-	-	-	-	-
30	11/12	12,595'	9.90	28	-	-	-	-	-	-	-	-	-	-	-	-	-	-
31	11/13	14,200'	9.90	28	-	-	-	-	-	-	-	-	-	-	-	-	-	-
32	11/14	15,750'	9.90	28	-	-	-	-	-	-	-	-	-	-	-	-	-	-
33	11/15	16,505'	9.90	28	-	-	-	-	-	-	-	-	-	-	-	-	-	-
34	11/16	18,020'	9.85	28	-	-	-	-	-	-	-	-	-	-	-	-	-	-
35	11/17	19,200'	9.85	27	-	-	-	-	-	-	-	-	-	-	-	-	-	-

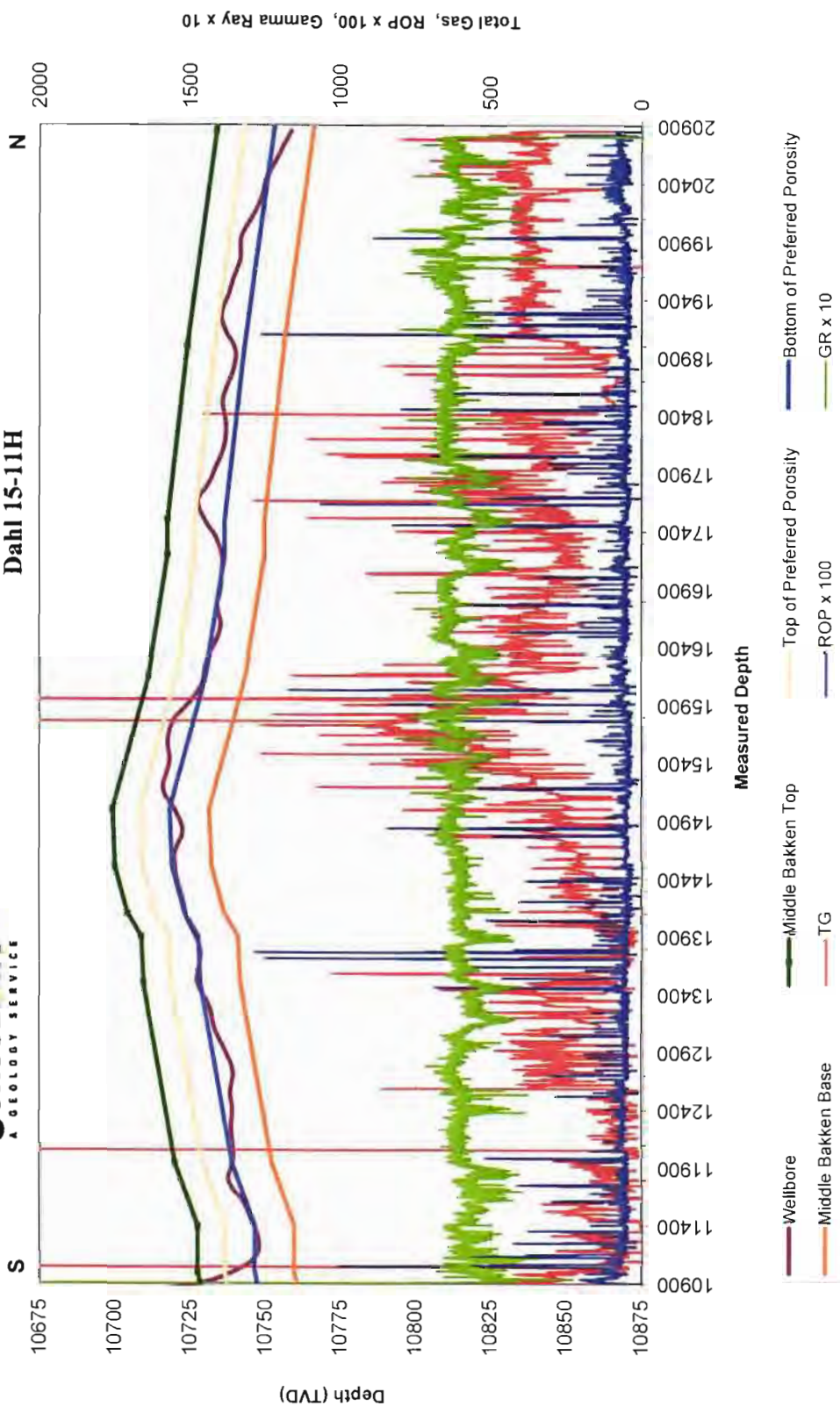
BIT RECORD

Bit #	Size	Type	Make	Model	Serial #	Jets	Depth In	Depth Out	Footage	Hours	Accum. Hours	Vert. Dev.
1	13 1/2	Tri-cone	Hughes	MXL-1X	7-1-02C8	3X20	80'	2,133'	2,092'	18.5	18.50	Surface
2	8 3/4	PDC	Hughes	DP506FX	7132487	6X16	2,133'	9,366'	7,233'	65	83.50	Vertical
3	8 3/4	PDC	Hughes	DP506FX	7129301	6X16	9,366'	10,286'	920'	36	119.50	Vertical
4	8 3/4	PDC	Security	FXD55M	11681292	6X17	10,286'	10,982'	696'	19.5	139.00	Curve
5	6	PDC	Hughes	Q505FX	7127699	5X16	10,982'	11,044'	62'	1	140.00	Lateral
6	6	PDC	Hughes	Q505FX	10982	5X16	11,044'	20,850'	9,806'	168	308.00	Lateral

PLAN VIEW

**SM Energy Company
Dahl 15-11H**





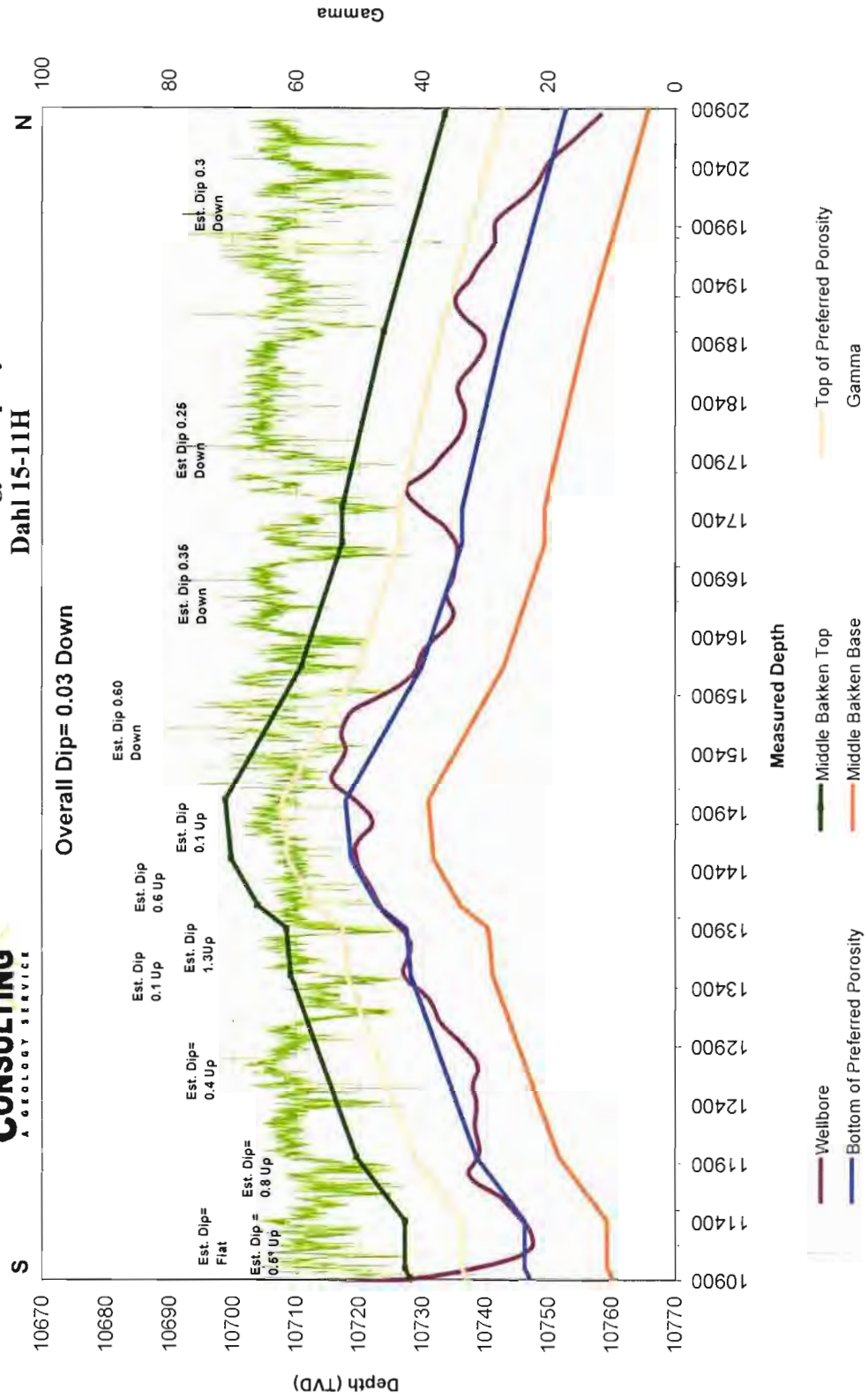
FORMATION MARKERS & DIP ESTIMATES

SM Energy Company - Dahl 15-11H

Dip Change Points	MD	TVD	TVD diff.	MD diff.	Dip	Dipping up/down	Type of Marker
Marker							
Cool Gamma Base Of Preferred Porosity	11,095'	10,744.77					
Cool Gamma Base Of Preferred Porosity	11,724'	10,738.67	-6.10	629.00	0.56	Up	Gamma
Cool Gamma Base Of Preferred Porosity	11,883'	10,737.98	-0.69	159.00	0.25	Up	Gamma
Cool Gamma Base Of Preferred Porosity	12,426'	10,738.47	0.49	543.00	-0.05	Down	Gamma
Cool Gamma Base Of Preferred Porosity	12,532'	10,737.94	-0.53	106.00	0.29	Up	Gamma
Cool Gamma Base Of Preferred Porosity	13,201'	10,731.87	-6.07	669.00	0.52	Up	Gamma
Cool Gamma Base Of Preferred Porosity	13,470'	10,727.46	-4.41	269.00	0.94	Up	Gamma
Cool Gamma Base Of Preferred Porosity	13,896'	10,726.49	-0.97	426.00	0.13	Up	Gamma
Cool Gamma Base Of Preferred Porosity	15,013'	10,718.86	-7.63	1117.00	0.39	Up	Gamma
Cool Gamma Upper Half Of Preferred Porosity	15,473'	10,717.53	10717.53	15473.00	-	-	Gamma
Cool Gamma Upper Half Of Preferred Porosity	15,950'	10,725.21	7.68	477.00	-0.92	Down	Gamma
Cool Gamma Base Of Preferred Porosity	16,148'	10,729.00	10729.00	16148.00	-	-	Gamma
Cool Gamma Base Of Preferred Porosity	16,387'	10,731.96	2.96	239.00	-0.71	Down	Gamma
Cool Gamma Base Of Preferred Porosity	17,140'	10,735.36	3.40	753.00	-0.26	Down	Gamma
Cool Gamma Base Of Preferred Porosity	19,066'	10,737.73	2.37	1926.00	-0.07	Down	Gamma
Cool Gamma Base Of Preferred Porosity	19,765'	10,741.38	3.65	699.00	-0.30	Down	Gamma
Gross Dip							
Initial Target Contact	10,908'	10,728.00					
Final Target Contact	20,850'	10,733.48	5.48	9942.00	-0.03	Down	

DIP PROFILE

SM Energy Company
Dahl 15-11H



SUNBURST CONSULTING, INC.

Operator:	SM Energy Company		
Well :	Dahl 15-11H		
County:	McKenzie	State:	ND
QQ:	SW SE	Section:	11
Township:	153	N/S:	N
Range:	101	E/W:	W
Footages:	22	FN/SL:	S
	2488	FE/WL:	E

Kick-off:	10/23/2011
Finish:	11/18/2011
Directional Supervision:	Ryan

Date: 11/21/2011
Time: 10:20

F9 to re-calculate

Proposed dir: 360

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		N-S	E-W	SECT	DLS/
			AZM	TVD				100
Tie	2113.00	0.00	0.00	2113.00	0.00	0.00	0.00	0.00
1	2167.00	1.20	203.90	2167.00	-0.52	-0.23	-0.52	2.22
2	2259.00	1.40	206.60	2258.97	-2.40	-1.12	-2.40	0.23
3	2351.00	1.60	215.00	2350.94	-4.46	-2.36	-4.46	0.32
4	2443.00	2.20	195.10	2442.89	-7.22	-3.56	-7.22	0.96
5	2534.00	2.60	190.70	2533.81	-10.93	-4.40	-10.93	0.48
6	2625.00	2.80	184.10	2624.71	-15.18	-4.94	-15.18	0.41
7	2716.00	3.00	189.70	2715.59	-19.74	-5.50	-19.74	0.38
8	2807.00	3.00	194.10	2806.47	-24.40	-6.48	-24.40	0.25
9	2897.00	2.50	192.80	2896.37	-28.60	-7.49	-28.60	0.56
10	2988.00	3.10	185.50	2987.26	-32.98	-8.17	-32.98	0.77
11	3079.00	3.00	185.40	3078.13	-37.80	-8.63	-37.80	0.11
12	3162.00	3.00	184.20	3161.01	-42.13	-8.99	-42.13	0.08
13	3245.00	3.00	183.90	3243.90	-46.46	-9.30	-46.46	0.02
14	3328.00	2.70	185.00	3326.80	-50.58	-9.61	-50.58	0.37
15	3411.00	2.40	185.30	3409.71	-54.25	-9.94	-54.25	0.36
16	3495.00	3.20	184.80	3493.61	-58.34	-10.30	-58.34	0.95
17	3578.00	2.70	184.90	3576.50	-62.60	-10.66	-62.60	0.60
18	3661.00	3.10	181.90	3659.40	-66.79	-10.91	-66.79	0.52
19	3744.00	3.00	183.90	3742.28	-71.20	-11.13	-71.20	0.18
20	3828.00	2.80	183.90	3826.17	-75.44	-11.42	-75.44	0.24
21	3910.00	2.50	183.70	3908.08	-79.22	-11.67	-79.22	0.37
22	3994.00	3.20	181.70	3991.98	-83.39	-11.86	-83.39	0.84
23	4077.00	3.30	182.80	4074.85	-88.09	-12.04	-88.09	0.14
24	4160.00	3.00	185.30	4157.72	-92.64	-12.36	-92.64	0.40
25	4244.00	2.30	191.50	4241.63	-96.48	-12.90	-96.48	0.90
26	4327.00	2.40	191.20	4324.56	-99.82	-13.57	-99.82	0.12
27	4401.00	2.00	197.50	4398.51	-102.57	-14.26	-102.57	0.63
28	4492.00	2.50	197.60	4489.44	-105.98	-15.33	-105.98	0.55
29	4576.00	2.50	202.30	4573.36	-109.42	-16.58	-109.42	0.24
30	4659.00	2.30	198.40	4656.28	-112.67	-17.80	-112.67	0.31
31	4742.00	2.40	197.20	4739.21	-115.92	-18.84	-115.92	0.13
32	4825.00	2.40	199.20	4822.14	-119.22	-19.92	-119.22	0.10
33	4908.00	2.50	196.30	4905.06	-122.60	-21.00	-122.60	0.19
34	4992.00	1.60	192.00	4989.01	-125.50	-21.76	-125.50	1.09
35	5075.00	2.20	193.70	5071.96	-128.18	-22.38	-128.18	0.73
36	5158.00	2.70	190.40	5154.89	-131.65	-23.11	-131.65	0.63
37	5241.00	2.10	184.10	5237.82	-135.09	-23.57	-135.09	0.79
38	5324.00	3.00	177.10	5320.73	-138.78	-23.57	-138.78	1.15
39	5408.00	3.00	182.10	5404.62	-143.17	-23.54	-143.17	0.31

SUNBURST CONSULTING, INC.

Operator:	SM Energy Company		
Well :	Dahl 15-11H		
County:	McKenzie	State:	ND
QQ:	SW SE	Section:	11
Township:	153	N/S:	N
Range:	101	E/W:	W
Footages:	22	FN/SL:	S
	2488	FE/WL:	E

Kick-off:	10/23/2011
Finish:	11/18/2011
Directional Supervision:	Ryan

Date: 11/21/2011

Time: 10:20

F9 to re-calculate

Proposed dir: 360

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		TVD	N-S	E-W	SECT	DLS/
			AZM						100
40	5490.00	2.10	184.80		5486.54	-146.81	-23.74	-146.81	1.11
41	5574.00	1.90	197.40		5570.48	-149.67	-24.29	-149.67	0.57
42	5657.00	1.60	203.20		5653.45	-152.05	-25.15	-152.05	0.42
43	5740.00	2.50	193.70		5736.39	-154.88	-26.04	-154.88	1.16
44	5823.00	2.50	188.50		5819.31	-158.43	-26.74	-158.43	0.27
45	5906.00	2.10	183.50		5902.25	-161.73	-27.10	-161.73	0.54
46	5989.00	1.60	204.20		5985.20	-164.31	-27.66	-164.31	1.00
47	6073.00	1.10	207.30		6069.18	-166.09	-28.51	-166.09	0.60
48	6156.00	0.50	198.20		6152.17	-167.15	-28.99	-167.15	0.74
49	6239.00	0.40	210.70		6235.17	-167.74	-29.25	-167.74	0.17
50	6322.00	0.30	225.80		6318.17	-168.14	-29.56	-168.14	0.16
51	6405.00	0.10	240.10		6401.17	-168.33	-29.78	-168.33	0.25
52	6488.00	0.20	8.20		6484.17	-168.22	-29.82	-168.22	0.33
53	6571.00	0.30	355.80		6567.17	-167.86	-29.81	-167.86	0.14
54	6655.00	0.20	2.70		6651.17	-167.49	-29.82	-167.49	0.12
55	6738.00	0.30	345.20		6734.16	-167.14	-29.87	-167.14	0.15
56	6821.00	0.30	328.20		6817.16	-166.75	-30.04	-166.75	0.11
57	6904.00	0.40	345.10		6900.16	-166.28	-30.23	-166.28	0.17
58	6987.00	0.60	337.80		6983.16	-165.60	-30.47	-165.60	0.25
59	7071.00	0.70	329.30		7067.15	-164.75	-30.90	-164.75	0.17
60	7154.00	0.60	2.40		7150.15	-163.88	-31.14	-163.88	0.46
61	7237.00	0.60	8.60		7233.14	-163.02	-31.06	-163.02	0.08
62	7320.00	0.70	19.50		7316.14	-162.11	-30.82	-162.11	0.19
63	7403.00	0.80	23.30		7399.13	-161.10	-30.42	-161.10	0.13
64	7486.00	0.80	19.30		7482.12	-160.02	-30.00	-160.02	0.07
65	7569.00	0.70	16.70		7565.12	-158.99	-29.66	-158.99	0.13
66	7653.00	0.70	17.30		7649.11	-158.00	-29.36	-158.00	0.01
67	7736.00	0.60	40.50		7732.10	-157.19	-28.93	-157.19	0.34
68	7819.00	1.10	43.10		7815.10	-156.28	-28.10	-156.28	0.60
69	7902.00	1.30	36.30		7898.08	-154.94	-27.00	-154.94	0.30
70	7985.00	1.00	32.30		7981.06	-153.57	-26.06	-153.57	0.37
71	8068.00	0.80	22.60		8064.05	-152.42	-25.45	-152.42	0.30
72	8152.00	0.90	28.10		8148.04	-151.30	-24.91	-151.30	0.15
73	8235.00	1.00	44.40		8231.03	-150.20	-24.10	-150.20	0.35
74	8318.00	0.90	30.30		8314.02	-149.12	-23.26	-149.12	0.31
75	8401.00	0.80	39.00		8397.01	-148.11	-22.57	-148.11	0.20
76	8484.00	0.80	31.10		8480.00	-147.16	-21.91	-147.16	0.13
77	8567.00	0.80	7.00		8562.99	-146.09	-21.54	-146.09	0.40
78	8650.00	0.80	351.50		8645.98	-144.94	-21.55	-144.94	0.26
79	8734.00	0.80	349.30		8729.98	-143.79	-21.75	-143.79	0.04

SUNBURST CONSULTING, INC.

Operator:	SM Energy Company		
Well :	Dahl 15-11H		
County:	McKenzie	State:	ND
QQ:	SW SE	Section:	11
Township:	153	N/S:	N
Range:	101	E/W:	W
Footages:	22	FN/SL:	S
	2488	FE/WL:	E

Kick-off:	10/23/2011
Finish:	11/18/2011
Directional Supervision:	Ryan

Date: 11/21/2011
Time: 10:20

F9 to re-calculate

Proposed dir: 360

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		N-S	E-W	SECT	DLS/ 100
			AZM	TVD				
80	8817.00	1.00	355.30	8812.97	-142.50	-21.91	-142.50	0.27
81	8900.00	1.00	355.80	8895.95	-141.05	-22.03	-141.05	0.01
82	8984.00	1.00	1.70	8979.94	-139.59	-22.06	-139.59	0.12
83	9066.00	1.10	352.10	9061.93	-138.09	-22.14	-138.09	0.25
84	9149.00	1.00	5.60	9144.91	-136.58	-22.18	-136.58	0.32
85	9233.00	1.30	2.60	9228.90	-134.90	-22.07	-134.90	0.36
86	9313.00	1.00	0.40	9308.88	-133.30	-22.02	-133.30	0.38
87	9396.00	1.10	9.10	9391.87	-131.79	-21.89	-131.79	0.23
88	9479.00	0.70	27.50	9474.86	-130.55	-21.53	-130.55	0.59
89	9562.00	0.80	117.30	9557.85	-130.37	-20.78	-130.37	1.28
90	9646.00	1.70	154.80	9641.83	-131.76	-19.73	-131.76	1.39
91	9729.00	3.00	182.40	9724.76	-135.05	-19.30	-135.05	2.03
92	9812.00	3.60	185.80	9807.62	-139.81	-19.65	-139.81	0.76
93	9895.00	3.40	187.10	9890.47	-144.85	-20.22	-144.85	0.26
94	9978.00	3.20	187.30	9973.33	-149.59	-20.82	-149.59	0.24
95	10061.00	3.10	196.70	10056.21	-154.03	-21.76	-154.03	0.63
96	10145.00	2.20	200.30	10140.12	-157.72	-22.97	-157.72	1.09
97	10228.00	1.40	205.90	10223.08	-160.13	-23.97	-160.13	0.99
98	10274.00	0.90	255.40	10269.07	-160.72	-24.56	-160.72	2.31
99	10316.00	4.90	358.20	10311.02	-159.01	-24.94	-159.01	12.32
100	10356.00	12.60	359.70	10350.52	-152.93	-25.01	-152.93	19.26
101	10398.00	18.50	359.40	10390.97	-141.68	-25.11	-141.68	14.05
102	10440.00	24.80	359.80	10429.98	-126.19	-25.21	-126.19	15.00
103	10481.00	28.50	359.40	10466.62	-107.81	-25.34	-107.81	9.03
104	10523.00	32.40	358.90	10502.82	-86.53	-25.66	-86.53	9.31
105	10564.00	35.30	359.90	10536.87	-63.69	-25.89	-63.69	7.20
106	10606.00	39.30	0.60	10570.27	-38.25	-25.77	-38.25	9.58
107	10648.00	43.60	1.70	10601.74	-10.46	-25.20	-10.46	10.38
108	10689.00	47.60	2.80	10630.43	18.80	-24.05	18.80	9.94
109	10731.00	52.40	4.80	10657.42	50.89	-21.89	50.89	12.00
110	10772.00	59.10	4.20	10680.48	84.66	-19.24	84.66	16.39
111	10814.00	65.80	4.40	10699.89	121.77	-16.45	121.77	15.96
112	10855.00	71.80	4.20	10714.71	159.87	-13.59	159.87	14.64
113	10897.00	77.50	1.30	10725.83	200.31	-11.66	200.31	15.12
114	10936.00	83.50	1.30	10732.26	238.74	-10.79	238.74	15.38
115	10988.00	84.80	0.20	10737.56	290.47	-10.11	290.47	3.27
116	11080.00	86.80	359.70	10744.30	382.22	-10.19	382.22	2.24
117	11172.00	89.60	0.80	10747.19	474.16	-9.79	474.16	3.27
118	11263.00	90.20	1.00	10747.35	565.15	-8.36	565.15	0.70
119	11355.00	90.70	359.80	10746.63	657.14	-7.72	657.14	1.41

SUNBURST CONSULTING, INC.

Operator:	SM Energy Company		
Well :	Dahl 15-11H		
County:	McKenzie	State:	ND
QQ:	SW SE	Section:	11
Township:	153	N/S:	N
Range:	101	E/W:	W
Footages:	22	FN/SL:	S
	2488	FE/WL:	E

Kick-off:	10/23/2011
Finish:	11/18/2011
Directional Supervision:	Ryan

Date: 11/21/2011
Time: 10:20

F9 to re-calculate

Proposed dir: 360

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		TVD	N-S	E-W	SECT	DLS/
			AZM						100
120	11447.00	91.20	0.10		10745.10	749.13	-7.80	749.13	0.63
121	11539.00	90.70	0.60		10743.58	841.11	-7.24	841.11	0.77
122	11630.00	91.60	0.90		10741.75	932.08	-6.05	932.08	1.04
123	11722.00	92.20	1.30		10738.70	1024.02	-4.28	1024.02	0.78
124	11814.00	89.60	1.40		10737.25	1115.97	-2.11	1115.97	2.83
125	11905.00	89.20	0.90		10738.21	1206.95	-0.29	1206.95	0.70
126	11996.00	89.80	0.30		10739.00	1297.94	0.67	1297.94	0.93
127	12087.00	90.50	0.00		10738.76	1388.94	0.90	1388.94	0.84
128	12178.00	90.00	0.60		10738.37	1479.93	1.38	1479.93	0.86
129	12268.00	90.10	0.60		10738.29	1569.93	2.32	1569.93	0.11
130	12359.00	89.60	359.90		10738.53	1660.93	2.72	1660.93	0.95
131	12449.00	90.50	0.10		10738.45	1750.93	2.72	1750.93	1.02
132	12533.00	90.20	1.40		10737.93	1834.92	3.82	1834.92	1.59
133	12616.00	89.40	0.40		10738.22	1917.90	5.12	1917.90	1.54
134	12699.00	89.80	1.10		10738.80	2000.89	6.21	2000.89	0.97
135	12782.00	90.90	0.60		10738.30	2083.88	7.44	2083.88	1.46
136	12865.00	90.90	0.90		10736.99	2166.86	8.53	2166.86	0.36
137	12948.00	91.00	0.70		10735.62	2249.84	9.69	2249.84	0.27
138	13032.00	91.40	0.80		10733.86	2333.82	10.78	2333.82	0.49
139	13115.00	90.40	0.60		10732.55	2416.80	11.80	2416.80	1.23
140	13198.00	90.50	0.30		10731.90	2499.80	12.45	2499.80	0.38
141	13281.00	90.60	1.00		10731.10	2582.79	13.39	2582.79	0.85
142	13364.00	91.40	1.30		10729.66	2665.76	15.06	2665.76	1.03
143	13447.00	91.30	1.00		10727.70	2748.72	16.72	2748.72	0.38
144	13530.00	89.90	359.00		10726.83	2831.71	16.72	2831.71	2.94
145	13614.00	89.50	359.20		10727.27	2915.69	15.40	2915.69	0.53
146	13697.00	89.60	359.10		10727.92	2998.68	14.17	2998.68	0.17
147	13780.00	90.30	0.00		10728.00	3081.68	13.52	3081.68	1.37
148	13863.00	90.90	359.80		10727.13	3164.67	13.38	3164.67	0.76
149	13946.00	91.30	0.40		10725.53	3247.66	13.52	3247.66	0.87
150	14029.00	90.90	359.90		10723.94	3330.64	13.74	3330.64	0.77
151	14113.00	90.40	358.40		10722.99	3414.62	12.49	3414.62	1.88
152	14196.00	90.60	358.40		10722.26	3497.59	10.17	3497.59	0.24
153	14279.00	90.50	357.20		10721.47	3580.52	6.99	3580.52	1.45
154	14362.00	90.90	358.30		10720.45	3663.45	3.73	3663.45	1.41
155	14445.00	90.10	357.40		10719.73	3746.39	0.62	3746.39	1.45
156	14528.00	90.40	357.80		10719.37	3829.31	-2.86	3829.31	0.60
157	14612.00	89.70	356.60		10719.29	3913.21	-6.96	3913.21	1.65
158	14695.00	88.70	356.70		10720.45	3996.06	-11.81	3996.06	1.21
159	14778.00	89.30	357.90		10721.90	4078.95	-15.72	4078.95	1.62

SUNBURST CONSULTING, INC.

Operator:	SM Energy Company		
Well :	Dahl 15-11H		
County:	McKenzie	State:	ND
QQ:	SW SE	Section:	11
Township:	153	N/S:	N
Range:	101	E/W:	W
Footages:	22	FN/SL:	S
	2488	FE/WL:	E

Kick-off:	10/23/2011
Finish:	11/18/2011
Directional Supervision:	Ryan

Date: 11/21/2011

Time: 10:20

F9 to re-calculate

Proposed dir: 360

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		N-S	E-W	SECT	DLS/
			AZM	TVD				100
160	14861.00	91.00	359.60	10721.68	4161.93	-17.53	4161.93	2.90
161	14944.00	90.90	359.50	10720.31	4244.91	-18.18	4244.91	0.17
162	15027.00	91.50	0.00	10718.57	4327.89	-18.55	4327.89	0.94
163	15111.00	91.60	359.40	10716.30	4411.86	-18.99	4411.86	0.72
164	15194.00	89.50	358.50	10715.50	4494.84	-20.51	4494.84	2.75
165	15277.00	89.40	358.40	10716.30	4577.80	-22.75	4577.80	0.17
166	15360.00	89.30	358.30	10717.24	4660.76	-25.14	4660.76	0.17
167	15443.00	90.00	359.30	10717.74	4743.74	-26.88	4743.74	1.47
168	15526.00	90.80	359.30	10717.16	4826.73	-27.89	4826.73	0.96
169	15609.00	89.20	358.60	10717.16	4909.72	-29.41	4909.72	2.10
170	15693.00	89.90	358.40	10717.82	4993.68	-31.61	4993.68	0.87
171	15776.00	88.40	357.40	10719.06	5076.62	-34.65	5076.62	2.17
172	15859.00	87.50	357.70	10722.02	5159.49	-38.20	5159.49	1.14
173	15942.00	88.40	358.00	10724.99	5242.37	-41.31	5242.37	1.14
174	16025.00	88.50	358.60	10727.24	5325.31	-43.77	5325.31	0.73
175	16108.00	89.40	359.00	10728.76	5408.27	-45.51	5408.27	1.19
176	16192.00	89.90	359.00	10729.27	5492.26	-46.98	5492.26	0.60
177	16275.00	89.20	358.60	10729.92	5575.24	-48.71	5575.24	0.97
178	16358.00	88.80	358.70	10731.37	5658.20	-50.67	5658.20	0.50
179	16441.00	88.90	358.40	10733.04	5741.16	-52.77	5741.16	0.38
180	16524.00	89.30	358.50	10734.34	5824.12	-55.01	5824.12	0.50
181	16607.00	90.00	358.60	10734.85	5907.09	-57.11	5907.09	0.85
182	16691.00	91.10	358.80	10734.04	5991.06	-59.02	5991.06	1.33
183	16774.00	89.40	359.10	10733.68	6074.04	-60.54	6074.04	2.08
184	16857.00	89.50	359.00	10734.48	6157.03	-61.92	6157.03	0.17
185	16940.00	89.90	358.80	10734.91	6240.01	-63.51	6240.01	0.54
186	17023.00	89.80	359.50	10735.13	6323.00	-64.74	6323.00	0.85
187	17106.00	89.80	359.00	10735.42	6405.99	-65.83	6405.99	0.60
188	17190.00	90.40	359.60	10735.27	6489.99	-66.85	6489.99	1.01
189	17273.00	90.90	0.00	10734.33	6572.98	-67.14	6572.98	0.77
190	17356.00	90.90	359.50	10733.03	6655.97	-67.51	6655.97	0.60
191	17439.00	92.20	0.00	10730.78	6738.94	-67.87	6738.94	1.68
192	17522.00	90.30	1.20	10728.97	6821.91	-67.00	6821.91	2.71
193	17605.00	91.60	1.60	10727.60	6904.87	-64.97	6904.87	1.64
194	17688.00	88.50	0.50	10727.52	6987.84	-63.45	6987.84	3.96
195	17772.00	88.30	0.60	10729.87	7071.81	-62.65	7071.81	0.27
196	17855.00	89.20	1.40	10731.68	7154.77	-61.20	7154.77	1.45
197	17938.00	89.10	1.90	10732.91	7237.73	-58.81	7237.73	0.61
198	18021.00	88.90	1.00	10734.36	7320.69	-56.71	7320.69	1.11
199	18104.00	89.40	359.60	10735.59	7403.68	-56.27	7403.68	1.79

SUNBURST CONSULTING, INC.

Operator:	SM Energy Company		
Well :	Dahl 15-11H		
County:	McKenzie	State:	ND
QQ:	SW SE	Section:	11
Township:	153	N/S:	N
Range:	101	E/W:	W
Footages:	22	FN/SL:	S
	2488	FE/WL:	E

Kick-off:	10/23/2011
Finish:	11/18/2011
Directional Supervision:	Ryan

Date: 11/21/2011
Time: 10:20

F9 to re-calculate

Proposed dir: **360**

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		TVD	N-S	E-W	SECT	DLS/
			AZM						100
200	18187.00	89.60	0.10		10736.32	7486.67	-56.49	7486.67	0.65
201	18271.00	90.00	359.30		10736.61	7570.67	-56.93	7570.67	1.06
202	18354.00	90.20	359.00		10736.46	7653.66	-58.16	7653.66	0.43
203	18437.00	90.70	359.10		10735.81	7736.65	-59.54	7736.65	0.61
204	18520.00	89.80	358.60		10735.45	7819.63	-61.20	7819.63	1.24
205	18603.00	89.20	359.00		10736.17	7902.61	-62.94	7902.61	0.87
206	18686.00	89.00	358.10		10737.48	7985.57	-65.04	7985.57	1.11
207	18770.00	89.30	358.10		10738.72	8069.51	-67.83	8069.51	0.36
208	18853.00	89.60	357.90		10739.52	8152.46	-70.72	8152.46	0.43
209	18936.00	90.00	357.10		10739.81	8235.38	-74.34	8235.38	1.08
210	19019.00	91.40	358.60		10738.80	8318.31	-77.46	8318.31	2.47
211	19102.00	91.20	359.30		10736.91	8401.27	-78.98	8401.27	0.88
212	19185.00	90.60	359.80		10735.61	8484.26	-79.63	8484.26	0.94
213	19269.00	90.10	0.60		10735.10	8568.26	-79.34	8568.26	1.12
214	19352.00	88.80	0.60		10735.89	8651.25	-78.47	8651.25	1.57
215	19435.00	89.20	0.60		10737.34	8734.23	-77.60	8734.23	0.48
216	19518.00	89.60	0.50		10738.21	8817.22	-76.80	8817.22	0.50
217	19601.00	89.00	358.70		10739.22	8900.21	-77.38	8900.21	2.29
218	19684.00	89.20	358.50		10740.53	8983.17	-79.41	8983.17	0.34
219	19767.00	89.60	359.10		10741.40	9066.15	-81.15	9066.15	0.87
220	19851.00	90.40	359.50		10741.40	9150.14	-82.17	9150.14	1.06
221	19934.00	89.40	359.80		10741.54	9233.14	-82.68	9233.14	1.26
222	20017.00	88.60	359.40		10742.99	9316.12	-83.26	9316.12	1.08
223	20100.00	88.70	359.40		10744.95	9399.10	-84.13	9399.10	0.12
224	20183.00	88.80	359.50		10746.76	9482.07	-84.92	9482.07	0.17
225	20266.00	89.40	359.40		10748.06	9565.06	-85.72	9565.06	0.73
226	20350.00	89.40	0.10		10748.94	9649.05	-86.09	9649.05	0.83
227	20433.00	89.40	359.90		10749.81	9732.05	-86.09	9732.05	0.24
228	20516.00	88.60	359.40		10751.26	9815.03	-86.59	9815.03	1.14
229	20599.00	88.80	0.10		10753.14	9898.01	-86.96	9898.01	0.88
230	20682.00	88.90	359.80		10754.81	9980.99	-87.03	9980.99	0.38
231	20765.00	88.80	0.20		10756.47	10063.98	-87.03	10063.98	0.50
232	20798.00	88.80	0.40		10757.16	10096.97	-86.86	10096.97	0.61
233	20850.00	88.80	0.40		10758.25	10148.96	-86.49	10148.96	0.00

FORMATION TOPS & STRUCTURAL RELATIONSHIPS

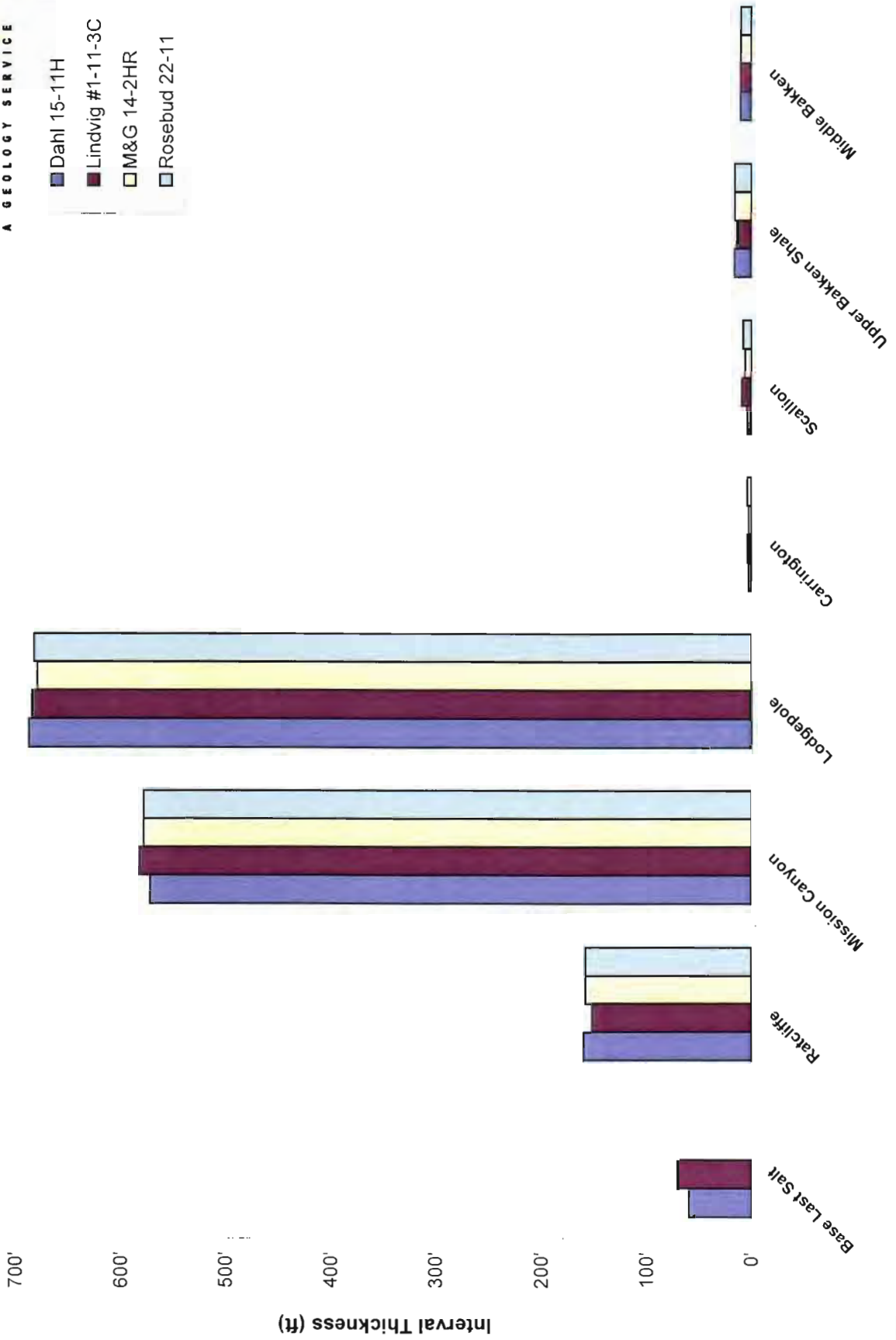
Operator: Well Name: Location: Elevation:	Subject Well:							Offset Wells:				
	SM Energy Dahl 15-11H 22' FSL & 2,488' FEL SW SE Section 11, T153N-R101W GL: 2,110' Sub: 17' KB: 2,127'											
Formation/ Zone	Prog. Top	Prog. Datum (MSL)	Est. Top (ROP)	E-Log Top	MWD GR Top	Datum (MSL)	Interval Thickness	Thickness to Target	Dip To Prognosis	Dip To Lindvig #1 11-3C	Dip To M&G 14- 2HR	Dip To Rosebud 22-11
Base Last Salt	9,212'	-7,085'	9,226'			-7,099'	60'	1,511'	-14'	2'		
Ratcliffe	9,283'	-7,156'	9,286'			-7,159'	160'	1,451'	-3'	13'	-50'	-33'
Mission Canyon	9,433'	-7,306'	9,446'		9,446'	-7,319'	572'	1,291'	-13'	5'	-52'	-35'
Lodgepole	10,017'	-7,890'	10,018'		10,018'	-7,891'	687'	719'	-1'	15'	-46'	-29'
Carrington	10,696'	-8,569'	10,705'		10,705'	-8,578'	3'	32'	-9'	12'	-54'	-34'
Scallion	10,705'	-8,578'	10,708'		10,708'	-8,581'	4'	29'	-3'	13'	-54'	-33'
Upper Bakken Shale	10,714'	-8,587'	10,712'		10,712'	-8,585'	16'	25'	2'	18'	-52'	-29'
Middle Bakken	10,727'	-8,600'	10,728'		10,728'	-8,601'	10'	9'	-1'	16'	-52'	-29'
Middle Bakken (Target)	10,737'	-8,610'	10,738'		10,738'	-8,611'			-1'	16'	-52'	-29'

CONTROL DATA

Operator: Well Name: Location:	SM Energy Company Lindvig 1-11HR SE SE 612 FSL 793 FEL McKenzie, ND 0.38 mi. NE of Dahl 15-11H KB: 2,108'	SM Energy Company M&G 14-2HR SW SW 566 FSL 575 FWL McKenzie, ND 1.18 mi. NW of Dahl 15-11H KB: 1,871'	SM Energy Company Rosebud 22-11 SE NW 1745 FNL 1755 FWL McKenzie County, ND 0.7 mi. NW to Dahl 15-11H KB: 1,872'									
Elevation:												
Formation/ Zone	E-Log Top	Datum (MSL)	Interval Thickness	Thickness to Target	E-Log Top	Datum (MSL)	Interval Thickness	Thickness to Target	E-Log Top	Datum (MSL)	Interval Thickness	Thickness to Target
Base Last Salt	9,209'	-7,101'	71'	1,526'	-	-	-	-	-	-	-	-
Ratcliffe	9,280'	-7,172'	152'	1,455'	8,980'	-7,109'	158'	1,450'	8,998'	-7,126'	158'	1,456'
Mission Canyon	9,432'	-7,324'	582'	1,303'	9,138'	-7,267'	578'	1,292'	9,156'	-7,284'	578'	1,298'
Lodgepole	10,014'	-7,906'	684'	721'	9,716'	-7,845'	679'	714'	9,734'	-7,862'	682'	720'
Carrington	10,698'	-8,590'	4'	37'	10,395'	-8,524'	3'	35'	10,416'	-8,544'	4'	38'
Scallion	10,702'	-8,594'	9'	33'	10,398'	-8,527'	6'	32'	10,420'	-8,548'	8'	34'
Upper Bakken Shale	10,711'	-8,603'	14'	24'	10,404'	-8,533'	16'	26'	10,428'	-8,556'	16'	26'
Middle Bakken	10,725'	-8,617'	10'	10'	10,420'	-8,549'	10'	10'	10,444'	-8,572'	10'	10'
Middle Bakken (Target)	10,735'	-8,627'	-	0'	10,430'	-8,559'	-	0'	10,454'	-8,582'	-	0'

INTERVAL THICKNESS

SM Energy - Dahl 15-11H

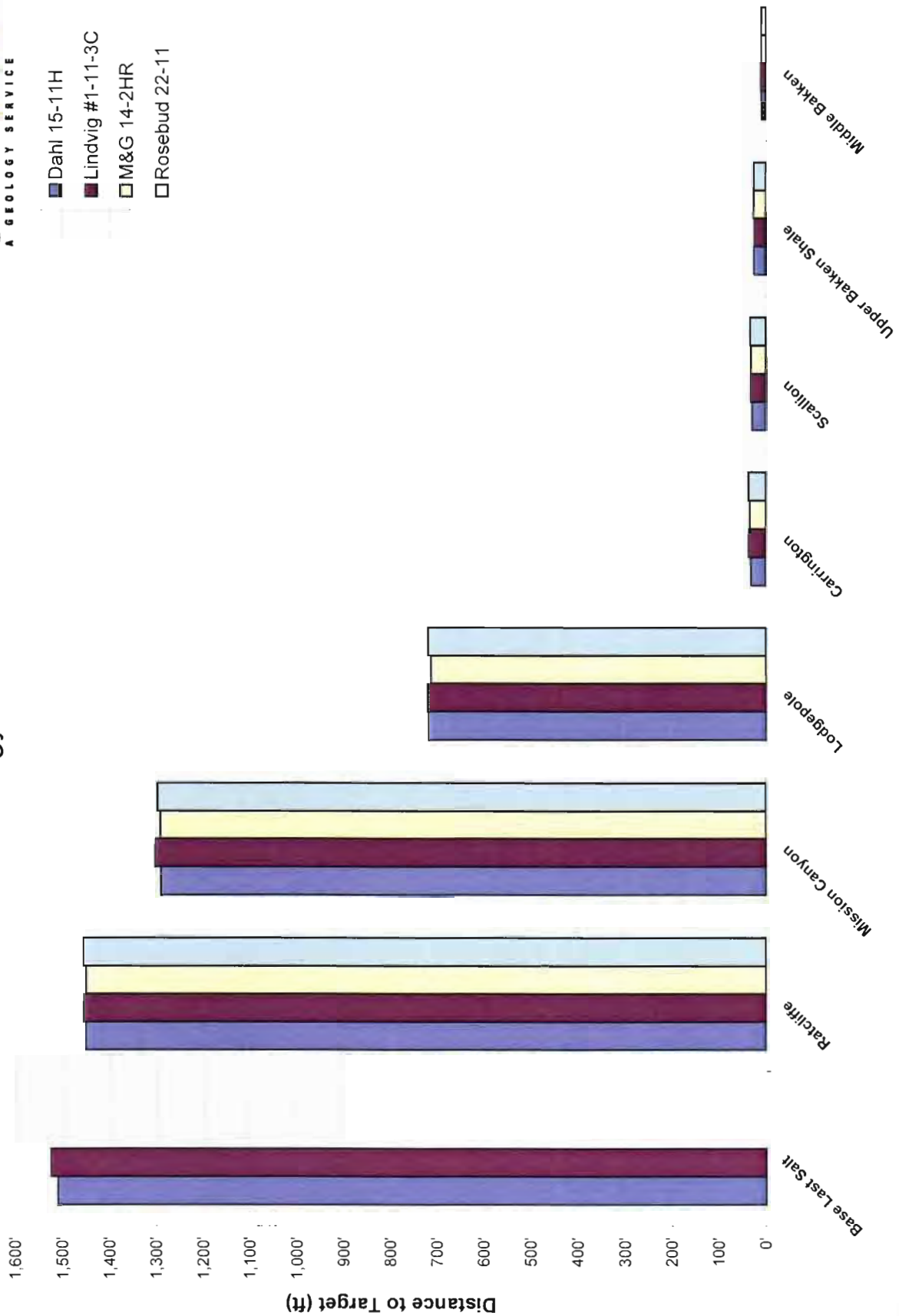


TARGET PROXIMATION

Formation/ Zone:	Proposed Top of Target From:				Average of Offset Wells
	Lindvig #1-11-3C	M&G 14-2HR	Rosebud 22-11		
Base Last Salt	10,752'	-	-		10,752'
Ratcliffe	10,741'	10,736'	10,742'		10,740'
Mission Canyon	10,749'	10,738'	10,744'		10,744'
Lodgepole	10,739'	10,732'	10,738'		10,736'
Carrington	10,742'	10,740'	10,743'		10,742'
Scallion	10,741'	10,740'	10,742'		10,741'
Upper Bakken Shale	10,736'	10,738'	10,738'		10,737'
Middle Bakken	10,738'	10,738'	10,738'		10,738'
Middle Bakken (Target)	10,738'	10,738'	10,738'		10,738'

ISOPACH TO TARGET

SM Energy - Dahl 15-11H



LITHOLOGY

Rig crews caught lagged samples in 30' intervals under the supervision of Sunburst geologists. A detailed list of sampling intervals is included in the well data summary page. Sample or gamma ray marker tops have been inserted in the sample descriptions below for reference. Samples were examined wet and dry under a binocular microscope. The drilling fluid was diesel-based invert from surface casing to 10,982'. The drilling fluid was saltwater from 10,982' to TD sample collection began at 9,120'.

9120-9150 LIMESTONE: wackestone cream to tan, dark gray, microcrystalline, firm to hard, earthy texture,

9150-9180 SALT: transparent to frosted, crystalline, hard, subhedral to occasional euhedral, crystalline texture; trace: LIMESTONE: wackestone cream to tan, dark gray, microcrystalline, firm to hard, earthy texture,

9180-9210 SALT: transparent to frosted, crystalline, hard, subhedral to occasional euhedral, crystalline texture; rare: LIMESTONE: wackestone cream to tan, dark gray, microcrystalline, firm to hard, earthy texture,

9210-9240 LIMESTONE: mudstone, cream to tan, light gray, trace brown to tan, microcrystalline, firm to hard, earthy texture, common ANHYDRITE: white, microcrystalline, soft, amorphous

Base Last Charles Salt

9,226' TVD -7,099' SS

9240-9270 LIMESTONE: mudstone, cream to tan, light gray, trace brown to tan, microcrystalline, firm to hard, earthy texture, common ANHYDRITE: white, microcrystalline, soft, amorphous

Ratcliffe

9,286' TVD -7,159' SS

9270-9300 LIMESTONE: wackestone cream to tan, light gray, trace brown to tan, microcrystalline, firm to hard, earthy texture, trace ANHYDRITE: white, microcrystalline, soft, amorphous

9300-9330 LIMESTONE: wackestone cream to tan, light gray, trace brown to tan, microcrystalline, firm to hard, earthy texture, trace DOLOMITE: mudstone, tan, microcrystalline, firm, blocky, trace ANHYDRITE: white, microcrystalline, soft, amorphous

9330-9360 LIMESTONE: wackestone cream to tan, light gray, trace brown to tan, microcrystalline, firm to hard, earthy texture, common DOLOMITE: mudstone, tan, microcrystalline, firm, blocky, common ANHYDRITE: white, microcrystalline, soft, amorphous

9360-9390 LIMESTONE: mudstone, medium to dark brown, common brown to tan, microcrystalline, firm to hard, earthy texture,

9390-9420 LIMESTONE: mudstone, to wackestone medium to dark brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace spotty brown oil to stain

9420-9450 LIMESTONE: mudstone, medium to light brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace spotty brown oil to stain

Mission Canyon Formation

9,446' TVD -7,319' SS

9420-9450 LIMESTONE: mudstone, medium to light brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace spotty brown oil to stain

9450-9480 LIMESTONE: mudstone, medium to dark brown, common brown to tan, microcrystalline, firm to hard, earthy texture, argillaceous in part, trace spotty brown oil to stain

9480-9510 LIMESTONE: mudstone, to wackestone medium to dark brown, common brown to tan, microcrystalline, firm to hard, earthy texture, argillaceous in part, trace spotty brown oil to stain

9510-9540 LIMESTONE: mudstone, to wackestone medium to dark brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace spotty brown oil to stain

9540-9570 LIMESTONE: mudstone, to wackestone medium to dark brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace spotty brown oil to stain

9570-9600 LIMESTONE: mudstone, to wackestone medium to dark brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace spotty brown oil to stain

9600-9630 LIMESTONE: mudstone, to wackestone medium to dark brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace spotty brown oil to stain

9630-9660 LIMESTONE: mudstone, to wackestone light to medium brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace spotty brown oil to stain, no visible porosity

9660-9690 LIMESTONE: mudstone, to wackestone light to medium brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace spotty brown oil to stain, no visible porosity

9690-9720 LIMESTONE: mudstone, to wackestone medium to dark brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace spotty brown oil to stain, no visible porosity

9720-9750 LIMESTONE: mudstone, to wackestone medium to dark brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace alga streaky, trace spotty brown oil to stain, no visible porosity

9750-9780 LIMESTONE: mudstone, to wackestone medium to dark brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace alga streaky, trace spotty brown oil to stain, no visible porosity

9780-9810 LIMESTONE: mudstone, to wackestone light to medium brown, rare brown to tan, microcrystalline, firm to hard, earthy texture, trace alga streaky, trace spotty brown oil to stain, no visible porosity

9810-9840 LIMESTONE: mudstone, to wackestone cream tan, trace brown to tan, microcrystalline, firm to hard, earthy texture, trace alga streaky, trace spotty brown oil to stain, no visible porosity

9840-9870 LIMESTONE: mudstone, to wackestone medium to dark brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace alga streaky, trace spotty brown oil to stain, no visible porosity

9870-9900 LIMESTONE: mudstone, to wackestone medium to dark brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace spotty brown oil to stain, no visible porosity

9900- 9930 LIMESTONE: mudstone, dark brown, rare brown to tan, microcrystalline, firm to hard, earthy texture, common dark brown dead oil to stain, no visible porosity

9930-9960 LIMESTONE: mudstone, medium to dark brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace spotty brown oil to stain, no visible porosity

9960-9990 LIMESTONE: mudstone, light to medium brown, common brown to tan, microcrystalline, firm to hard, earthy texture, no visible porosity

9990-10020 LIMESTONE: mudstone, medium to dark brown, common brown to tan, microcrystalline, firm to hard, earthy texture, trace dead dark brown oil to stain, no visible porosity

Lodgepole Formation

10,018' TVD -7,891' SS

10020-10050 ARGILLACEOUS LIMESTONE: mudstone, medium to light brown, common tan, microcrystalline, firm, earthy texture, no visible porosity

10050-10080 ARGILLACEOUS LIMESTONE: mudstone, medium to light brown, common tan, microcrystalline, firm, earthy texture, no visible porosity

10080-10110 ARGILLACEOUS LIMESTONE: mudstone, medium to light brown, common tan, microcrystalline, firm, earthy texture, no visible porosity

10110-10140 ARGILLACEOUS LIMESTONE: mudstone, medium to light brown, common tan, microcrystalline, firm, earthy texture, no visible porosity

10140-10170 ARGILLACEOUS LIMESTONE: mudstone, medium to light brown, common tan, microcrystalline, firm, earthy texture, no visible porosity

10170-10200 ARGILLACEOUS LIMESTONE: mudstone, medium to light brown, common tan, microcrystalline, firm, earthy texture, no visible porosity

10200-10230 ARGILLACEOUS LIMESTONE: mudstone, medium to light brown, common tan, microcrystalline, firm, earthy texture, no visible porosity, trace disseminated pyrite

10230-10260 ARGILLACEOUS LIMESTONE: mudstone, medium to light brown, common tan, microcrystalline, firm, earthy texture, no visible porosity, trace disseminated pyrite

10260-10290 No Sample

10190-10220 No Sample

10290-10320 ARGILLACEOUS LIMESTONE: mudstone, medium to light brown, common brown tan, microcrystalline, firm, earthy texture, no visible porosity, trace disseminated pyrite

10320-10350 ARGILLACEOUS LIMESTONE: mudstone, medium to light brown, common brown tan, microcrystalline, firm, earthy texture, no visible porosity, trace disseminated pyrite

10350-10380 ARGILLACEOUS LIMESTONE: mudstone, medium to light brown, common brown tan, microcrystalline, firm, earthy texture, no visible porosity, trace disseminated pyrite

10380-10410 ARGILLACEOUS LIMESTONE: mudstone, medium to light brown, common brown tan, microcrystalline, firm, dense, earthy texture, no visible porosity, trace disseminated pyrite

10410-10440 ARGILLACEOUS LIMESTONE: mudstone, medium to light brown, common brown tan, microcrystalline, firm, dense, earthy texture, no visible porosity, trace disseminated pyrite

10440-10470 ARGILLACEOUS LIMESTONE: mudstone, medium brown, common brown tan, microcrystalline, firm, dense, earthy texture, no visible porosity, trace disseminated pyrite

10470-10500 ARGILLACEOUS LIMESTONE: mudstone, medium to light brown, common brown tan, microcrystalline, firm, dense, earthy texture, no visible porosity, trace disseminated pyrite

10500-10530 ARGILLACEOUS LIMESTONE: mudstone, medium to light brown, common brown tan, microcrystalline, firm to hard, dense, earthy texture, no visible porosity, trace disseminated pyrite

10530-10560 ARGILLACEOUS LIMESTONE: mudstone, medium brown, common brown tan, microcrystalline, firm to hard, dense, earthy texture, no visible porosity, trace disseminated pyrite

10560-10590 ARGILLACEOUS LIMESTONE: mudstone, medium brown, common brown tan, microcrystalline, firm to hard, dense, earthy texture, no visible porosity, trace disseminated pyrite

10590-10620 ARGILLACEOUS LIMESTONE: mudstone, medium brown, common brown tan, microcrystalline, firm to hard, dense, earthy texture, no visible porosity, trace disseminated pyrite

10620-10650 ARGILLACEOUS LIMESTONE: mudstone, medium brown, common brown tan, microcrystalline, firm to hard, dense, earthy texture, no visible porosity, trace disseminated pyrite

10650-10680 ARGILLACEOUS LIMESTONE: mudstone, medium brown, common brown tan, microcrystalline, firm to hard, dense, earthy texture, no visible porosity, trace disseminated pyrite

10680-10710 ARGILLACEOUS LIMESTONE: mudstone, medium brown, common brown tan, microcrystalline, firm to hard, dense, earthy texture, no visible porosity, trace disseminated pyrite

10710-10740 ARGILLACEOUS LIMESTONE: mudstone, medium brown, common brown tan, microcrystalline, firm to hard, dense, earthy texture, no visible porosity, common disseminated pyrite

10740-10770 ARGILLACEOUS LIMESTONE: mudstone, medium brown, common brown tan, microcrystalline, firm to hard, dense, earthy texture, no visible porosity, common disseminated pyrite

10770-10800 ARGILLACEOUS LIMESTONE: mudstone, medium brown, common brown tan, microcrystalline, firm to hard, dense, earthy texture, no visible porosity, common disseminated pyrite

10800-10830 ARGILLACEOUS LIMESTONE: mudstone, medium brown, common brown tan, microcrystalline, firm to hard, dense, earthy texture, no visible porosity, common disseminated pyrite

Carrington

10,705' TVD -8,578'SS

Scallion

10,708' TVD -8,581'SS

10830-10860 ARGILLACEOUS LIMESTONE: mudstone, medium brown, common brown tan, microcrystalline, firm to hard, dense, earthy texture, no visible porosity, common disseminated pyrite;
SHALES: black, hard to firm, blocky, earthy texture, petroliferous, rare disseminated pyrite

Bakken Formation, Upper Member

10,712' TVD -8,585'SS

10860-10890 SHALES: black, hard to firm, blocky, earthy texture, petroliferous, rare disseminated pyrite

Bakken Formation, Middle Member

10,728' TVD -8,601'SS

10890-10920 SHALES: black, hard to firm, blocky, earthy texture, petroliferous, rare disseminated pyrite;
abundant SILTSTONE: medium gray, friable, splintery, calcareous cement, poorly cemented, possible intergranular porosity; trace LIMESTONE: mudstone, light tan to gray, microcrystalline, firm to hard, dense, earthy texture, no visible porosity

10920-10950 SILTSTONE: medium gray, common brown to tan, trace white to off white, friable, splintery, calcareous cement, poorly cemented, possible intergranular porosity, trace white calcareous, abundant disseminated and nodules pyrite

10950-10980 SILTSTONE: medium gray, common brown to tan, trace white to off white, friable, splintery, calcareous cement, poorly cemented, possible intergranular porosity, trace white calcareous, abundant disseminated and nodules pyrite

10980-11010 heavy contaminated SILTSTONE: medium to light gray brown, blocky, earthy, occasional
SHALES: black, firm, platy, earthy, petroliferous, common disseminated pyrite, carbonaceous

11010-11040 heavy contaminated SILTSTONE: medium to light gray brown, blocky, earthy, occasional
SHALES: black, firm, platy, earthy, petroliferous, common disseminated pyrite, carbonaceous

11460-11490 SILTY SANDSTONE: light gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, common disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

11490-11520 SILTY SANDSTONE: medium to light gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, common disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

11520-11550 SILTY SANDSTONE: medium to light gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, common disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

11550-11580 SILTY SANDSTONE: medium to light gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, common disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

11580-11610 SILTY SANDSTONE: medium to light gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, common disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

11610-11640 SILTY SANDSTONE: medium to light gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, common disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

11640-11670 SILTY SANDSTONE: light gray, abundant white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

11670-11700 SILTY SANDSTONE: light gray, abundant white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

11700-11730 no sample

11730-11760 SILTY SANDSTONE: light gray, abundant white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

11760-11790 SILTY SANDSTONE: light gray, abundant white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

11790-11820 SILTY SANDSTONE: light gray, abundant white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

11820-11850 SILTY SANDSTONE: light gray, abundant white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

11850-11880 SILTY SANDSTONE: light gray, abundant white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain, no cut fluorescence

11880-11910 SILTY SANDSTONE: light gray, abundant white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain,

11910-11940 SILTY SANDSTONE: light gray, abundant white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

11940-11970 SILTY SANDSTONE: light brown, common gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain, no cut fluorescence

11970-12000 SILTY SANDSTONE: light brown, common gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

12000-12030 SILTY SANDSTONE: light brown, common gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

12030-12060 SILTY SANDSTONE: light brown, common gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain, moderately yellow to green blooming cut fluorescence

12060-12090 SILTY SANDSTONE: light brown, common gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

12090-12120 SILTY SANDSTONE: light brown, common gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

12120-12150 SILTY SANDSTONE: light brown, common gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain, moderately yellow to green streaming cut fluorescence

12150-12180 SILTY SANDSTONE: light brown, common gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

12180-12210 SILTY SANDSTONE: light brown, common gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

12210-12240 SILTY SANDSTONE: light brown, medium to light gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

12240-12270 SILTY SANDSTONE: medium to light gray, abundant light brown, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

12270-12300 SILTY SANDSTONE: light brown, medium to light gray, common white to off white, trace light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13170-13200 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13200-13230 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13230-13260 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13260-13290 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain, slow streaming yellow to green cut fluorescence

13290-13320 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13320-13350 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13350-13380 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain, no cut fluorescence

13380-13410 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13410-13440 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13440-13470 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain, fast streaming to diff yellow to green cut fluorescence

13470-13500 SILTY SANDSTONE: medium to light gray, common light brown, occasional white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13500-13530 SILTY SANDSTONE: medium to light gray, common light brown, occasional white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13530-13560 SILTY SANDSTONE: medium to light gray, common light brown, occasional white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain, slow streaming to diff yellow to green cut fluorescence

13560-13590 SILTY SANDSTONE: medium to light gray, common light brown, occasional white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13590-13620 SILTY SANDSTONE: medium to light gray, common light brown, occasional white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13620-13650 SILTY SANDSTONE: medium to light gray, common light brown, occasional white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13650-13680 SILTY SANDSTONE: medium to light gray, common light brown, occasional white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13680-13710 SILTY SANDSTONE: medium to light gray, common light brown, occasional white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13710-13740 SILTY SANDSTONE: medium to light gray, common light brown, occasional white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13740-13770 SILTY SANDSTONE: medium to light gray, common light brown, occasional white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13770-13800 SILTY SANDSTONE: medium to light gray, common light brown, occasional white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13800-13830 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13830-13860 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain, fast streaming to diff yellow to green cut fluorescence

13860-13890 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain, fast streaming to diff yellow to green cut fluorescence

13890-13920 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13920-13950 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13950-13980 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

13980-14010 SILTY SANDSTONE: white to off white, medium to light gray, common light brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

15660-15690 SILTY SANDSTONE: medium to light brown, common dark gray, common light brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

15690-15720 SILTY SANDSTONE: medium to light brown, common dark gray, common light brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

15720-15750 SILTY SANDSTONE: medium to light brown, common dark gray, common light brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

15750-15780 SILTY SANDSTONE: medium to light brown, common dark gray, common light brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, rare light brown spotty oil stain

15780-15810 SILTY SANDSTONE: medium gray, common brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, occasional light brown spotty oil stain

15810-15840 SILTY SANDSTONE: medium gray, common brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, occasional light brown spotty oil stain

15840-15870 SILTY SANDSTONE: medium gray, common brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, occasional light brown spotty oil stain

15870-15900 SILTY SANDSTONE: medium gray, common brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, occasional light brown spotty oil stain

15900-15930 SILTY SANDSTONE: medium gray, common brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, occasional light brown spotty oil stain

15930-15960 SILTY SANDSTONE: medium gray, common brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, occasional light brown spotty oil stain

15960-15990 SILTY SANDSTONE: medium gray, common brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, occasional light brown spotty oil stain

15990-16020 SILTY SANDSTONE: medium gray, common brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, occasional light brown spotty oil stain

16020-16050 SILTY SANDSTONE: medium gray, common brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, occasional light brown spotty oil stain

16890-16920 SILTY SANDSTONE: medium gray, common brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

16920-16950 SILTY SANDSTONE: medium gray, common brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

16950-16980 SILTY SANDSTONE: medium gray, common brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

16980-17010 SILTY SANDSTONE: medium gray, common brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17010-17040 No Sample

17040-17070 No Sample

17070-17100 SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17100-17130 SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17130-17160 SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17160-17190 SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17190-17220 SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17220-17250 slightly contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17250-17280 slightly contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17280-17310 slightly contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17310-17340 slightly contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17340-17370 slightly contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17370-17400 slightly contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17400-17430 slightly contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17430-17460 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17460-17490 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17490-17520 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17520-17550 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17550-17580 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17580-17610 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17610-17640 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17640-17670 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17670-17700 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17700-17730 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17730-17760 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17760-17790 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17790-17820 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17820-17850 slightly contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17850-17880 slightly contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17880-17910 slightly contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17910-17940 slightly contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17940-17970 slightly contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

17970-18000 slightly contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18000-18030 slightly contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18030-18060 SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18060-18090 SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18090-18120 SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18120-18150 SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18150-18180 SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18180-18210 SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18210-18240 SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18240-18270 SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18270-18300 SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18300-18330 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, are disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18330-18360 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18360-18390 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18390-18420 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18420-18450 No Sample

18450-18480 No Sample

18480-18510 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18510-18540 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18540-18570 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18570-18600 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18600-18630 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18630-18660 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18660-18690 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18690-18720 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18720-18750 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18750-18780 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, occasional brown, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18780-18810 SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

18810-18840 SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19200-19230 SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19230-19260 SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19260-19290 SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19290-19320 SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19320-19350 SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19350-19380 SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19380-19410 SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19410-19440 heavy contaminated with lubricant SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19440-19470 heavy contaminated with lubricant SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19470-19500 heavy contaminated with lubricant SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19500-19530 heavy contaminated with lubricant SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19530-19560 heavy contaminated with lubricant SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19560-19590 heavy contaminated with lubricant SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19590-19620 heavy contaminated with lubricant SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19620-19650 moderately contaminated with lubricant SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19650-19680 moderately contaminated with lubricant SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19680-19710 moderately contaminated with lubricant SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19710-19740 moderately contaminated with lubricant SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19740-19770 moderately contaminated with lubricant SILTY SANDSTONE: light to medium gray, occasional dark gray, occasional brown, trace white to off white, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19770-19800 moderately contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19800-19830 moderately contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19830-19860 moderately contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19860-19890 moderately contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19890-19920 moderately contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19920-19950 moderately contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19950-19980 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

19980-20010 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20010-20040 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20040-20070 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20070-20100 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20100-20130 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20130-20160 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20160-20190 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20190-20220 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20220-20250 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20250-20280 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20280-20310 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20310-20340 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20340-20370 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20370-20400 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20400-20430 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20430-20460 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20460-20490 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20490-20520 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20520-20550 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20550-20580 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20580-20610 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20610-20640 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20640-20670 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20670-20700 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20700-20730 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20730-20760 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20760-20790 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20790-20820 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain

20820-20850 heavy contaminated with lubricant SILTY SANDSTONE: white to off white, light to medium gray, trace brown, fine grain, friable to firm, sub-angular to angular, moderately sorted, calcareous cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 6

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
21266

TH

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date	☐ Drilling Prognosis	☐ Spill Report
☒ Report of Work Done	Date Work Completed November 4, 2011	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☒ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☒ Other	Drilling Progress

Well Name and Number Dahl 15-11H			
Footages	Qtr-Qtr	Section	Township
22 F S L 2488 F E L SWSE	11	153 N	101 W
Field	Pool	County	
Baker	Bakken	McKenzie	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Nabors Drilling USA LP			
Address	City	State	Zip Code
P. O. Box 2396	Williston	ND	58802

DETAILS OF WORK

Continued drilling the 8 3/4" vertical hole from 7,453' - 10,286' KOP. Drilled the 8 3/4" curve from 10,286' - 10,982', then ran 7" 29-32# L-80 intermediate casing to 10,952' and cemented it with 922 sks of cement. Drilled the 6" lateral from 10,982' - 11,044' and had to shut down for rig repairs. Drilling operations are expected to resume next week as soon as repairs are completed.

Company SM Energy Company		Telephone Number (406) 245-6248	
Address P. O. Box 7168			
City Billings	State MT	Zip Code 59103-7168	
Signature <i>Joy Torgerson</i>	Printed Name Joy Torgerson		
Title Engineering Tech	Date November 4, 2011		
Email Address jtorgerson@sm-energy.com			

FOR STATE USE ONLY	
☒ Received	☐ Approved
Date <i>November 10, 2011</i>	
By <i>Dan G. Melcher</i>	
Title PETROLEUM ENGINEER	

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4
SFN 5749

1. Approval shall be obtained prior to perforating or recompleting a well in a reservoir other than the reservoir in which the well is currently completed, prior to plug back of a well, prior to temporary abandonment of a well, prior to abandonment of a well, prior to reclamation of a well site, prior to reclamation of a reserve pit, and prior to beginning a workover project, which may qualify for a tax exemption pursuant to NDCC Section 57-51.1-03. Please refer to Section 43-02-03-16 of the North Dakota Administrative Code (NDAC) regarding recompleting a well in a reservoir other than the reservoir in which the well is currently completed or plugging back of a well, to Section 43-02-03-55 NDAC regarding temporary abandonment of a well, to Section 43-02-03-33 or Section 43-02-05-08 NDAC regarding abandonment of wells, to Section 43-02-03-19 NDAC regarding reclamation, and to Section 43-02-09-03 NDAC regarding workover projects.

2. Upon the completion of any remedial work, or attempted remedial work such as plugging back, drilling deeper, acidizing, shooting, formation fracturing, squeezing operations, setting liner, fishing operations, repair work, perforating, reperforating, or other similar operations not specifically covered herein, a report on the operation shall be filed on a Sundry Notice - Form 4 (SFN 5749) with the Director. The report shall present a detailed account of all work done and the date of such work; the daily production of oil, gas, and water both prior to and after the operation; the shots per foot, size, and depth of perforations; the quantity of sand, crude, chemical, or other materials employed in the operation; and any other pertinent information or operations which affect the original status of the well and are not specifically covered herein. Please refer to Section 43-02-03-31 NDAC.

3. Upon the completion of a workover project, which may qualify for a tax exemption pursuant to NDCC Section 57-51.1-03, a report on the operation shall be filed on a Sundry Notice - Form 4 (SFN 5749) detailing the work done. Include the dates during which the workover rig was in service actually performing work on the workover project and the date the workover was completed, a detailed list identifying all labor, services, and materials used and equipment replaced during the workover project, the cost of each item, and whether the replacement equipment was new or used. The value of all equipment removed from service must be listed. The average daily oil production from the well during the first two months after completion of the project must be included if the costs of the project did not exceed sixty-five thousand dollars. All gauge tickets of oil produced in incomplete months during the first two months after completion of the workover and the volume of oil stored on the well premises immediately prior to commencement of the workover project must also be included. Please refer to Section 43-02-09-04 NDAC.

4. Upon the initial installation of pumping equipment, or change in type or depth of pumping equipment designed to increase productivity in a well, the operator shall file a Sundry Notice - Form 4 (SFN 5749) of such installation. The notice shall include all pertinent information on the pump and the operation thereof including the date of such installation, and the daily production of the well prior to and after the pump has been installed. Please refer to Section 43-02-03-31 NDAC.

5. The well file number, well name and number, well location, field, pool, and county shall coincide with the official records on file with the Commission.

6. The original and one copy of this report shall be filed with the Industrial Commission of North Dakota, Oil and Gas Division, 600 East Boulevard, Dept. 405, Bismarck, ND 58505-0840.

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.

21266

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date	☐ Drilling Prognosis	☐ Spill Report
☒ Report of Work Done	Date Work Completed October 19, 2011	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☒ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☒ Other	Notice of Spud & Drilling Progress

Well Name and Number Dahl 15-11H					
Footages	Qtr-Qtr	Section	Township	Range	
22 F S L 2488 F E L	SWSE	11	153 N	101 W	
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Nabors Drilling USA LP			
Address P. O. Box 2396	City Williston	State ND	Zip Code 58802

DETAILS OF WORK

The Dahl 15-11H was spud at 12:30 AM CDT on October 15, 2011. Drilled a 13 1/2" surface hole to 2,113', ran 9 5/8" 36# J-55 surface casing to 2,113' and cemented it with 736 sks of cement with 100 Bbls cement returns to the surface. Drilled the 8 3/4" vertical hole from 2,113', reaching 7,453' as of 6:00 AM 10/19/11.

Company SM Energy Company		Telephone Number (406) 245-6248	
Address P. O. Box 7168			
City Billings		State MT	Zip Code 59103-7168
Signature <i>Joy Torgerson</i>	Printed Name Joy Torgerson		
Title Engineering Tech	Date October 19, 2011		
Email Address jtorgerson@sm-energy.com			

FOR STATE USE ONLY

☒ Received	☐ Approved
Date <i>October 21, 2011</i>	
By <i>Dei J. McInnes</i>	
Title PETROLEUM ENGINEER	

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4
SFN 5749

1. Approval shall be obtained prior to perforating or recompleting a well in a reservoir other than the reservoir in which the well is currently completed, prior to plug back of a well, prior to temporary abandonment of a well, prior to abandonment of a well, prior to reclamation of a well site, prior to reclamation of a reserve pit, and prior to beginning a workover project, which may qualify for a tax exemption pursuant to NDCC Section 57-51.1-03. Please refer to Section 43-02-03-16 of the North Dakota Administrative Code (NDAC) regarding recompleting a well in a reservoir other than the reservoir in which the well is currently completed or plugging back of a well, to Section 43-02-03-55 NDAC regarding temporary abandonment of a well, to Section 43-02-03-33 or Section 43-02-05-08 NDAC regarding abandonment of wells, to Section 43-02-03-19 NDAC regarding reclamation, and to Section 43-02-09-03 NDAC regarding workover projects.

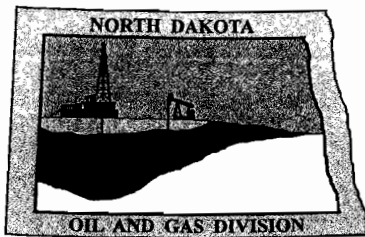
2. Upon the completion of any remedial work, or attempted remedial work such as plugging back, drilling deeper, acidizing, shooting, formation fracturing, squeezing operations, setting liner, fishing operations, repair work, perforating, reperforating, or other similar operations not specifically covered herein, a report on the operation shall be filed on a Sundry Notice - Form 4 (SFN 5749) with the Director. The report shall present a detailed account of all work done and the date of such work; the daily production of oil, gas, and water both prior to and after the operation; the shots per foot, size, and depth of perforations; the quantity of sand, crude, chemical, or other materials employed in the operation; and any other pertinent information or operations which affect the original status of the well and are not specifically covered herein. Please refer to Section 43-02-03-31 NDAC.

3. Upon the completion of a workover project, which may qualify for a tax exemption pursuant to NDCC Section 57-51.1-03, a report on the operation shall be filed on a Sundry Notice - Form 4 (SFN 5749) detailing the work done. Include the dates during which the workover rig was in service actually performing work on the workover project and the date the workover was completed, a detailed list identifying all labor, services, and materials used and equipment replaced during the workover project, the cost of each item, and whether the replacement equipment was new or used. The value of all equipment removed from service must be listed. The average daily oil production from the well during the first two months after completion of the project must be included if the costs of the project did not exceed sixty-five thousand dollars. All gauge tickets of oil produced in incomplete months during the first two months after completion of the workover and the volume of oil stored on the well premises immediately prior to commencement of the workover project must also be included. Please refer to Section 43-02-09-04 NDAC.

4. Upon the initial installation of pumping equipment, or change in type or depth of pumping equipment designed to increase productivity in a well, the operator shall file a Sundry Notice - Form 4 (SFN 5749) of such installation. The notice shall include all pertinent information on the pump and the operation thereof including the date of such installation, and the daily production of the well prior to and after the pump has been installed. Please refer to Section 43-02-03-31 NDAC.

5. The well file number, well name and number, well location, field, pool, and county shall coincide with the official records on file with the Commission.

6. The original and one copy of this report shall be filed with the Industrial Commission of North Dakota, Oil and Gas Division, 600 East Boulevard, Dept. 405, Bismarck, ND 58505-0840.



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

RANDY CARLSON
SM ENERGY COMPANY
PO BOX 7168
BILLINGS, MT 59103-7168 USA

Date: 8/8/2011

RE: **CORES AND SAMPLES**

Well Name: **DAHL 15-11H** Well File No.: **21266**
Location: **SWSE 11-153-101** County: **MCKENZIE**
Permit Type: **Development - HORIZONTAL**
Field: **BAKER** Target Horizon: **BAKKEN**

Dear RANDY CARLSON:

North Dakota Century Code (NDCC) Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for the NDCC Section 38-08-04 and North Dakota Administrative Code 43-02-03-38.1.
- 2) Samples shall include all cuttings from:

Base of the Last Charles Salt

Samples of cuttings shall be taken at 30' maximum intervals through all vertical, build and horizontal sections. Samples must be washed, dried, packed in sample envelopes in correct order with labels showing operator, well name, location and depth, and forwarded in standard boxes to the State Geologist within 30 days of the completion of drilling operations.

- 3) Cores: ALL CORES cut shall be preserved in correct order, properly boxed, and forwarded to the State Geologist within 90 days of completion of drilling operations. Any extension of time must have written approval from the State Geologist.
- 4) All cores, core chips, and samples must be shipped, prepaid, to the State Geologist at the following address:
**ND Geological Survey Core Library
Campus Road and Cornell
Grand Forks, ND 58202**
- 5) NDCC Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

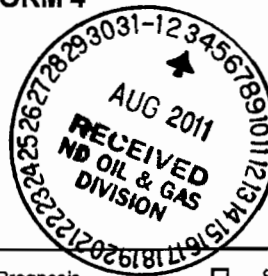
Sincerely

Richard A. Suggs
Geologist



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (03-2004)



Well File No. **21266**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date August 12, 2011
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	
Approximate Start Date	

☒ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Request for waiver of open hole logs

Well Name and Number DAHL 15-11H					
Footages	Qtr-Qtr	Section	Township	Range	
22 F S L 2488 F E L	SWSE	11	153 N	101 W	
Field BAKER	Pool BAKKEN	County MCKENZIE			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

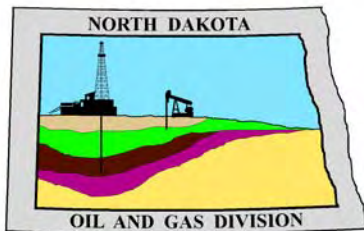
DETAILS OF WORK

SM ENERGY COMPANY PLANS TO DRILL THE DAHL 15-11H AS A HORIZONTAL BAKKEN WELL ON A 1280-ACRE DRILLING UNIT. SM ENERGY REQUESTS A WAIVER FROM RUNNING OPEN HOLE LOGS ON THE WELL DUE TO THE EXISTENCE OF OPEN HOLE LOGS RUN ON THE LINDVIG 1-11HR (NO. 9309) LOCATED 612' FSL, 793' FEL IN SESE SEC. 11, T153N-R101W.

**SM Energy must run a CBL on the production casing and a Gamma Ray log to ground level.*

Company SM ENERGY COMPANY		Telephone Number 406-869-8716	
Address P.O. BOX 7168			
City BILLINGS	State MT	Zip Code 59103	
Signature <i>RCC</i>	Printed Name RANDY CARLSON		
Title DRILLING MANAGER	Date August 2, 2011		

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 8-3-2011	
By <i>*RCA</i>	
Title Richard A. Suggs	
Geologist	



Oil and Gas Division

Lynn D. Helms - Director Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

August 3, 2011

RANDY CARLSON
DRILLING MANAGER
SM ENERGY COMPANY
P.O. Box 7168
Billings, MT 59103-7168

**RE: HORIZONTAL WELL
DAHL 15-11H
SWSE Section 11-153N-101W
McKenzie County
Well File # 21266**

Dear RANDY:

Pursuant to Commission Order No. 14938, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be no closer than the **200' setback** from the north & south boundaries and **1220' setback** (per Commission policy) from the east & west boundaries within the 1280 acre drilling unit consisting of Sections 2 & 11-T153N-R101W.

PERMIT STIPULATIONS: PRECAUTIONS MUST BE TAKEN WHILE DESIGNING THE FRAC JOB FOR THE DAHL 15-11H WELL AS TO NOT ADVERSELY AFFECT THE SM ENERGY LINDVIG 1-11HR WELL. Due to river adjacent to the well site, a dike is required surrounding the entire location. Also, a Closed Mud System may be required, although the disposal of drill cuttings is contingent upon site specific conditions to be determined by an NDIC Field Inspector. SM ENERGY CO must contact NDIC Field Inspector Kevin Connors at 701-220-5989 prior to location construction.

Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Permit Fee & Notification

Payment was received in the amount of \$100 via credit card. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

RANDY CARLSON

August 3, 2011

Page 2

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov. Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Confidential status

Your request for confidential status of all information furnished to the Director, or his representatives, is hereby granted. Such information, except production runs, shall remain confidential for six months commencing on the date the well is spud.

Confidential status notwithstanding, the Director and his representatives shall have access to all well records wherever located. Your company personnel, or any person performing work for your company shall permit the Director and his representatives to come upon any lease, property, well, or drilling rig operated or controlled by them, complying with all safety rules, and to inspect the records and operation of such wells and to have access at all times to any and all records of wells. The Commission's field personnel periodically inspect producing and drilling wells. Any information regarding such wells shall be made available to them at any time upon request. The information so obtained by the field personnel shall be maintained in strict confidence and shall be available only to the Commission and its staff.

Drilling pit

Please be advised approval for use of a drilling reserve pit or drill cuttings pit on this site is contingent on site specific conditions and special pit closure requirements may apply.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of a Cement Bond Log from which the presence of cement can be determined in every well in which production or intermediate casing has been set and a Gamma Ray Log must be run from total depth to ground level elevation of the well bore. All logs must be submitted as one paper copy and one digital copy in LAS (Log ASCII) format, or a format approved by the Director. Image logs that include, but are not limited to, Mud Logs, Cement Bond Logs, and Cyberlook Logs, cannot be produced in their entirety as LAS (Log ASCII) files. To create a solution and establish a standard format for industry to follow when submitting image logs, the Director has given approval for the operator to submit an image log as a TIFF (*.tif) formatted file. The TIFF (*.tif) format will be accepted only when the log cannot be produced in its entirety as a LAS (Log ASCII) file format. The digital copy may be submitted on a standard CD, or attached to an email sent to digitallogs@nd.gov. Thank you for your cooperation.

Sincerely,

David Tabor
Engineering Technician IV



APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work New Location	Type of Well Oil & Gas	Approximate Date Work Will Start 8 / 8 / 2011	Confidential Status Yes
Operator SM ENERGY COMPANY			Telephone Number {406} 245-6248
Address P.O. Box 7168		City Billings	State MT Zip Code 59103-7168

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name DAHL				Well Number 15-11H			
Surface Footages 22 F S L 2488 F E L		Qtr-Qtr SWSE	Section 11	Township 153 N	Range 101 W	County McKenzie	
Longstring Casing Point Footages 451 F S L 2489 F E L		Qtr-Qtr SWSE	Section 11	Township 153 N	Range 101 W	County McKenzie	
Longstring Casing Point Coordinates From Well Head 429 N From WH 1 W From WH		Azimuth 359.8 °	Longstring Total Depth 10991 Feet MD 10737 Feet TVD				
Bottom Hole Footages From Nearest Section Line 223 F N L 2518 F E L		Qtr-Qtr NWNE	Section 2	Township 153 N	Range 101 W	County McKenzie	
Bottom Hole Coordinates From Well Head 10164 N From WH 30 W From WH		KOP Lateral 1 10328 Feet MD	Azimuth Lateral 1 359.8 °	Estimated Total Depth Lateral 1 20725 Feet MD 10737 Feet TVD			
Latitude of Well Head 48 ° 04 ' 56.38 "		Longitude of Well Head -103 ° 38 ' 09.24 "		NAD Reference NAD83	Description of (Subject to NDIC Approval) Drilling Unit: Sections 2 & 11-T153N-R101W		
Ground Elevation 2112 Feet Above S.L.	Acres in Spacing/Drilling Unit 1280	Spacing/Drilling Unit Setback Requirement 200 Feet N/S 1220 Feet E/W			Industrial Commission Order 14938		
North Line of Spacing/Drilling Unit 5281 Feet		South Line of Spacing/Drilling Unit 5279 Feet		East Line of Spacing/Drilling Unit 10417 Feet		West Line of Spacing/Drilling Unit 10402 Feet	
Objective Horizons BAKKEN						Pierre Shale Top 1977	
Proposed Surface Casing	Size 9 - 5/8 "	Weight 36 Lb./Ft.	Depth 2100 Feet	Cement Volume 715 Sacks	NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.		
Proposed Longstring Casing	Size 7 - 0 "	Weight(s) 29-32 Lb./Ft.	Longstring Total Depth 10991 Feet MD 10737 Feet TVD		Cement Volume 770 Sacks	Cement Top 4900 Feet	Top Dakota Sand 5473 Feet
Base Last Charles Salt (If Applicable) 9212 Feet		NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.					
Proposed Logs MUD LOG {CBL & GR ON COMPLETION} SEE ATTACHMENT							
Drilling Mud Type (Vertical Hole - Below Surface Casing) Invert				Drilling Mud Type (Lateral) Brine			
Survey Type in Vertical Portion of Well MWD Every 100 Feet		Survey Frequency: Build Section 30 Feet		Survey Frequency: Lateral 90 Feet		Survey Contractor RYAN ENERGY	

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.
See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

GEOLOGIC TOPS, PLATS, DIRECTIONAL, DRILLING PLAN, LOG WAIVER

Lateral 2

KOP Lateral 2 Feet MD	Azimuth Lateral 2 °	Estimated Total Depth Lateral 2 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 3

KOP Lateral 3 Feet MD	Azimuth Lateral 3 °	Estimated Total Depth Lateral 3 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 4

KOP Lateral 4 Feet MD	Azimuth Lateral 4 °	Estimated Total Depth Lateral 4 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 5

KOP Lateral 5 Feet MD	Azimuth Lateral 5 °	Estimated Total Depth Lateral 5 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.		Date 08 / 02 / 2011
ePermit	Printed Name RANDY CARLSON	Title DRILLING MANAGER

FOR STATE USE ONLY

Permit and File Number 21266	API Number 33 - 053 - 03703
Field BAKER	
Pool BAKKEN	Permit Type DEVELOPMENT

FOR STATE USE ONLY

Date Approved 8 / 3 / 2011
By David Tabor
Title Engineering Technician IV

WELL LOCATION PLAT

SM Energy Company
P.O. Box 7168 Billings, MT 59103

Dahl 15-11H

22 feet from the south line and 2488 feet from the east line (surface location)

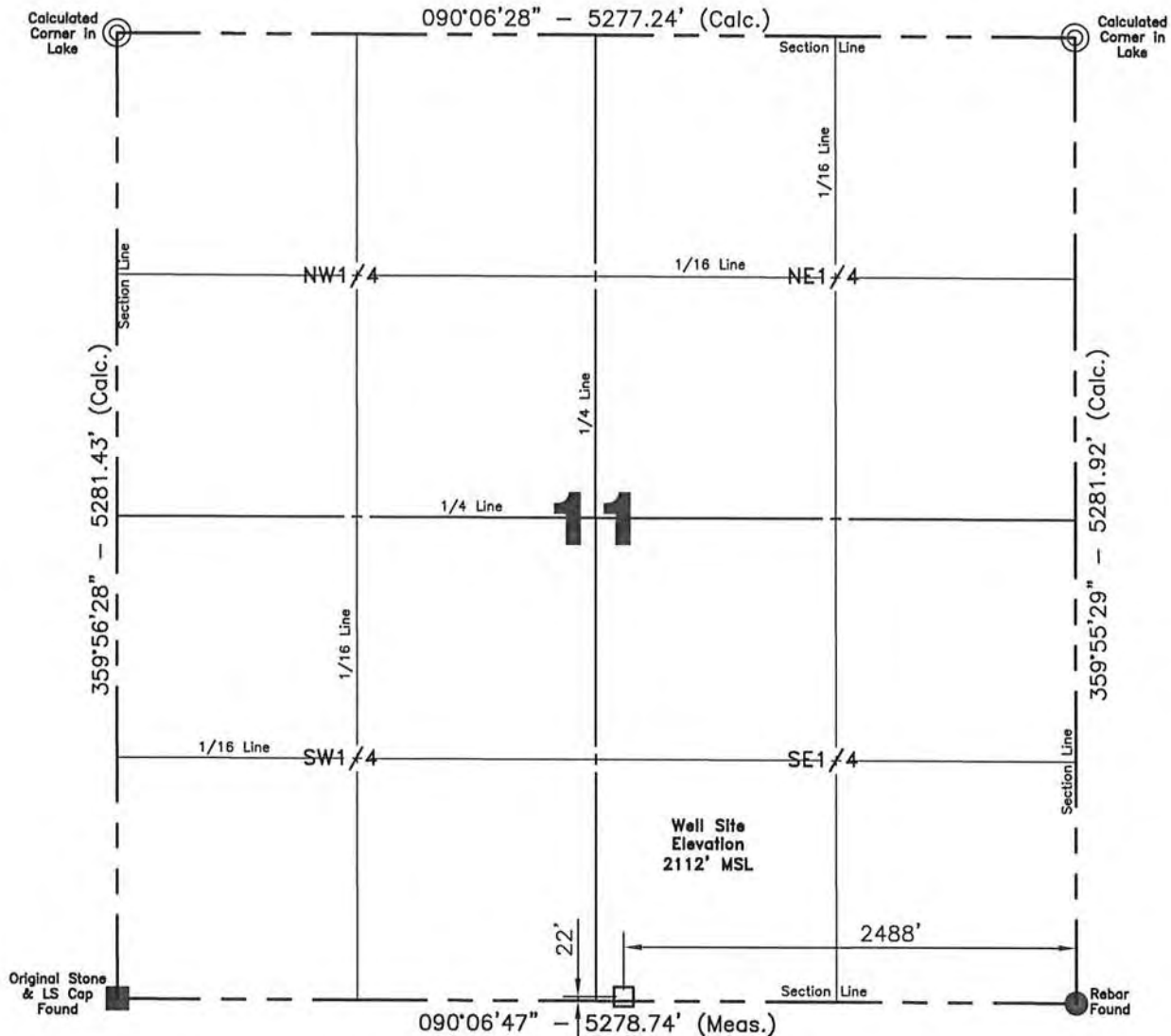
Section 11, T. 153 N., R. 101 W., 5th P.M.

McKenzie County, North Dakota

Surface owner @ well site - Sharon D. Eilers et al

Latitude 48°04'56.376" North; Longitude 103°38'09.239" West (surface location)

[Derived from OPUS Solution NAD-83(CORS96)]



Confidentiality Notice: The information contained on this plat is legally privileged and confidential information intended only for the use of recipients. If you are not the intended recipients, you are hereby notified that any use, dissemination, distribution or copying of this information is strictly prohibited.

NOTE: All land corners are assumed unless otherwise noted. The well location shown hereon is not an as-built location.

Scale 1"=1000'

I, Donald Leischner, Professional Land Surveyor, N.D. No. 7103, do hereby certify that the survey plat shown hereon was made by me, or under my direction, from notes made in the field, and the same is true and correct to the best of my knowledge and belief.

Jeff McElhiney

Surveyed By

06/14/2011

Date

Vertical Control Datum Used
Sea-Level Datum of NAVD 88
Based on elevation derived from OPUS Solution on GPS* Q462 Located a distance of 9501.84' on an azimuth of 203°09'37" from the SW corner of Section 11 T.153N., R.101W., 5th P.M. being at 2099.99' Elevation MSL.

Project No. 8711180

Book 0-89 Pg. 74-79 Staking

Professional Consulting Engineers
and Surveyors
Registered in
North Dakota, South Dakota
Montana, Wyoming & Minnesota
Tele-Fax No. 701-572-2019
Bus. Phone No. 701-572-6352
222 Airport Road
Williston, North Dakota 58801-2976
Certificate of Authorization #C-061

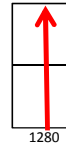
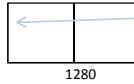


Kadrmass
Lee &
Jackson
Engineers Surveyors
Planners



GEOLOGIC PROGNOSIS

Well Name: **Dahl 15-11H**
Prospect: Baker Field
County: McKenzie
State: North Dakota
Twp / Rge: **153N-101W**



Offset Wells

- 1 SME Lindvig #1-11-3C 2105' k.b.
se/se sec.11-153N-101W
- 2

GL elev.: **2110** GL
KB elev. Datum: **2127** KB
Xtreme 18

		Geologic Tops					
		Est.			Actual		
Fm Name/Poss Pay	Depth	thickness	Subsea	Depth	thickness	Subsea	
Greenhorn	4628	845	-2501				
Dakota ss	5473	394	-3346				
Jm	5867	1022	-3740				
Dunham Salt	6889	69	-4762				
Trsp	6958	314	-4831				
Pine Salt	7272	52	-5145				
base Pine	7324	37	-5197				
Minnekahta	7361	20	-5234				
Opeche Salt	7381	8	-5254				
base Opeche	7389	438	-5262				
Tyler	7827	539	-5700				
Kibbey L.S.	8366	147	-6239				
Charles	8513	699	-6386				
BLS	9212	71	-7085				
Ratcliffe	9283	150	-7156				
Mission Canyon	9433	584	-7306				
Lodgepole	10017	679	-7890				
Carrington Sh	10696	9	-8569				
Scallion L.S.	10705	9	-8578				
Mb Upper Sh	10714	13	-8587				
Mb Middle	10727	10	-8600				
Target	10737	30	-8610				
Mb Lower Shale	10767	30	-8640				
Dtf							
Target							
Dtf1-mbb							

Offset Wells					
#1			#2		
Depth	Thickness	Subsea	Depth	Thickness	Subsea
4625	845	-2520			
5470	394	-3365			
5864	1022	-3759			
6886	69	-4781			
6955	314	-4850			
7269	52	-5164			
7321	37	-5216			
7358	20	-5253			
7378	8	-5273			
7386	438	-5281			
7824	539	-5719			
8363	147	-6258			
8510	699	-6405			
9209	71	-7104			
9280	150	-7175			
9430	584	-7325			
10014	679	-7909			
10693	9	-8588			
10702	9	-8597			
10711	13	-8606			
10724	10	-8619			
10734	30	-8629			
10764	30	-8659			
10794		-8689			

Mudlogging 2-man unit with chromatograph

KOP @ XXXXX to EOL

Openhole Logs none - wavier request from NDIC filed w/APD

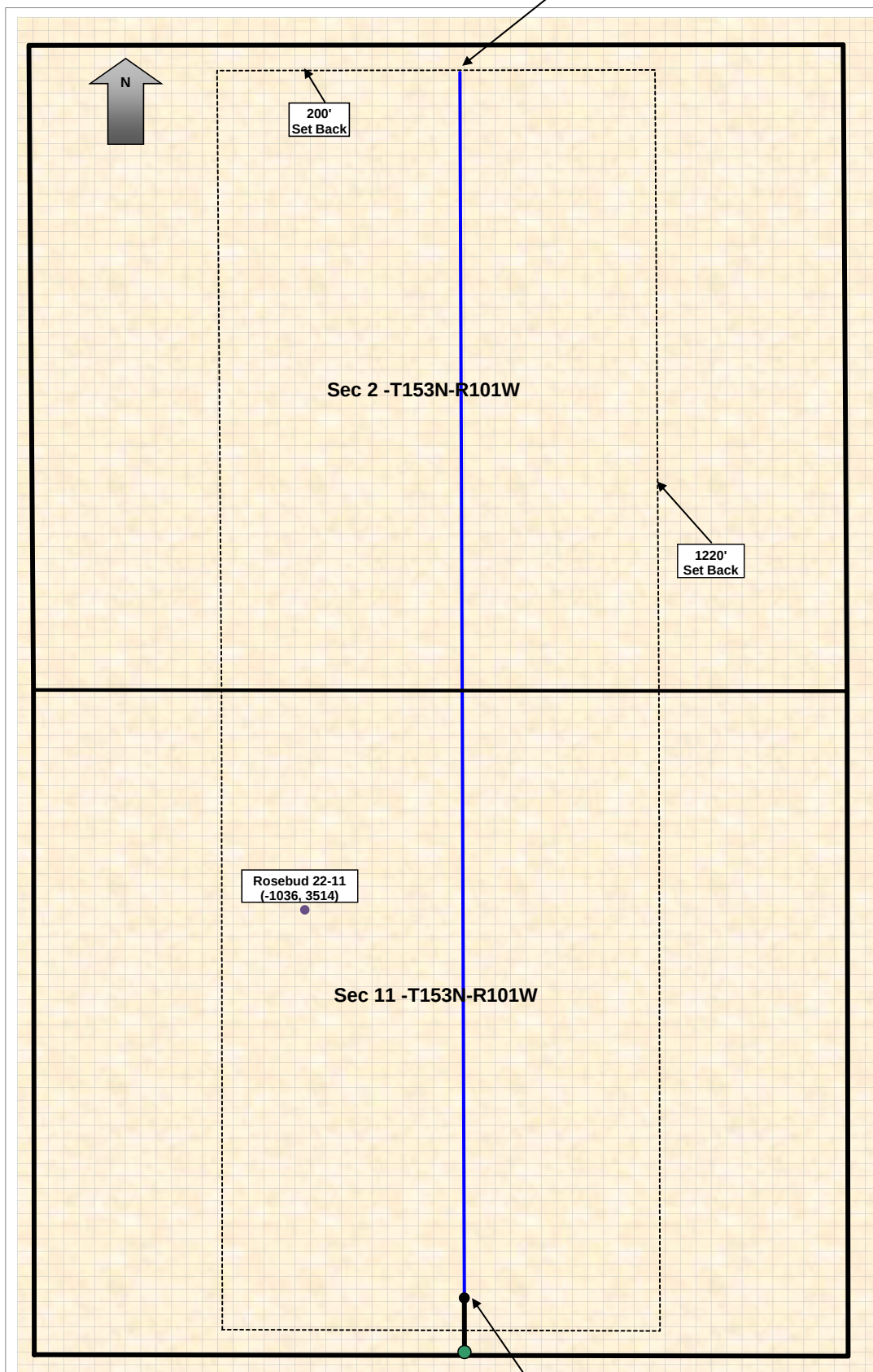
Core / DST none

Geologist Mike Bryant Bob Bachman
406-255-8625 office 406-255-8621 office
406-860-7970 cell 406-861-8836 cell

Engineer Greg Hilton
406-869-8744 office
406-861-0566 cell



Est. TD: 20,725 FT MD
FT of Pay 9,735 FT
BH Target Coord: 30 W 10164 N
BH Target: 2518 FEL & 215 FNL



Dahl 15-11H
Location: SWSE Sec 11 - T153N - R101W
Footage: 2488 FEL & 22 FSL
Elev: Graded Pad 2110, 2127 KB

7" Casing: 10,991 FT MD
Csg Shoe Coord: 1 W 429 N
BHL: 2489 FEL & 451 FSL

Dahl 15-11H

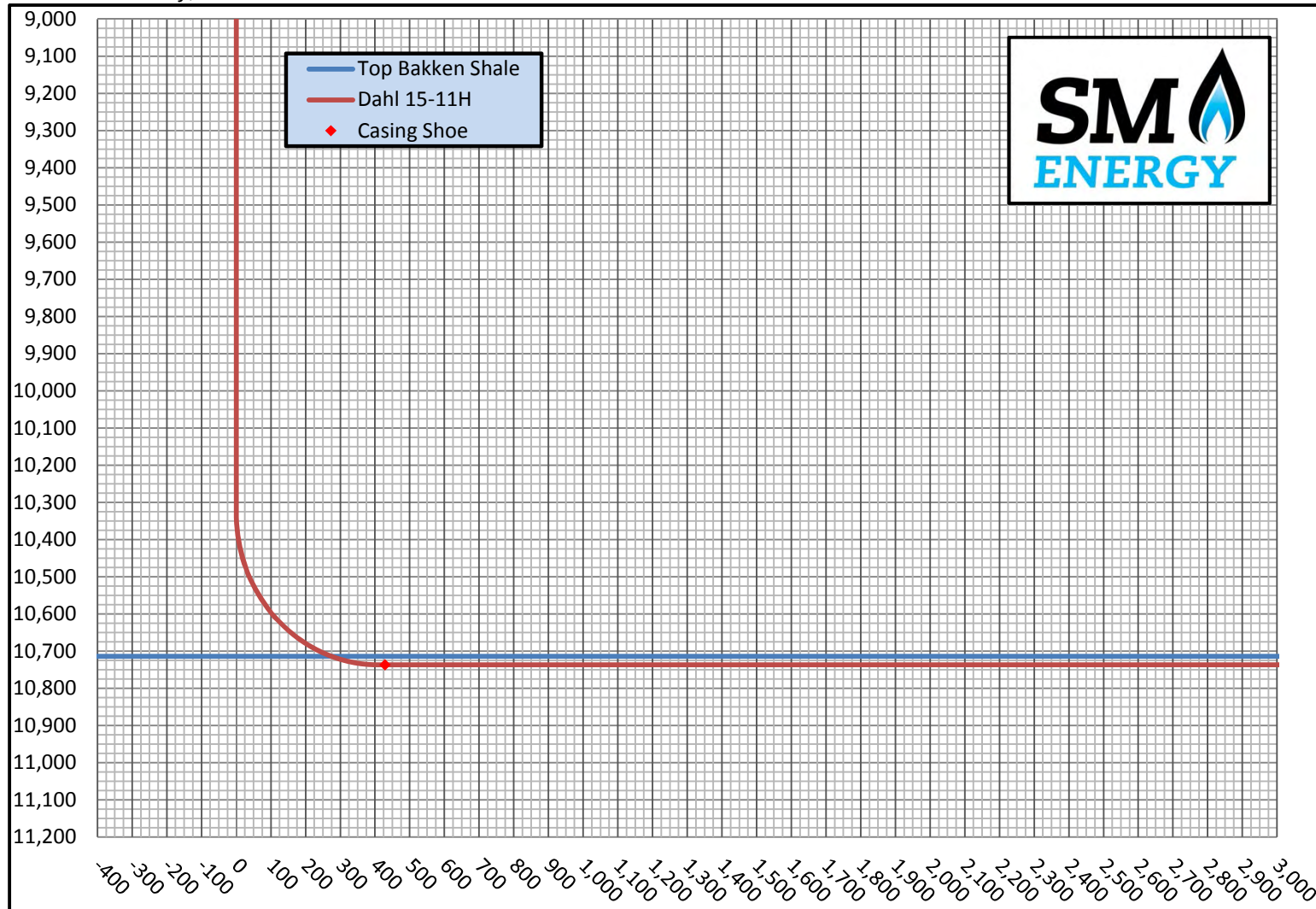
Location: SWSE Sec 11 - T153N - R101W

Footage: 2488 FEL & 22 FSL

Elev: Graded Pad 2110, 2127 KB

McKenzie County, ND

TVD VERSUS VERTICAL SECTION (AZ = 359.83)



		Company SM Energy						Tgt. Inc. 90					
		Well Dahl 15-11H						Mag. Decl. _____					
		Build Rate 14.00						Target TVD 10,737					
		*NOTE: ALL AZIMUTHS RELATIVE TO TRUE NORTH						Target Az. 359.8308866		10164 N		10,164 VS	
Tgt. Coord. 30 W													
Date	No.	DEPTH	INC.	*AZM	C.L.	T.V.D.	V.S.	N/S	E/W	DLS	B./D.	Walk	BRN
	1			359.83									
SURF CSG SHOE	2	2,100		359.83	2,100.00	2100.00							0.66
KOP-START CURVE	3	10,328		359.83	8,227.74	10327.74							14.00
	4	10,341	1.80	359.83	12.86	10340.60	0.20	0.20	0.00	14.00	14.00		14.00
	5	10,353	3.60	359.83	12.86	10353.44	0.81	0.81	0.00	14.00	14.00		14.00
	6	10,366	5.40	359.83	12.86	10366.26	1.82	1.82	-0.01	14.00	14.00		14.00
	7	10,379	7.20	359.83	12.86	10379.04	3.23	3.23	-0.01	14.00	14.00		14.00
	8	10,392	9.00	359.83	12.86	10391.77	5.04	5.04	-0.01	14.00	14.00		14.00
	9	10,405	10.80	359.83	12.86	10404.43	7.25	7.25	-0.02	14.00	14.00		14.00
	10	10,418	12.60	359.83	12.86	10417.02	9.86	9.86	-0.03	14.00	14.00		14.00
	11	10,431	14.40	359.83	12.86	10429.52	12.86	12.86	-0.04	14.00	14.00		14.00
	12	10,443	16.20	359.83	12.86	10441.92	16.25	16.25	-0.05	14.00	14.00		14.00
	13	10,456	18.00	359.83	12.86	10454.21	20.03	20.03	-0.06	14.00	14.00		14.00
	14	10,469	19.80	359.83	12.86	10466.37	24.19	24.19	-0.07	14.00	14.00		14.00
	15	10,482	21.60	359.83	12.86	10478.40	28.74	28.74	-0.08	14.00	14.00		14.00
	16	10,495	23.40	359.83	12.86	10490.28	33.66	33.66	-0.10	14.00	14.00		14.00
	17	10,508	25.20	359.83	12.86	10502.00	38.95	38.95	-0.11	14.00	14.00		14.00
	18	10,521	27.00	359.83	12.86	10513.54	44.61	44.61	-0.13	14.00	14.00		14.00
	19	10,533	28.80	359.83	12.86	10524.90	50.62	50.62	-0.15	14.00	14.00		14.00
	20	10,546	30.60	359.83	12.86	10536.07	56.99	56.99	-0.17	14.00	14.00		14.00
	21	10,559	32.40	359.83	12.86	10547.03	63.71	63.71	-0.19	14.00	14.00		14.00
	22	10,572	34.20	359.83	12.86	10557.78	70.77	70.77	-0.21	14.00	14.00		14.00
	23	10,585	36.00	359.83	12.86	10568.30	78.16	78.16	-0.23	14.00	14.00		14.00
	24	10,598	37.80	359.83	12.86	10578.58	85.88	85.88	-0.25	14.00	14.00		14.00
	25	10,611	39.60	359.83	12.86	10588.61	93.92	93.92	-0.28	14.00	14.00		14.00
	26	10,623	41.40	359.83	12.86	10598.39	102.27	102.27	-0.30	14.00	14.00		14.00
	27	10,636	43.20	359.83	12.86	10607.90	110.92	110.92	-0.33	14.00	14.00		14.00
	28	10,649	45.00	359.83	12.86	10617.13	119.87	119.87	-0.35	14.00	14.00		14.00
	29	10,662	46.80	359.83	12.86	10626.08	129.10	129.10	-0.38	14.00	14.00		

HORIZONTAL SECTION PLAT

SM Energy Company
P.O. Box 7168 Billings, MT 59103

Dahl 15-11H

22 feet from the south line and 2488 feet from the east line (surface location)

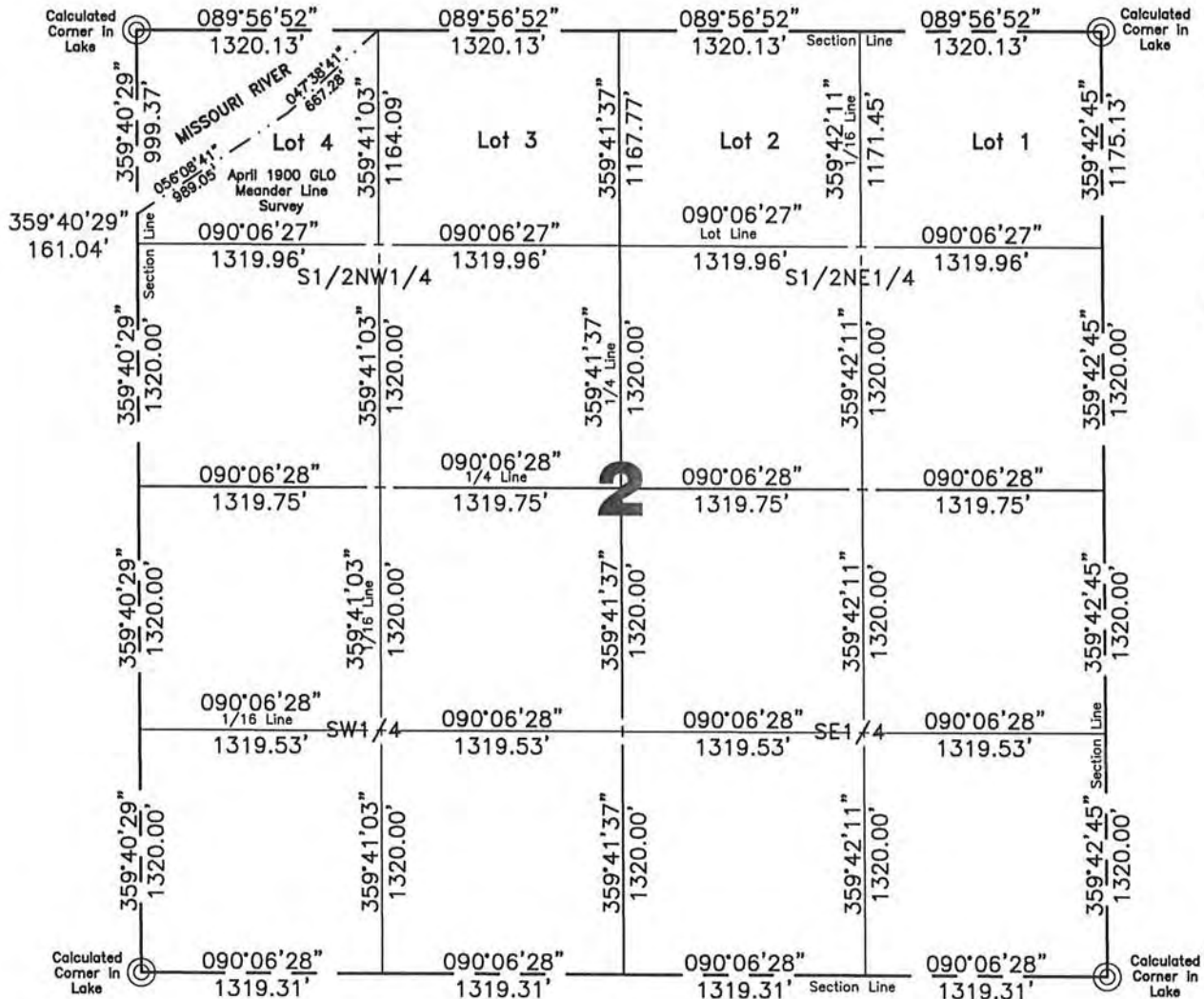
Section 11, T. 153 N., R. 101 W., 5th P.M.

McKenzie County, North Dakota

Surface owner @ well site - Sharon D. Eilers et al

Latitude 48°04'56.376" North; Longitude 103°38'09.239" West (surface location)

[Derived from OPUS Solution NAD-83(CORS96)]

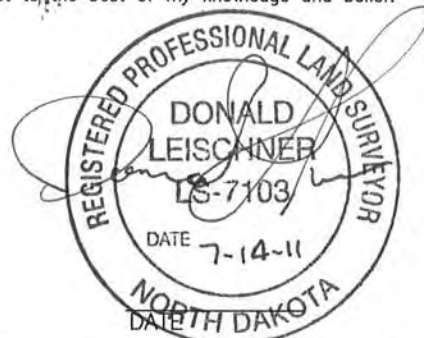


Confidentiality Notice: The information contained on this plat is legally privileged and confidential information intended only for the use of recipients. If you are not the intended recipients, you are hereby notified that any use, dissemination, distribution or copying of this information is strictly prohibited.

Scale 1"=1000'

I, Donald Leischner, Professional Land Surveyor, N.D. No. 7103, do hereby certify that the survey plat shown hereon was made by me, or under my direction, from notes made in the field, and the same is true and correct to the best of my knowledge and belief.

All corners shown on this plat were found in the field during SM Energy, Dahl 15-11H oil well survey on June 14, 2010. Distances to all others are calculated. All azimuths are based on the south line of Section 11, being on an azimuth of 090°06'47".



Kadrmass
Lee &
Jackson
Engineers Surveyors
Planners

Surveyed By	Field Book
J. McElhiney	0-89
Computed & Drawn By	Project No.
M. Meismer	8711180

HORIZONTAL SECTION PLAT

SM Energy Company
P.O. Box 7168 Billings, MT 59103

Dahl 15-11H

22 feet from the south line and 2488 feet from the east line (surface location)

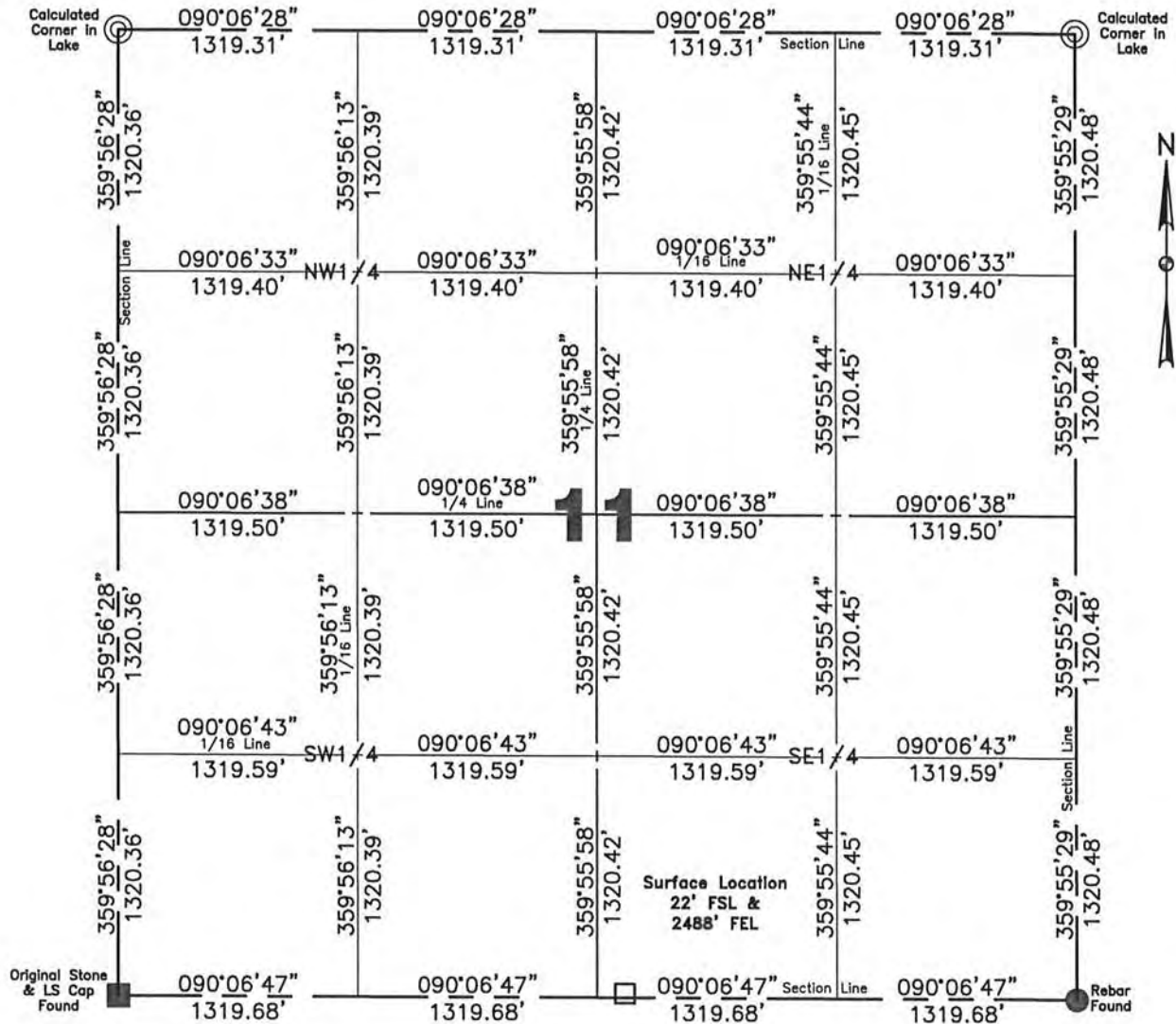
Section 11, T. 153 N., R. 101 W., 5th P.M.

McKenzie County, North Dakota

Surface owner @ well site - Sharon D. Eilers et al

Latitude 48°04'56.376" North; Longitude 103°38'09.239" West (surface location)

[Derived from OPUS Solution NAD-83(CORS96)]

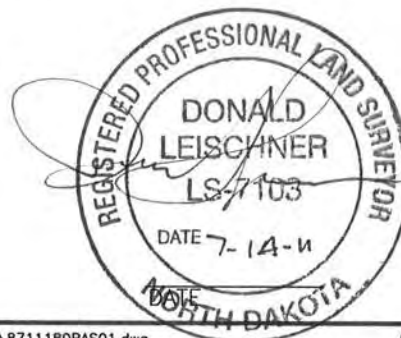


Confidentiality Notice: The information contained on this plat is legally privileged and confidential information intended only for the use of recipients. If you are not the intended recipients, you are hereby notified that any use, dissemination, distribution or copying of this information is strictly prohibited.

Scale 1"=1000'

I, Donald Leischner, Professional Land Surveyor, N.D. No. 7103, do hereby certify that the survey plat shown hereon was made by me, or under my direction, from notes made in the field, and the same is true and correct to the best of my knowledge and belief.

All corners shown on this plat were found in the field during SM Energy, Dahl 15-11H oil well survey on June 14, 2010. Distances to all others are calculated. All azimuths are based on the south line of Section 11, being on an azimuth of 090°06'47".



Kadmas
Lee &
Jackson
Engineers Surveyors
Planners

Surveyed By J. McElhiney	Field Book 0-89
Computed & Drawn By M. Meismer	Project No. 8711180

BOTTOM HOLE LOCATION PLAT

SM Energy Company
P.O. Box 7168 Billings, MT 59103

Dahl 15-11H

22 feet from the south line and 2488 feet from the east line (surface location)

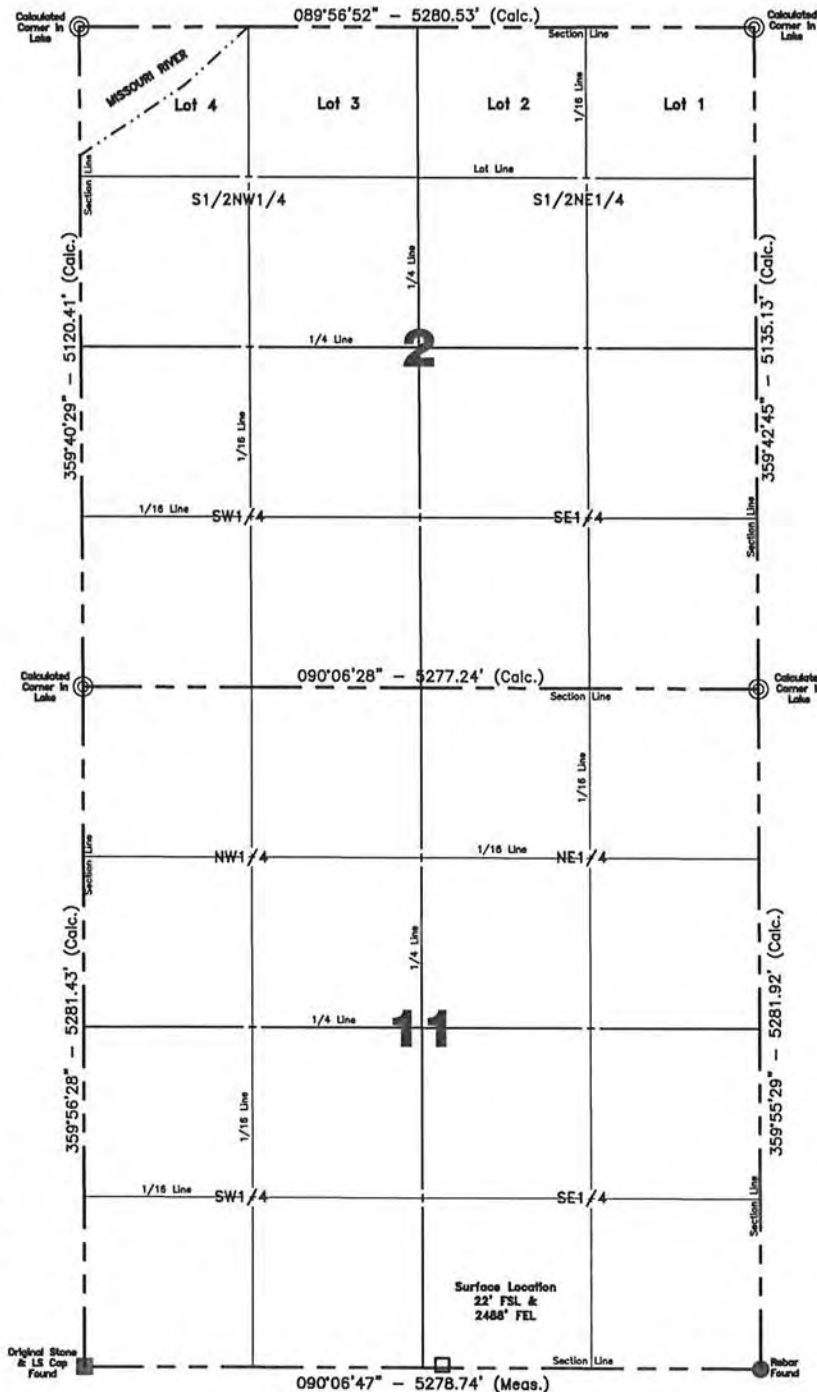
Section 11, T. 153 N., R. 101 W., 5th P.M.

McKenzie County, North Dakota

Surface owner @ well site - Sharon D. Eilers et al

Latitude 48°04'56.376" North; Longitude 103°38'09.239" West (surface location)

[Derived from OPUS Solution NAD-83(CORS96)]



Scale 1"=1500'

Confidentiality Notice:
The information contained on this plat is legally privileged and confidential information intended only for the use of recipients. If you are not the intended recipients, you are hereby notified that any use, dissemination, distribution or copying of this information is strictly prohibited.

Computed & Drawn By M. Meismer	Surveyed By J. McElhiney	Approved By D. Leischner	Scale 1"=1500'	Date 07/01/2011
Field Book O-89	Material B.H. Layout	Revised -	Project No. 8711180	Drawing No. 4

Kadrmass
Lee &
Jackson
Engineers Surveyors
Planners

SM Energy Company
Dahl 15-11H
Section 11, T 153 N, R 101 W, 5th P.M.
McKenzie County, North Dakota

Well Site Elevation 2111.7' MSL
Well Pad Elevation 2109.9' MSL

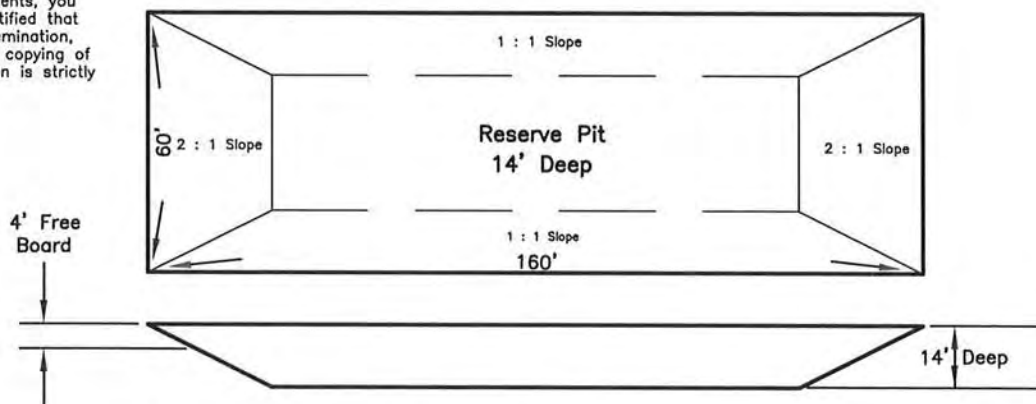
Excavation	6,210 C.Y.
Plus Pit	3,150 C.Y.
	9,360 C.Y.
Embankment	2,275 C.Y.
Plus Shrinkage (+30%)	685 C.Y.
	2,960 C.Y.
Stockpile Pit	3,150 C.Y.
Stockpile Top Soil (6")	2,910 C.Y.
Production Rehabilitation	0 C.Y.
Road Embankment & Stockpile from Pad	340 C.Y.
Disturbed Area From Pad	3.61 Acres

NOTE :
All cut end slopes are designed at 1:1 slopes &
all fill end slopes are designed at 1 1/2:1 slopes

Well Site Location

22' FSL
2488' FEL

Confidentiality Notice:
The information contained
on this plat is legally
privileged and confidential
information intended only
for the use of recipients.
If you are not the
intended recipients, you
are hereby notified that
any use, dissemination,
distribution or copying of
this information is strictly
prohibited.



Drawn By M. Meismer	Surveyed By J. McElhiney	Approved By D. Leischner	Scale None	Date 07/07/2011
Field Book O-89	Material Quantities	Revised —	Project No. 8711180	Drawing No. 5

Kadrmass
Lee &
Jackson
Engineers Surveyors
Planners

Dahl 15-11H Pad Layout

NE1/4NW1/4
Section 14
T. 153 N., R. 101 W.,
5th P.M.

SE1/4SW1/4 Section 11
T. 153 N., R. 101 W., 5th P.M.

SW1/4SE1/4 Section 11
T. 153 N., R. 101 W., 5th P.M.

NW1/4NE1/4 Section 14
T. 153 N., R. 101 W., 5th P.M.

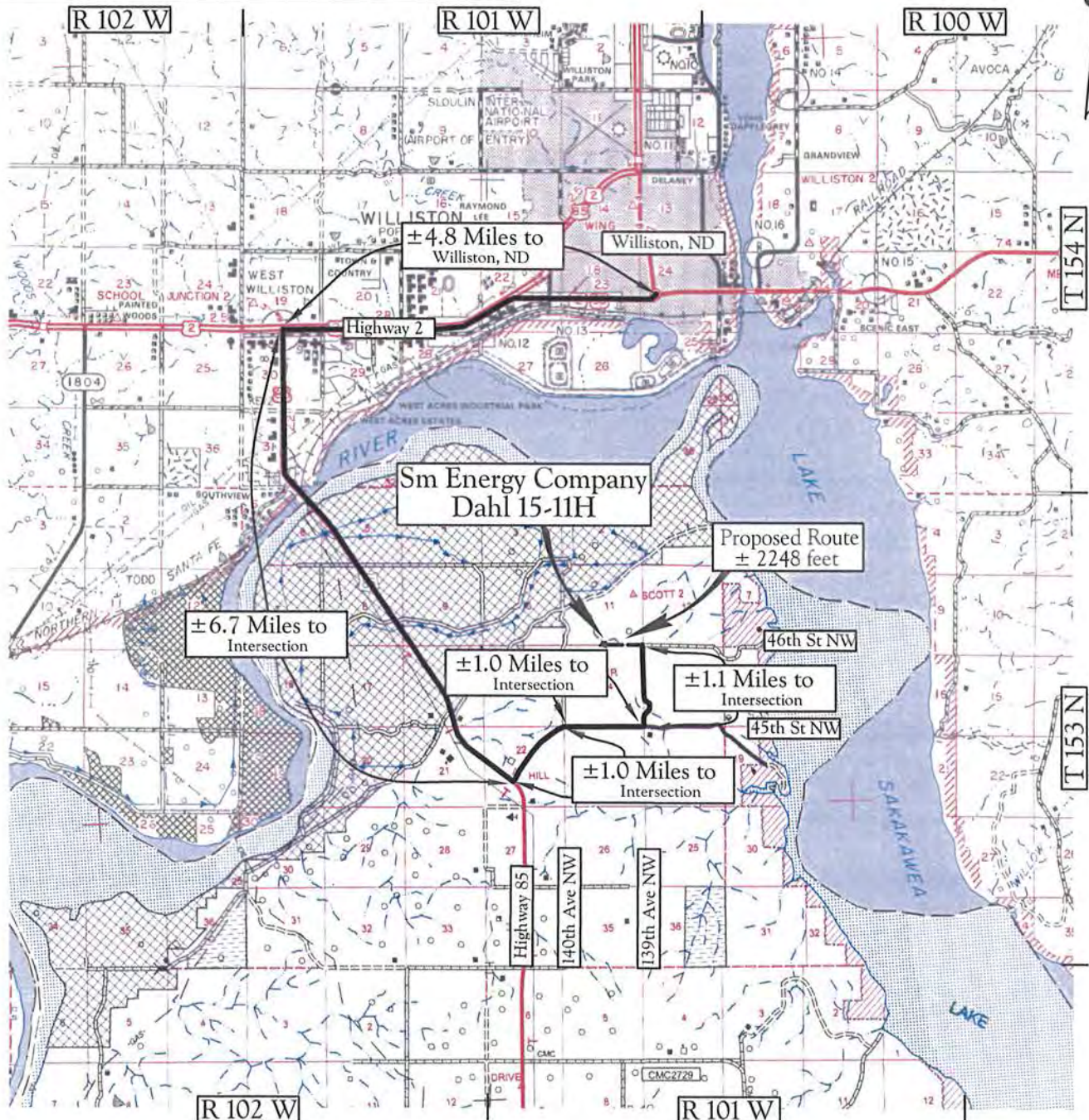
Confidentiality Notice: The information contained on this plot is legally privileged and confidential information intended only for the use of recipients. If you are not the intended recipients, you are hereby notified that any use, dissemination, distribution or copying of this information is strictly prohibited.

Drawn By M. Meisner	Surveyed By J. McElhiney	Approved By D. Leischner	Scale 1" = 80'	Date 07/07/2011
Field Book O-89	Material Pad Layout	Revised —	Project No. 8711180	Drawing No. 6

Kadmas
Lee &
Jackson
Engineers Surveyors
Planners

SM Energy Company
 Dahl 15-11H
 22' FSL & 2488' FEL
 SW1/4SE1/4 Section 11
 T. 153 N., R.101W., 5th P.M.
 McKenzie County, ND

Confidentiality Notice: The information contained on this plot is legally privileged and confidential information intended only for the use of recipients. If you are not the intended recipients, you are hereby notified that any use, dissemination, distribution or copying of this information is strictly prohibited.



Scale 1" = 2 Miles

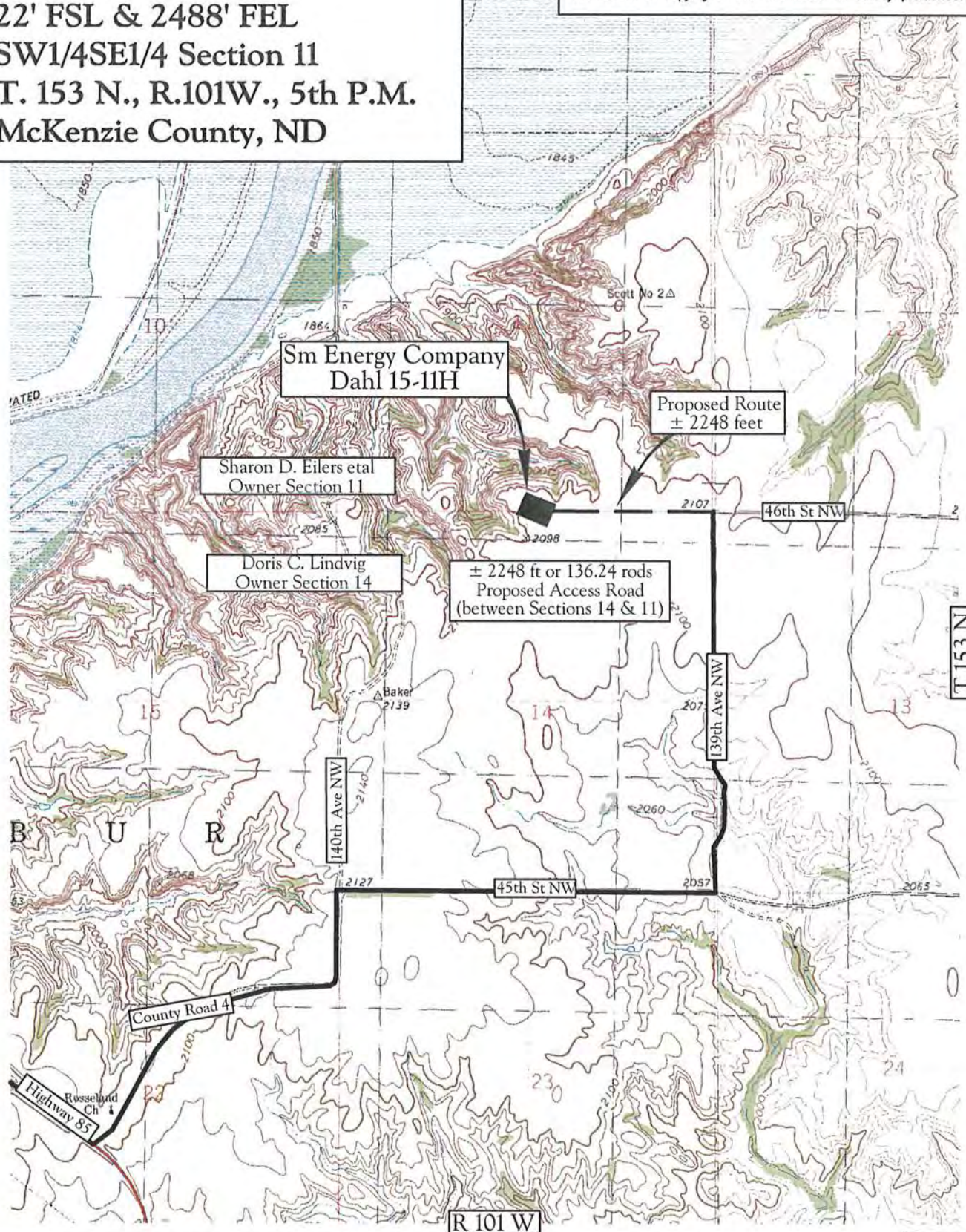
Map "A"
 County Access Route

Legend
 Existing Roads ———
 Proposed Roads - - - - -

Kadmas
 Lee &
 Jackson
 Engineers Surveyors
 Planners

SM Energy Company
Dahl 15-11H
22' FSL & 2488' FEL
SW1/4SE1/4 Section 11
T. 153 N., R.101W., 5th P.M.
McKenzie County, ND

Confidentiality Notice: The information contained on this plot is legally privileged and confidential information intended only for the use of recipients. If you are not the intended recipients, you are hereby notified that any use, dissemination, distribution or copying of this information is strictly prohibited.



Map "B"
Quad Access Route

Legend
Existing Roads —————
Proposed Roads - - - - -

Scale 1"=2000'

Kadmas
Lee &
Jackson
Engineers Surveyors
Planners